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Bang et al.

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(54) **METHOD, APPARATUS AND STORAGE MEDIUM FOR IMAGE ENCODING/DECODING USING SUBPICTURE**
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This patent is subject to a terminal disclaimer.

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US 2024/0040112 A1 Feb. 1, 2024

Related U.S. Application Data
(63) Continuation of application No. 17/955,278, filed on Sep. 28, 2022, now Pat. No. 11,812,013, which is a (Continued)

(30) **Foreign Application Priority Data**
Apr. 3, 2020 (KR) 10-2020-0040765
Mar. 31, 2021 (KR) 10-2021-0041924

(51) **Int. Cl.**
H04N 19/105 (2014.01)
H04N 19/169 (2014.01)
(Continued)

(52) **U.S. Cl.**
CPC **H04N 19/105** (2014.11); **H04N 19/174** (2014.11); **H04N 19/188** (2014.11); **H04N 19/70** (2014.11); **H04N 19/46** (2014.11)
(58) **Field of Classification Search**
CPC .. H04N 19/105; H04N 19/174; H04N 19/188; H04N 19/70; H04N 19/46
See application file for complete search history.

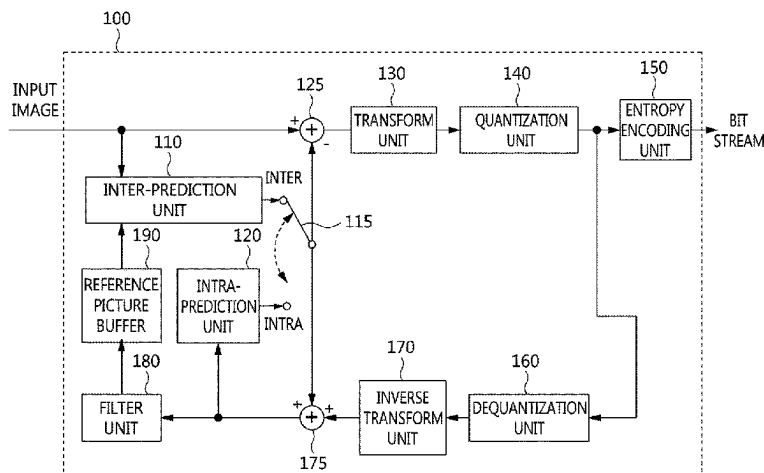
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(57) **ABSTRACT**
Disclosed herein are a method, an apparatus and a storage medium for performing encoding/decoding on an image using a subpicture. A picture may be partitioned into various units such as subpictures and slices, and various syntax elements and structures may be used to specify the units. In an embodiment, there is a method for configuring a reference picture list for motion compensation in the case where the location of a reference subpicture to be referred to for motion compensation is different from the location of a target subpicture. Further, in an embodiment, there is pro-
(Continued)



vided a method for providing the configuration of a reference picture list and an override on the number of activated reference indexes.

17 Claims, 42 Drawing Sheets

Related U.S. Application Data

continuation of application No. 17/221,635, filed on Apr. 2, 2021, now Pat. No. 11,496,730.

(51) **Int. Cl.**
H04N 19/174 (2014.01)
H04N 19/70 (2014.01)
H04N 19/46 (2014.01)

(56)

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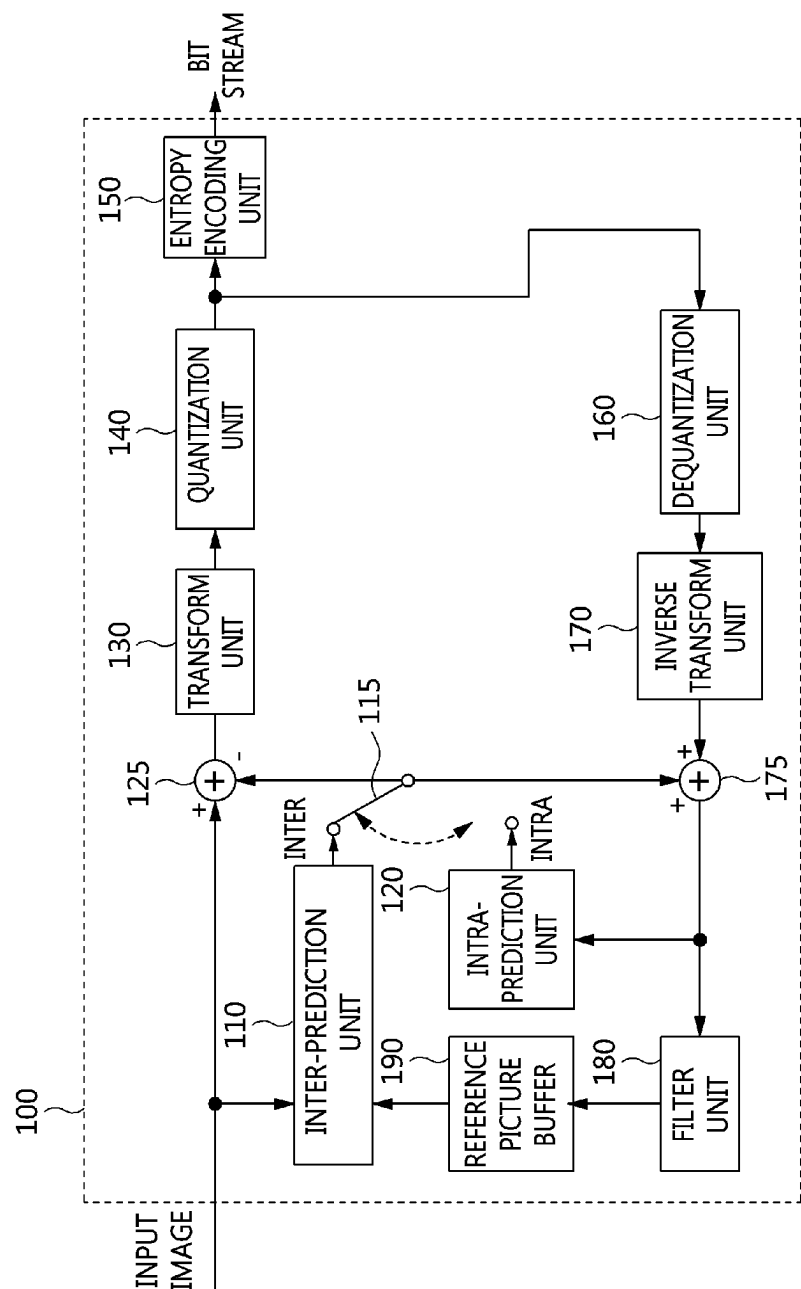


FIG. 1

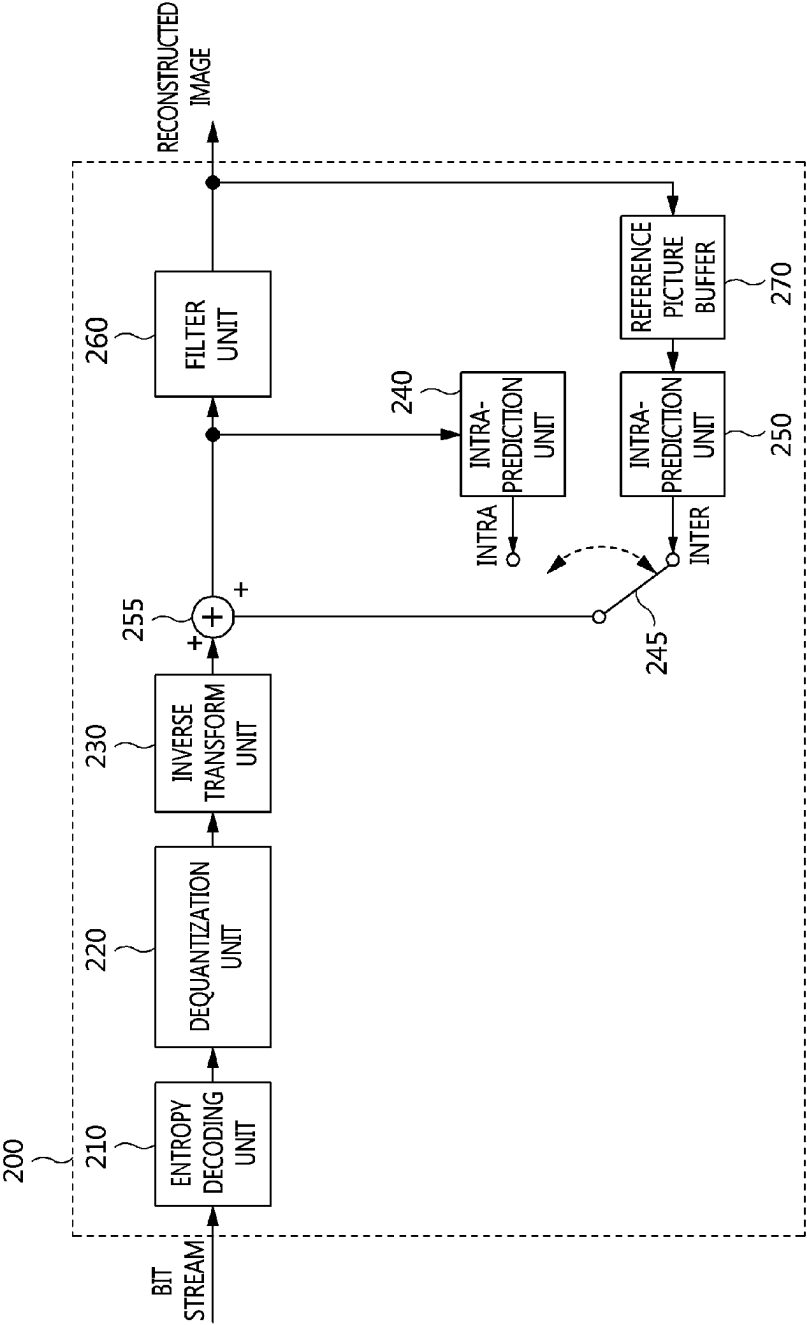


FIG. 2

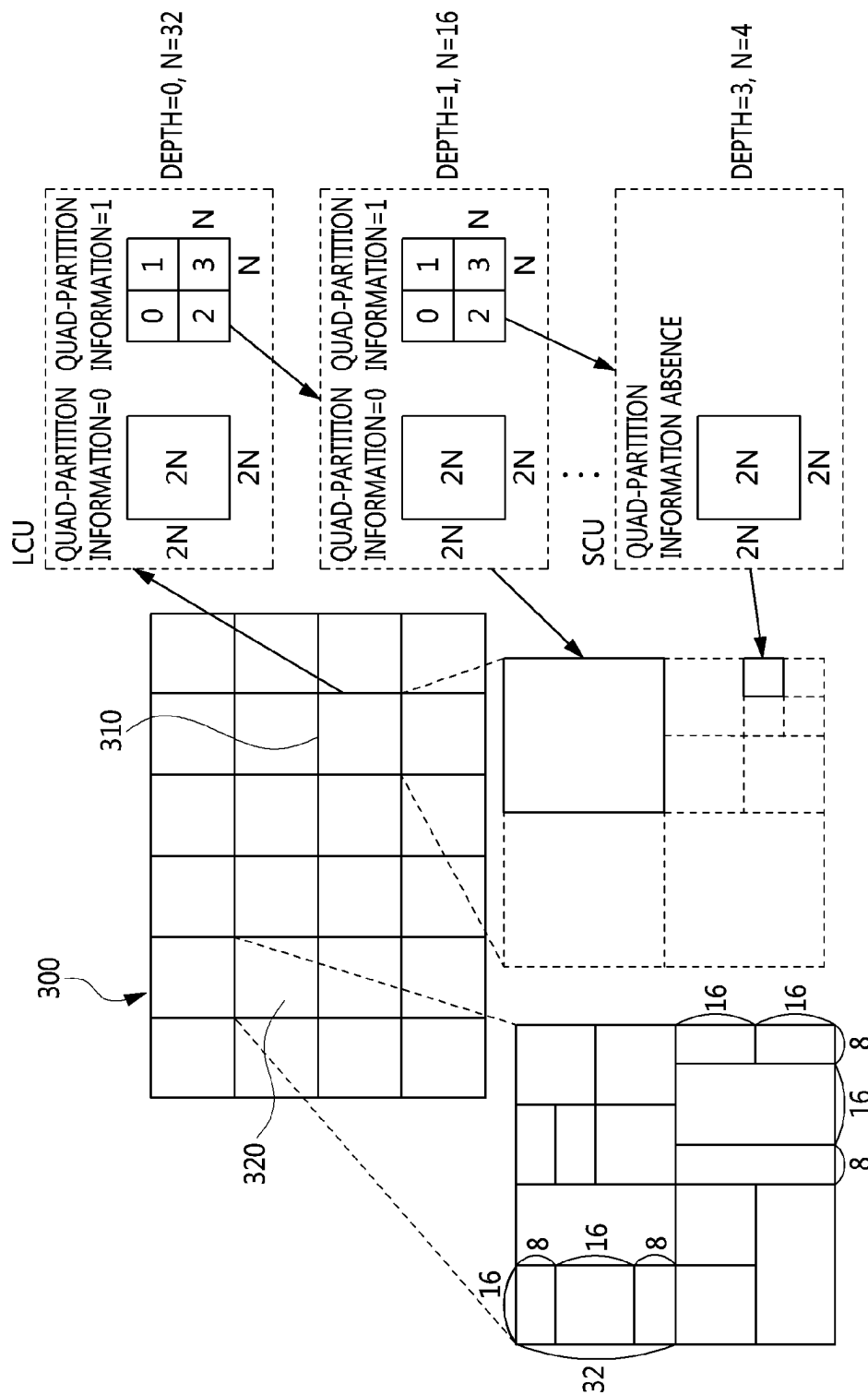
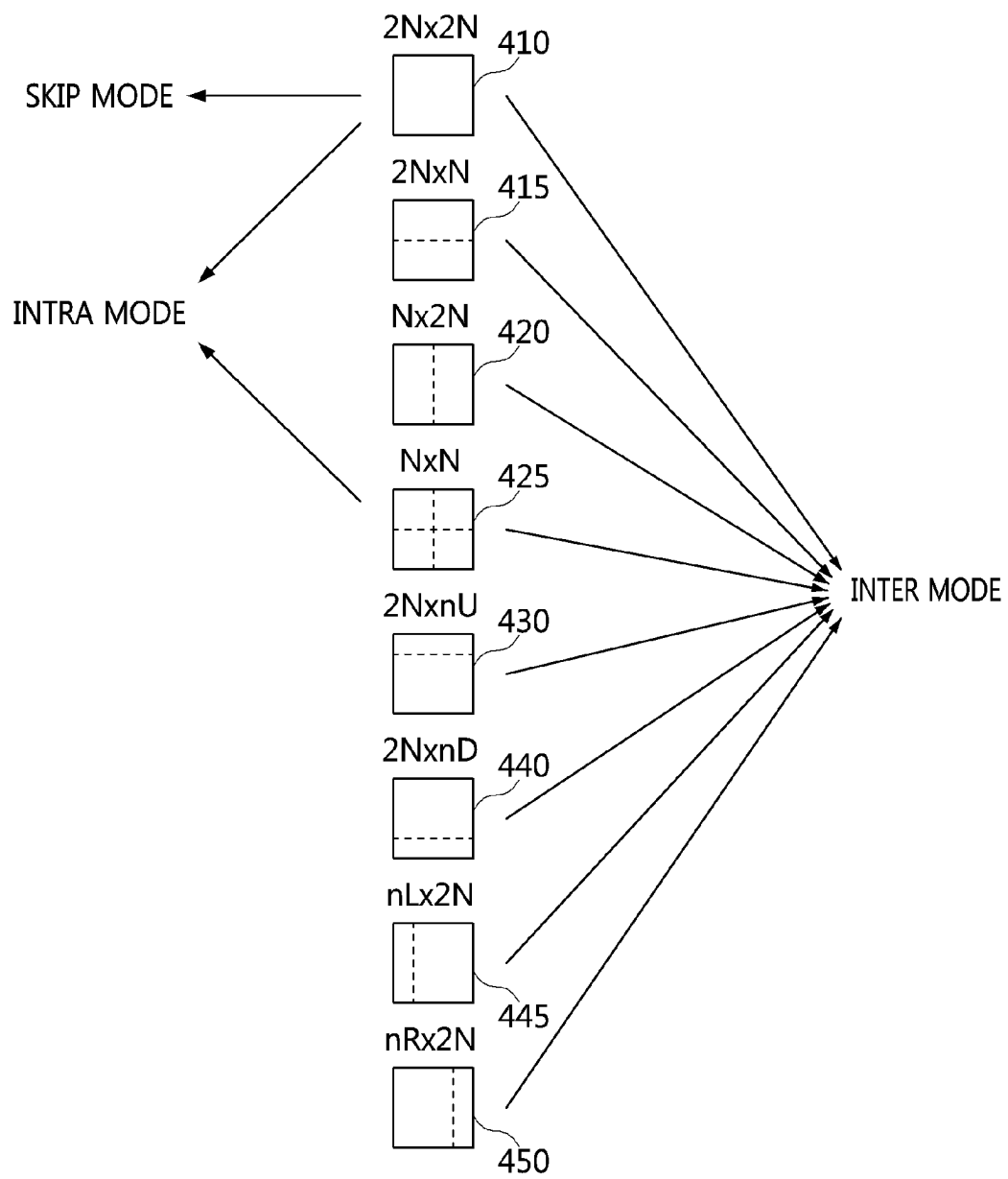


FIG. 3

**FIG. 4**

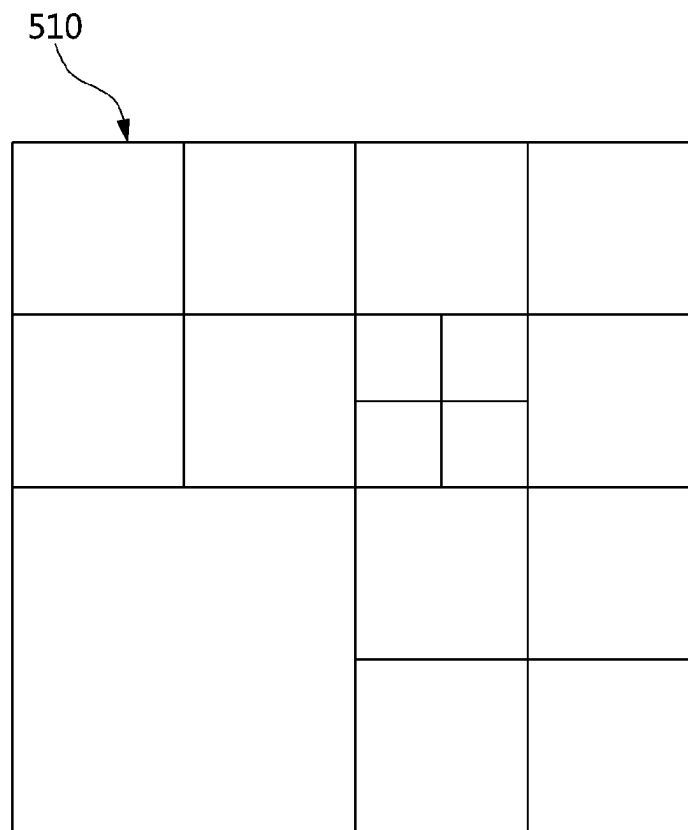


FIG. 5

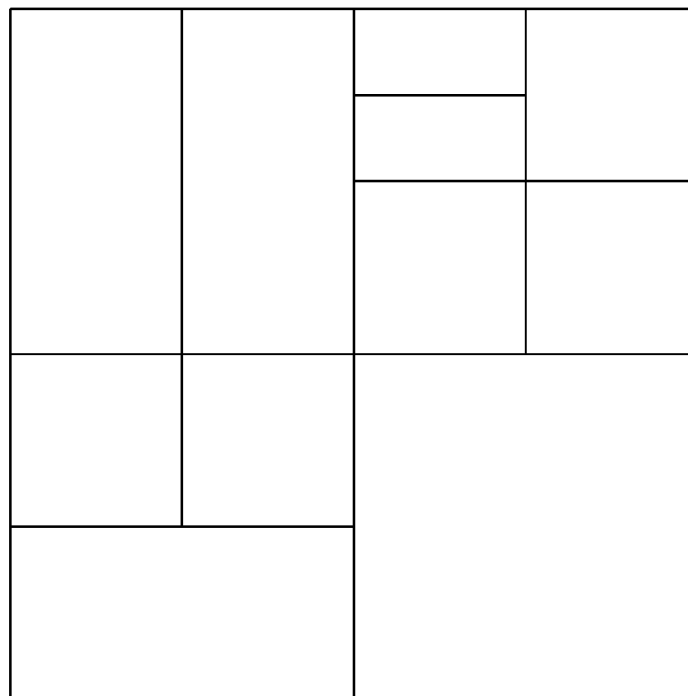


FIG. 6

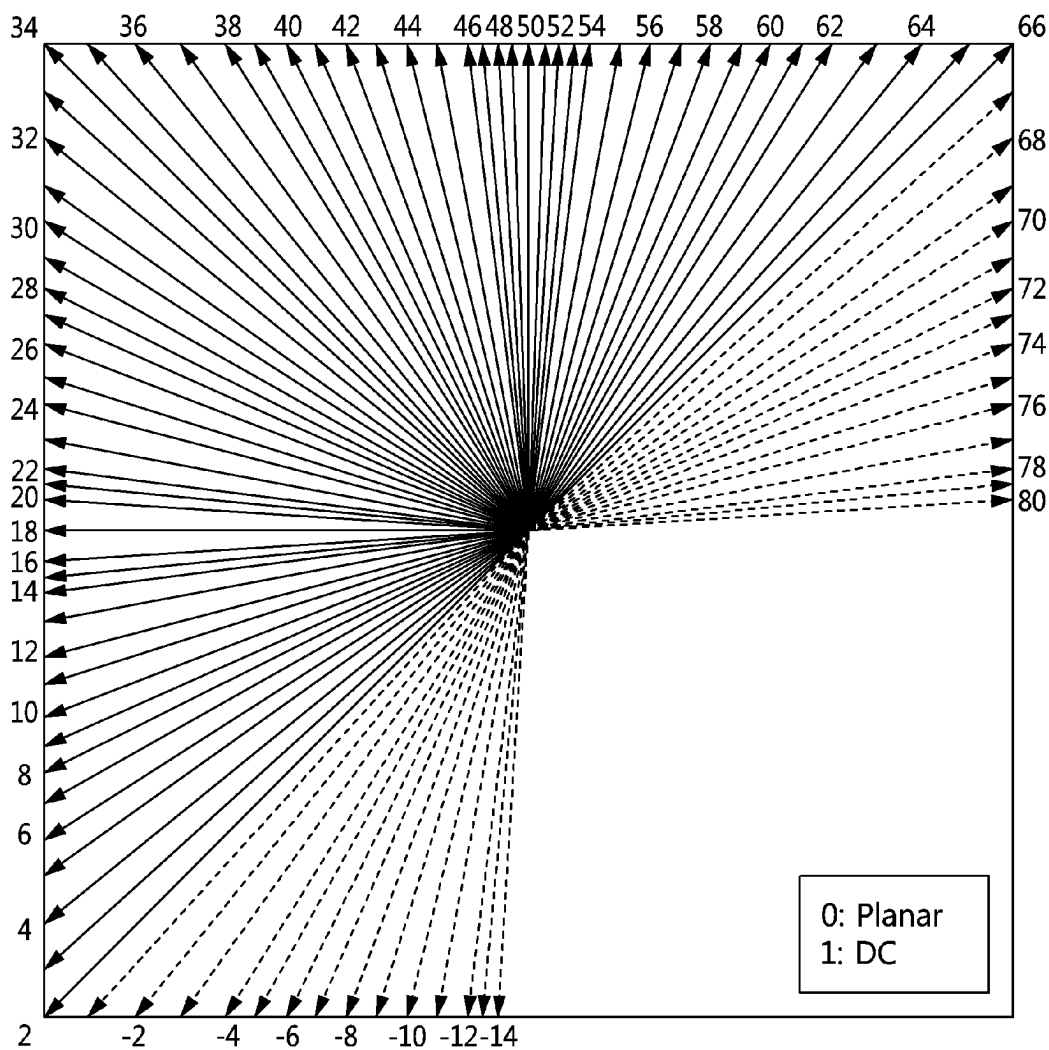


FIG. 7

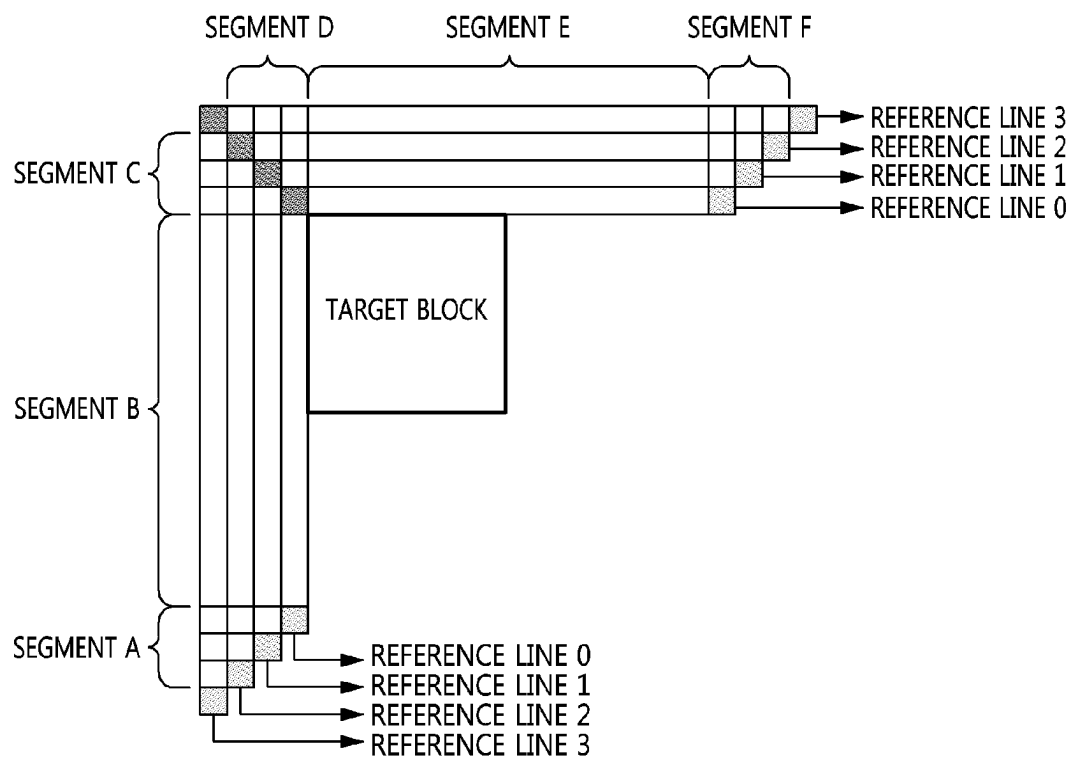


FIG. 8

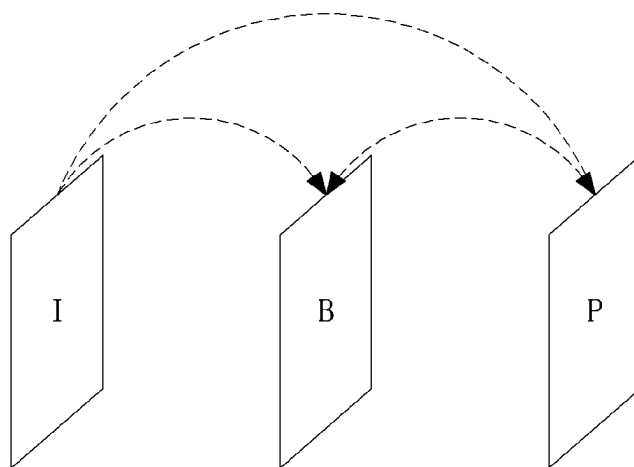
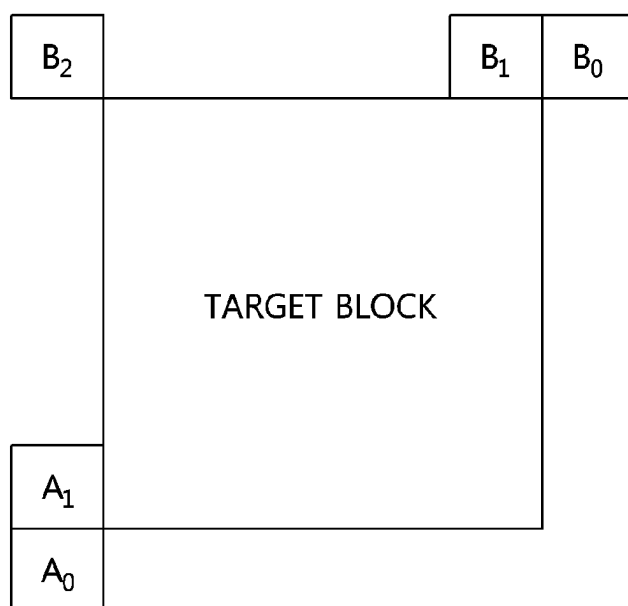


FIG. 9

**FIG. 10**

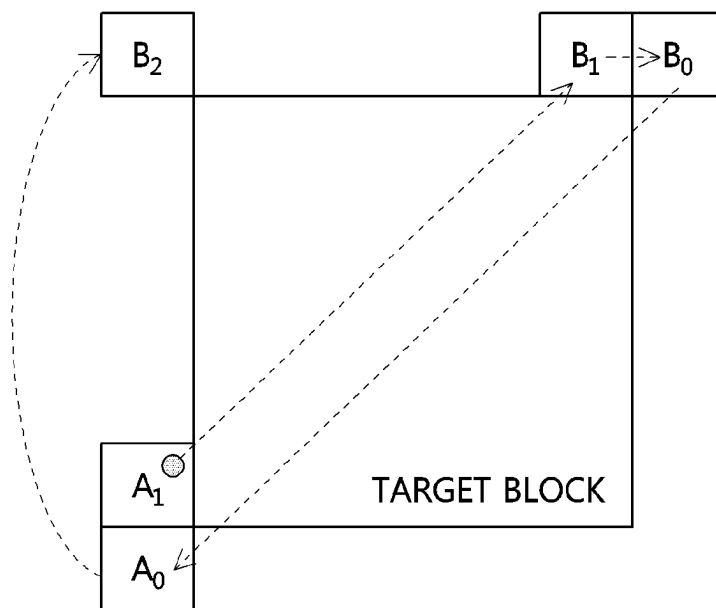


FIG. 11

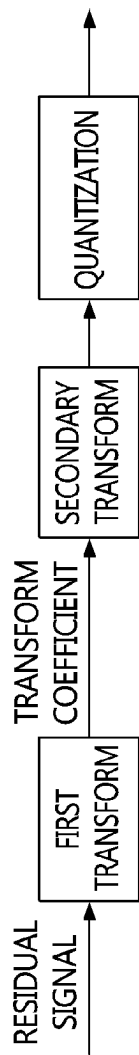
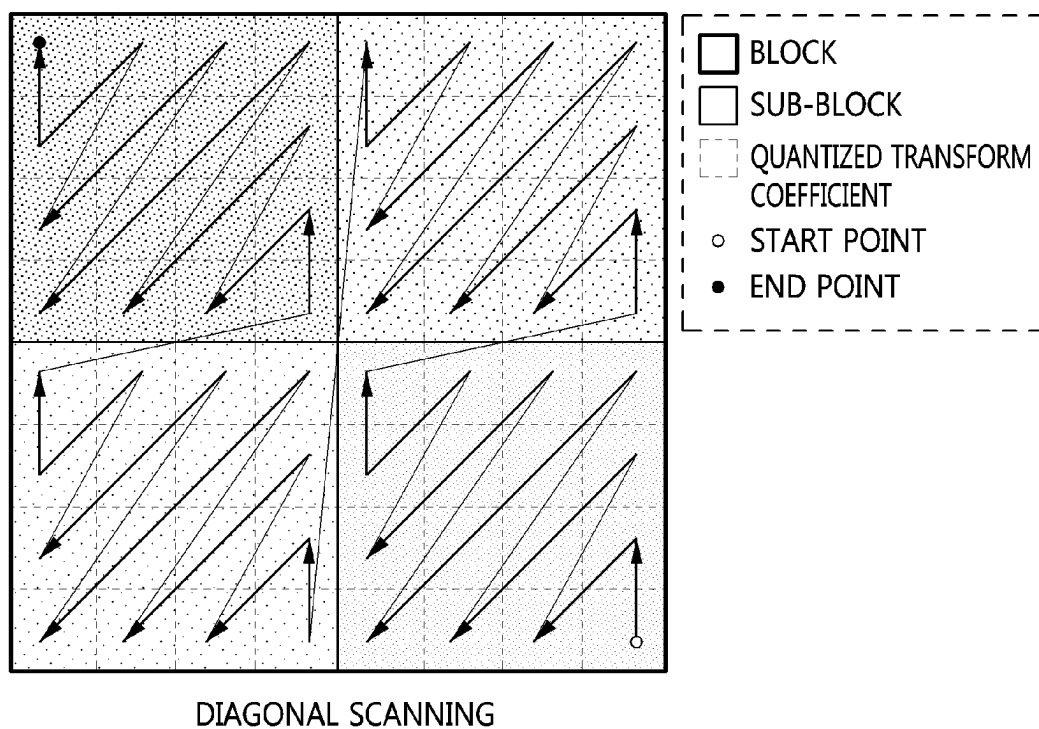
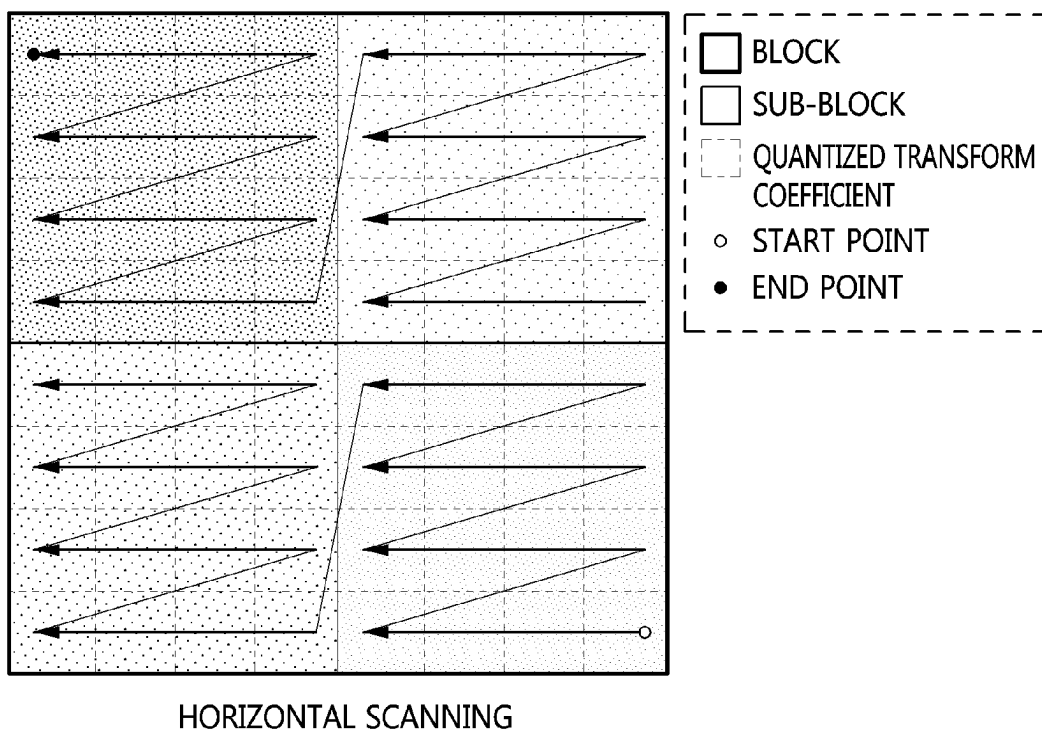


FIG. 12

**FIG. 13**

**FIG. 14**

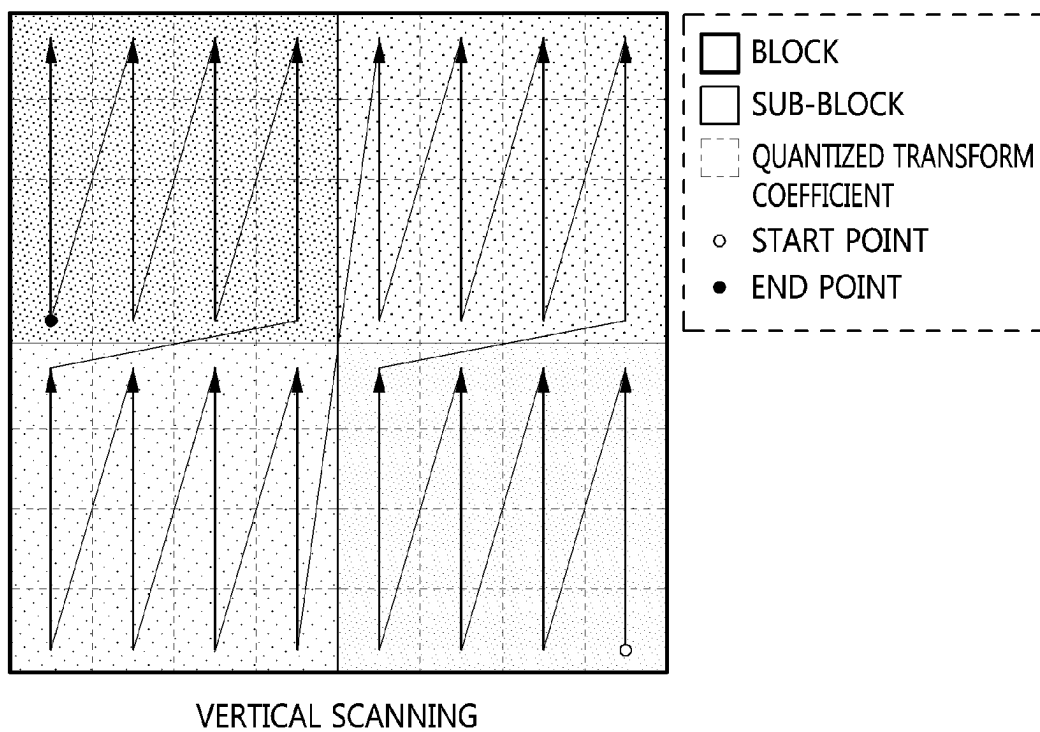
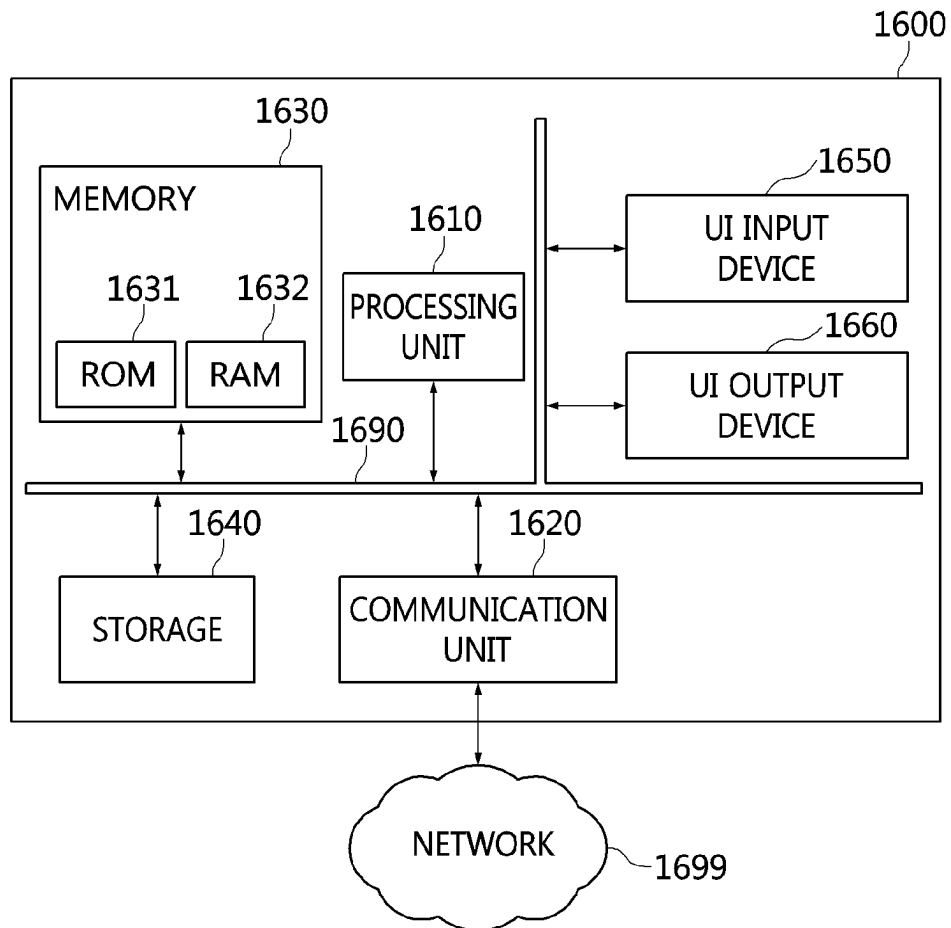
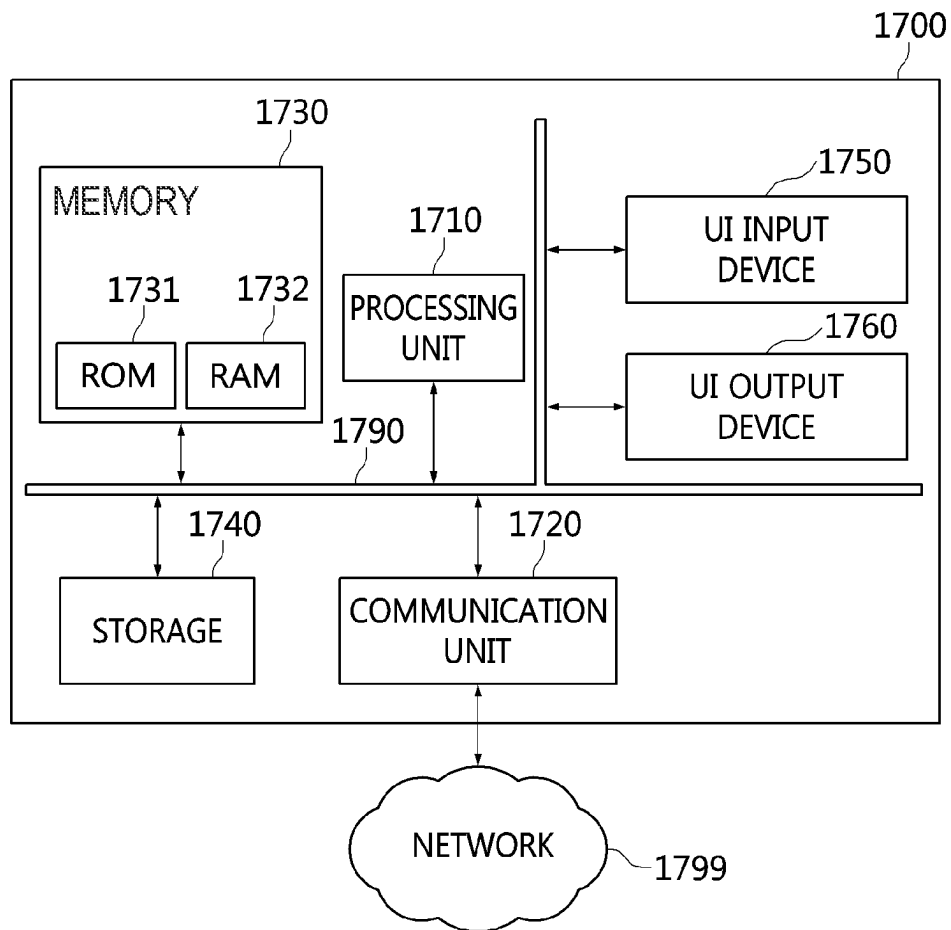
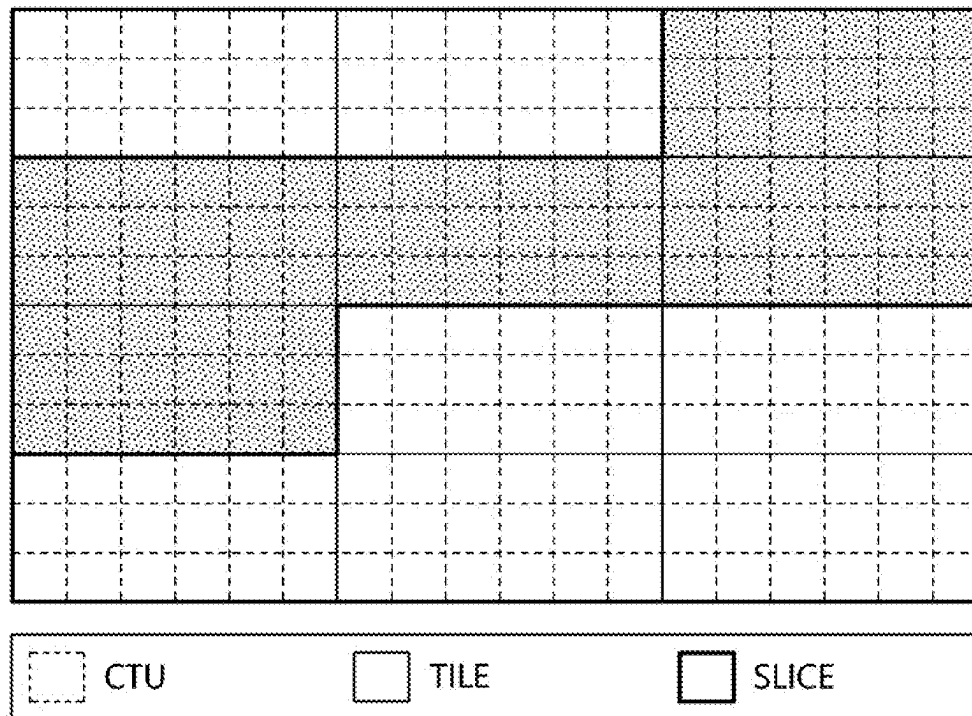


FIG. 15

**FIG. 16**

**FIG. 17**

**FIG. 18**

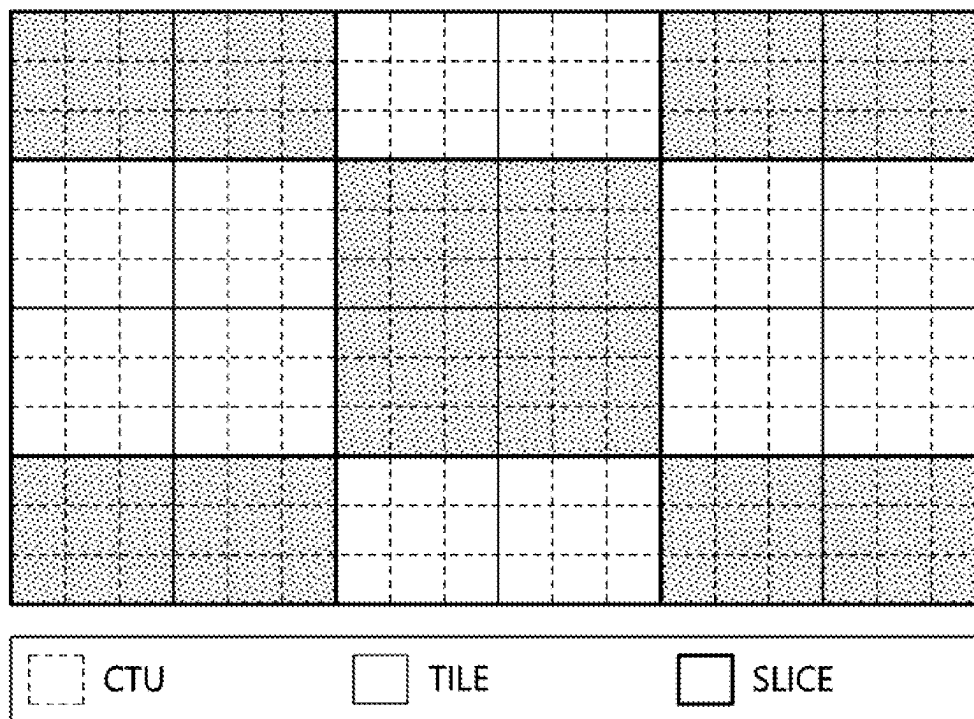


FIG. 19

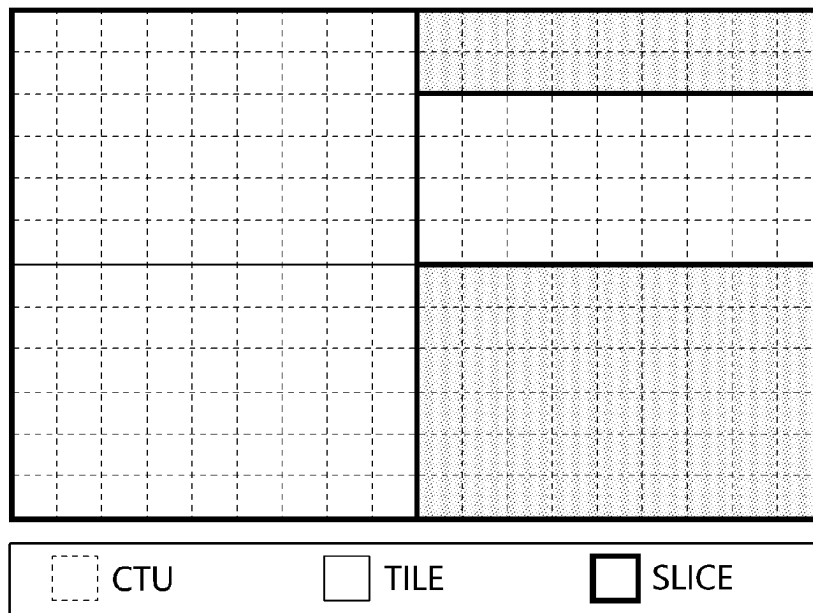


FIG. 20

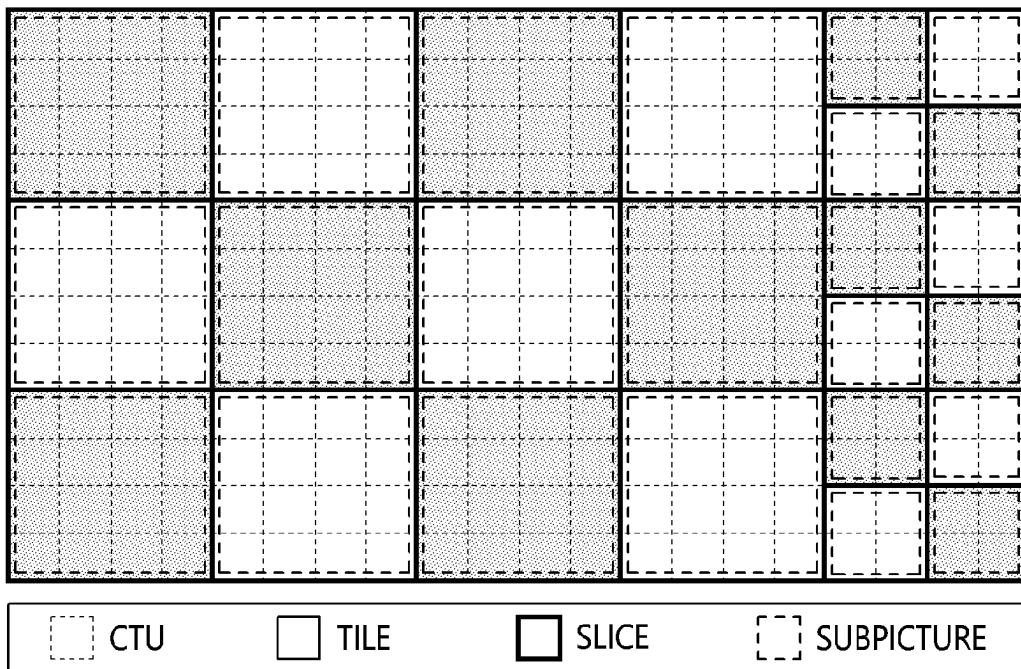


FIG. 21

```
for( i = 0; i <= sps_num_subpics_minus1; i++ )  
    if( subpic_id_mapping_explicitly_signalled_flag )  
        SubpicIdVal[ i ] = subpic_id_mapping_in_pps_flag ? pps_subpic_id[ i ] : sps_subpic_id[ i ]  
    else  
        SubpicIdVal[ i ] = i
```

FIG. 22

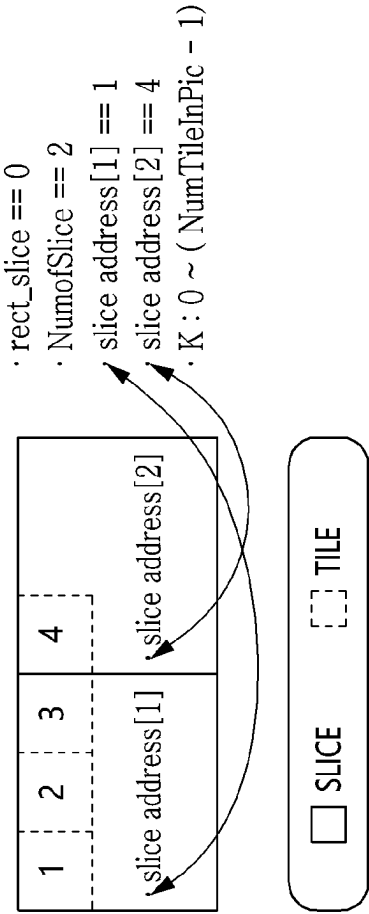


FIG. 23

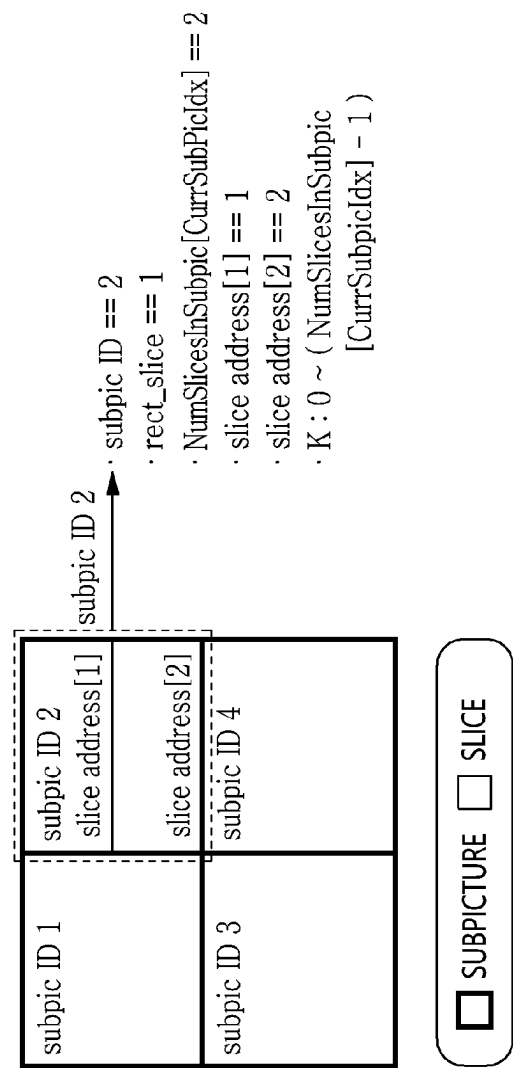


FIG. 24

seq_parameter_set_rbsp() {	Desc.
...	
subpic_id_mapping_explicitly_signalled_flag	
if(subpic_id_mapping_explicitly_signalled_flag) {	
subpic_id_mapping_in_sps_flag	u(1)
if(subpic_id_mapping_in_sps_flag)	
for(i = 0; i <= sps_num_subpics_minus1; i++)	
sps_subpic_id[i]	u(v)
}	
}	
...	

FIG. 25

```
for( i=0;i<=sps_num_subpics_minus1;i++)  
    sps_subpic_id[i] = i
```

FIG. 26

```
function AddCtbToSlice(sliceldx, startX, stopX, startY, stopY):  
    for( ctbY = startY; ctbY < stopY; ctbY++ )  
        for( ctbX = startX; ctbX < stopX; ctbX++ ) {  
            CtbAddrInSlice[ sliceldx ][ NumCtusInSlice[ sliceldx ] ] = ctbY * PicWidthInCtbsY + ctbX  
            NumCtusInSlice[ sliceldx ]++  
        }  
}
```

FIG. 27

```
if( rect_slice_flag ) {  
    picLevelSliceIdx = slice_address  
    for( j = 0; j < CurrSubpicIdx; j++ )  
        picLevelSliceIdx += NumSlicesInSubpic[ j ]  
    NumCtusInCurrSlice = NumCtusInSlice[ picLevelSliceIdx ]  
    for( i = 0; i < NumCtusInCurrSlice; i++ )  
        CtbAddrInCurrSlice[ i ] = CtbAddrInSlice[ picLevelSliceIdx ][ i ]  
} else {...
```

FIG. 28

pic_parameter_set_rbsp() {	Desc.
...	
subpic_id_mapping_explicitly_signalled_pps_flag	u(1)
if(subpic_id_mapping_explicitly_signalled_pps_flag) {	
subpic_id_mapping_in_pps_flag	ue(v)
if(subpic_id_mapping_in_pps_flag)	
{	
pps_num_subpics_minus1	
pps_subpic_id_len_minus1	ue(v)
for(i = 0; i <= pps_num_subpic_minus1; i++)	
pps_subpic_id[i]	u(v)
}	
}	
...	u(1)

FIG. 29

```
for( i = 0; i <= sps_num_subpics_minus1; i++ )
{
    if( subpic_id_mapping_explicitly_signalled_flag || subpic_id_mapping_explicitly_signalled_pps_flag )
        SubpicIdVal[ i ] = subpic_id_mapping_in_pps_flag ? pps_subpic_id[ i ] : sps_subpic_id[ i ]
    else
        SubpicIdVal[ i ] = i
}
```

FIG. 30

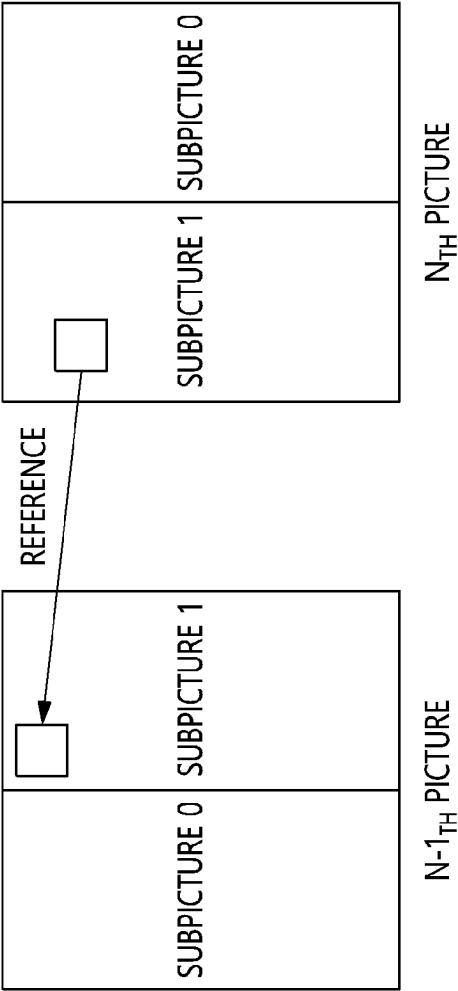


FIG. 31

seq_parameter_set_rbsp() {	Descriptor
sps_seq_parameter_set_id	u(4)
sps_video_parameter_set_id	u(4)
...	
sps_idr_rpl_present_flag	u(1)

FIG. 32

pic_parameter_set_rbsp() {	Descriptor
pps_pic_parameter_set_id	ue(v)
pps_seq_parameter_set_id	u(4)
mixed_nalu_types_in_pic_flag	u(1)
pic_width_in_luma_samples	ue(v)

FIG. 33

slice_header() {	Descriptor
picture_header_in_slice_header_flag	u(1)
...	
if(!rpl_info_in_ph_flag && ((nal_unit_type != IDR_W_RADL && nal_unit_type != IDR_N_LP) (sps_idr_rpl_present_flag && mixed_nalu_types_in_pic_flag)))	
ref_pic_lists()	
if((rpl_info_in_ph_flag ((nal_unit_type != IDR_W_RADL && nal_unit_type != IDR_N_LP) (sps_idr_rpl_present_flag && mixed_nalu_types_in_pic_flag))) && (slice_type != I && num_ref_entries[0][RplIdx[0]] > 1) (slice_type == B && num_ref_entries[1][RplIdx[1]] > 1)) {	
num_ref_idx_active_override_flag	u(1)
if(num_ref_idx_active_override_flag)	
for(i = 0; i < (slice_type == B ? 2 : 1); i++)	
if(num_ref_entries[i][RplIdx[i]] > 1)	
num_ref_idx_active_minus1[i]	ue(v)

FIG. 34

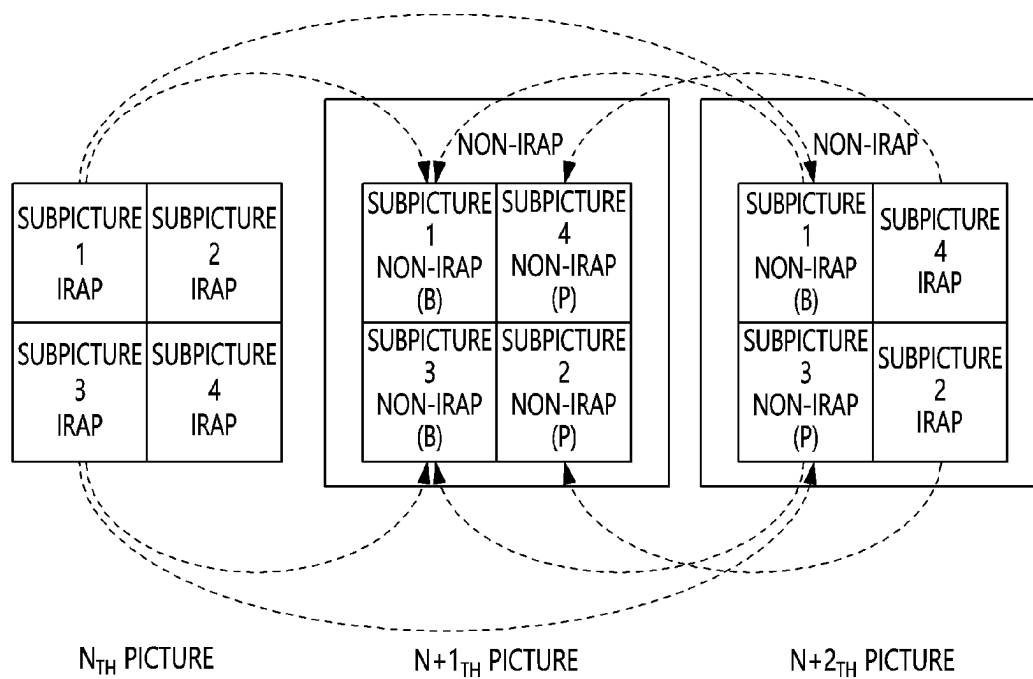


FIG. 35

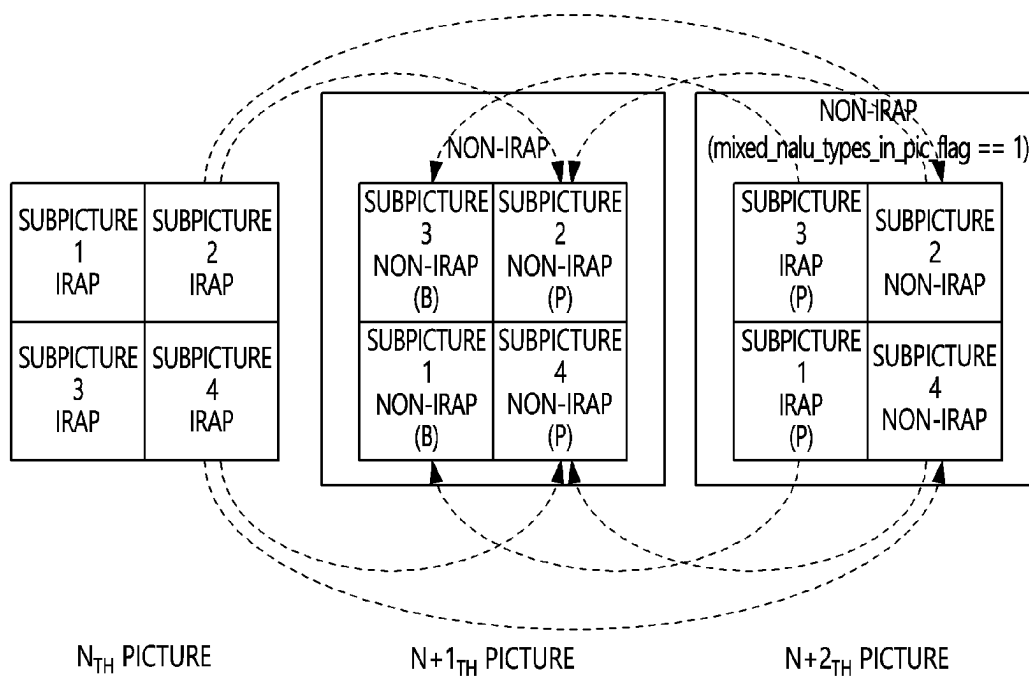


FIG. 36

slice_header() {	Descriptor
picture_header_in_slice_header_flag	u(1)
...	
if(!rpl_info_in_ph_flag && ((nal_unit_type != IDR_W_RADL && nal_unit_type != IDR_N_LP) mixed_nalu_types_in_pic_flag))	
ref_pic_lists()	
if((rpl_info_in_ph_flag ((nal_unit_type != IDR_W_RADL && nal_unit_type != IDR_N_LP) mixed_nalu_types_in_pic_flag)) && (slice_type != I && num_ref_entries[0][RplIdx[0]] > 1) (slice_type == B && num_ref_entries[1][RplIdx[1]] > 1)) {	
num_ref_idx_active_override_flag	u(1)
if(num_ref_idx_active_override_flag)	
for(i = 0; i < (slice_type == B ? 2 : 1); i++)	
if(num_ref_entries[i][RplIdx[i]] > 1)	
num_ref_idx_active_minus1[i]	ue(v)

FIG. 37

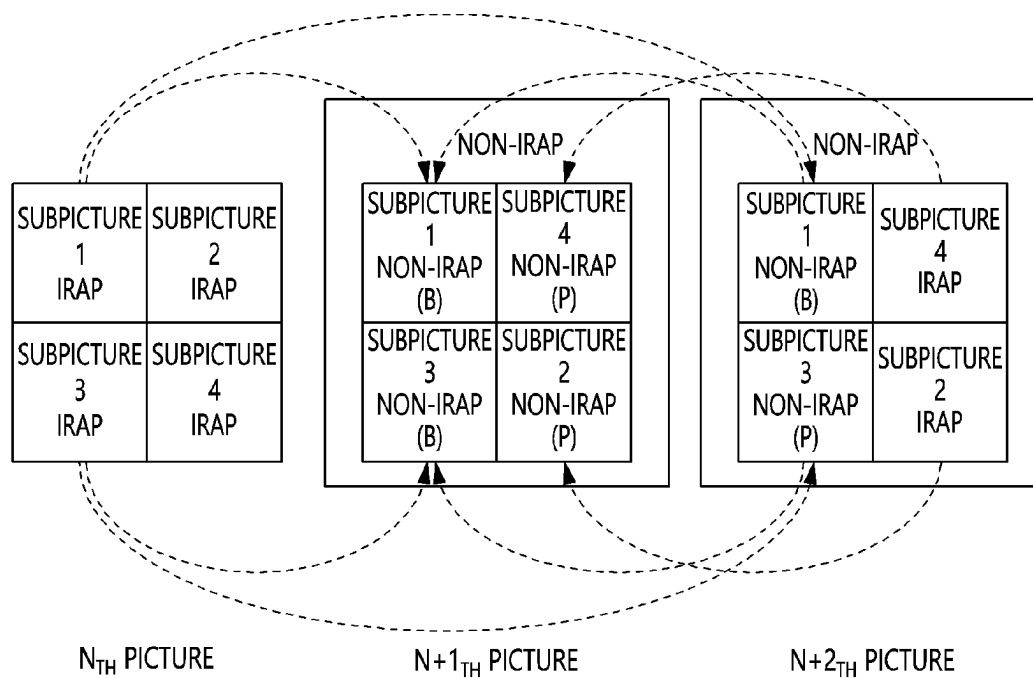


FIG. 38

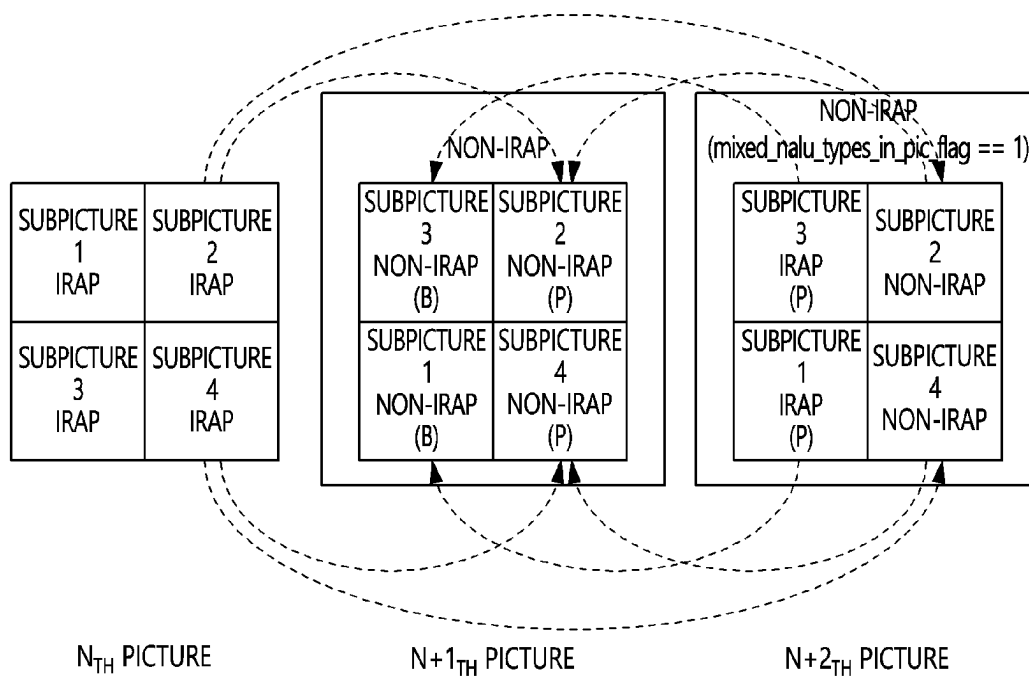
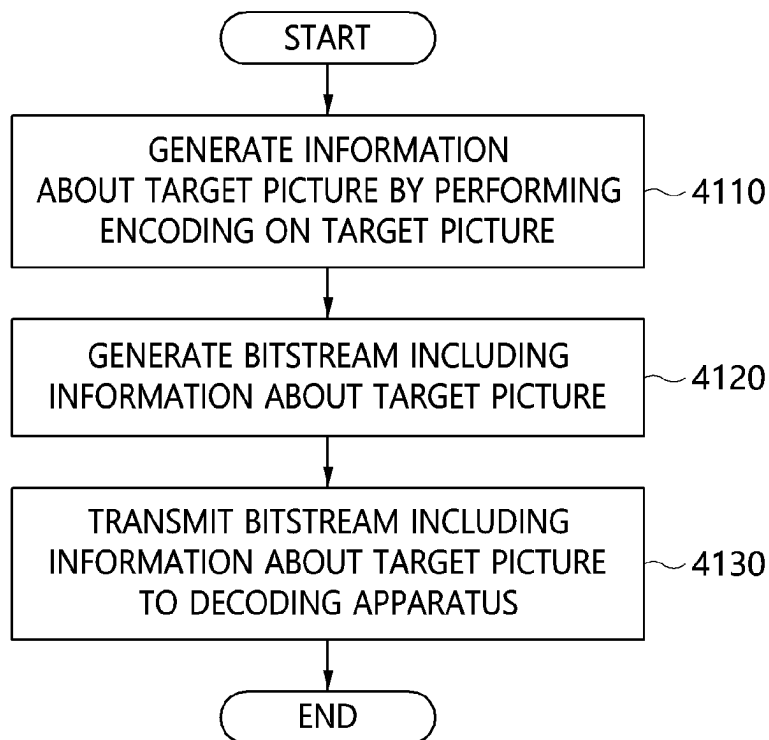
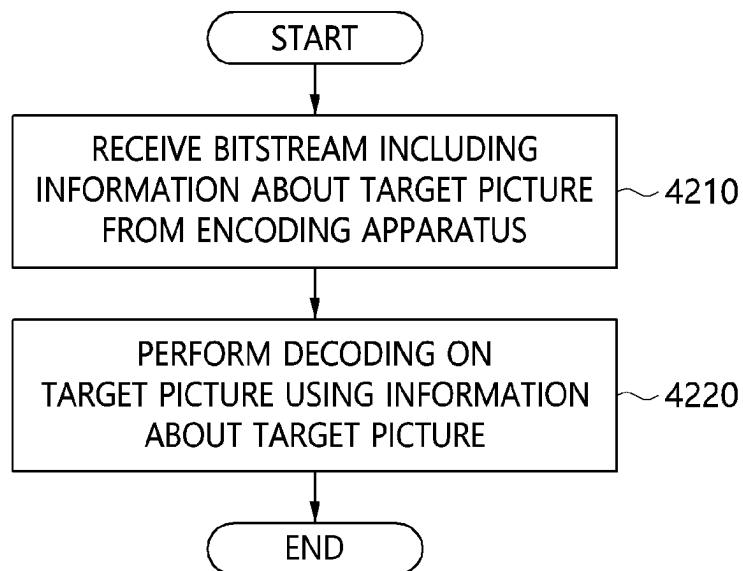


FIG. 39

slice_header() {	Descriptor
picture_header_in_slice_header_flag	u(1)
...	
if(!rpl_info_in_ph_flag && ((nal_unit_type != IDR_W_RADL && nal_unit_type != IDR_N_LP) sps_idr_rpl_present_flag mixed_nalu_types_in_pic_flag))	
ref_pic_lists()	
if((rpl_info_in_ph_flag ((nal_unit_type != IDR_W_RADL && nal_unit_type != IDR_N_LP) sps_idr_rpl_present_flag mixed_nalu_types_in_pic_flag)) && (slice_type != I && num_ref_entries[0][RplIdx[0]] > 1) (slice_type == B && num_ref_entries[1][RplIdx[1]] > 1)) {	
num_ref_idx_active_override_flag	u(1)
if(num_ref_idx_active_override_flag)	
for(i = 0; i < (slice_type == B ? 2 : 1); i++)	
if(num_ref_entries[i][RplIdx[i]] > 1)	
num_ref_idx_active_minus1[i]	ue(v)

FIG. 40

**FIG. 41**

**FIG. 42**

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METHOD, APPARATUS AND STORAGE MEDIUM FOR IMAGE ENCODING/DECODING USING SUBPICTURE

CROSS REFERENCE TO RELATED APPLICATIONS

The present application is a continuation application of U.S. patent application Ser. No. 17/955,278, filed Sep. 28, 2022, which is a continuation application of U.S. patent application Ser. No. 17/221,635, filed Apr. 2, 2021, and claims priority under 35 U.S.C. § 119(a) to Korean application numbers 10-2020-0040765, filed Apr. 3, 2020 and 10-2021-0041924, filed Mar. 31, 2021, in the Korean Intellectual Property Office, which is incorporated herein by reference in its entirety.

BACKGROUND OF THE INVENTION

1. Technical Field

The present disclosure relates generally to a method, an apparatus and a storage medium for image encoding/decoding. More particularly, the present disclosure relates to a method, an apparatus and a storage medium for performing encoding/decoding on an image using a subpicture.

2. Description of the Related Art

With the continuous development of the information and communication industries, broadcasting services supporting High-Definition (HD) resolution have been popularized all over the world. Through this popularization, a large number of users have become accustomed to high-resolution and high-definition images and/or video.

To satisfy users' demand for high definition, many institutions have accelerated the development of next-generation imaging devices. Users' interest in UHD TVs, having resolution that is more than four times as high as that of Full HD (FHD) TVs, as well as High-Definition TVs (HDTV) and FHD TVs, has increased. As interest therein has increased, image encoding/decoding technology for images having higher resolution and higher definition is currently required.

As image compression technology, there are various technologies, such as inter-prediction technology, intra-prediction technology, transform, quantization technology, and entropy coding technology.

Inter-prediction technology is technology for predicting the value of a pixel included in a current picture using a picture previous to and/or a picture subsequent to the current picture. Intra-prediction technology is technology for predicting the value of a pixel included in a current picture using information about pixels in the current picture. Transform and quantization technology may be technology for compressing the energy of a residual signal. The entropy coding technology is technology for assigning a short codeword to a frequently occurring value and assigning a long codeword to a less frequently occurring value.

By utilizing this image compression technology, data about images may be effectively compressed, transmitted, and stored.

SUMMARY OF THE INVENTION

An embodiment is intended to provide a method, an apparatus and a storage medium that perform encoding/decoding on an image using a subpicture.

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An embodiment is intended to provide a method, an apparatus and a storage medium that configure a reference picture list for motion compensation in the case where the location of a reference subpicture to be referred to for motion compensation is different from the location of a target subpicture.

An embodiment is intended to provide a method, an apparatus and a storage medium that configure a reference picture list and provide an override for the number of activated reference indexes.

In accordance with an aspect of the present invention to accomplish the above object, there is provided a decoding method, including acquiring information about a target picture from a bitstream; and performing decoding on the target picture using the information about the target picture, wherein the decoding includes a determination of a reference picture list for the target picture and a prediction based on the reference picture list, and wherein the information about the target picture includes a slice header for a slice of the target picture.

The determination of the reference picture list in the slice header may be performed when a Network Abstraction Layer (NAL) unit type of the slice is not a specific type.

The bitstream may include a picture header for the target picture.

The determination of the reference picture list in the slice header may be performed when information related to the reference picture list is not present in the picture header for the target picture.

The bitstream may include first information.

The decoding may include a determination of whether an override is to be performed on a number of activated reference indexes in the slice header.

The determination of whether the override is to be performed may be performed regardless of a value of the first information.

The first information may be information indicating whether reference picture list syntax elements are capable of being present in slice headers of slices for which a NAL unit type is a specific type.

The bitstream may further include second information.

The override may include decoding on the second information.

The second information may be information indicating whether an override is to be performed on the number of activated reference indexes for the reference picture list.

When the second information indicates that the override is to be performed, the information indicating the number of activated reference indexes for the reference picture list may be decoded.

The specific type may be an Instantaneous Decoding Refresh (IDR)_N_LP or IDR_W_Random Access Decodable Leading (RADL) type.

In accordance with another aspect of the present invention to accomplish the above object, there is provided an encoding method, including generating information about a target picture by performing encoding on the target picture; and generating a bitstream including the information about the target picture, wherein the information about the target picture includes a determination of a reference picture list for the target picture and information for a prediction based on the reference picture list, and wherein the information about the target picture includes a slice header for a slice of the target picture.

The determination of the reference picture list in the slice header may be performed when a Network Abstraction Layer (NAL) unit type of the slice is not a specific type.

The bitstream may include a picture header for the target picture.

The determination of the reference picture list in the slice header may be performed when information related to the reference picture list is not present in the picture header.

The bitstream may include first information.

The information about the target picture may further include information required to determine whether an override is to be performed on a number of activated reference indexes in the slice header.

The determination of whether the override is to be performed may be performed regardless of a value of the first information.

The first information may be information indicating whether reference picture list syntax elements are capable of being present in slice headers of slices for which a NAL unit type is a specific type.

The bitstream may further include second information.

The second information may be information indicating whether an override is to be performed on the number of activated reference indexes for the reference picture list.

When the second information indicates that the override is to be performed, the information indicating the number of activated reference indexes for the reference picture list may be encoded.

The specific type may be an Instantaneous Decoding Refresh (IDR)_N_LP or IDR_W_Random Access Decodable Leading (RADL) type.

In accordance with a further aspect of the present invention to accomplish the above object, there is provided a computer-readable storage medium storing a bitstream, wherein the bitstream includes information about a target picture, the information about the target picture is acquired from the bitstream, decoding on the target picture is performed using the information about the target picture, the decoding includes a determination of a reference picture list for the target picture and a prediction based on the reference picture list, and the information about the target picture includes a slice header for a slice of the target picture.

The determination of the reference picture list in the slice header may be performed when a NAL unit type of the slice is not a specific type.

The bitstream may include a picture header for the target picture.

The determination of the reference picture list in the slice header may be performed when information related to the reference picture list is not present in the picture header of the target picture.

The bitstream may include first information.

The decoding may include a determination of whether an override is to be performed on a number of activated reference indexes in the slice header.

The determination of whether the override is to be performed may be performed regardless of a value of the first information.

The first information may be information indicating whether reference picture list syntax elements are capable of being present in slice headers of slices for which a NAL unit type is a specific type.

The bitstream may further include second information.

The override may include decoding on the second information.

The second information may be information indicating whether an override is to be performed on the number of activated reference indexes for the reference picture list.

When the second information indicates that the override is to be performed, the information indicating the number of activated reference indexes for the reference picture list may be decoded.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and other objects, features and advantages of the present disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

FIG. 1 is a block diagram illustrating the configuration of an embodiment of an encoding apparatus to which the present disclosure is applied;

FIG. 2 is a block diagram illustrating the configuration of an embodiment of a decoding apparatus to which the present disclosure is applied;

FIG. 3 is a diagram schematically illustrating the partition structure of an image when the image is encoded and decoded;

FIG. 4 is a diagram illustrating the form of a Prediction Unit (PU) that a Coding Unit (CU) can include;

FIG. 5 is a diagram illustrating the form of a Transform Unit (TU) that can be included in a CU;

FIG. 6 illustrates splitting of a block according to an example;

FIG. 7 is a diagram for explaining an embodiment of an intra-prediction procedure;

FIG. 8 is a diagram illustrating reference samples used in an intra-prediction procedure;

FIG. 9 is a diagram for explaining an embodiment of an inter-prediction procedure;

FIG. 10 illustrates spatial candidates according to an embodiment;

FIG. 11 illustrates the order of addition of motion information of spatial candidates to a merge list according to an embodiment;

FIG. 12 illustrates a transform and quantization process according to an example;

FIG. 13 illustrates diagonal scanning according to an example;

FIG. 14 illustrates horizontal scanning according to an example;

FIG. 15 illustrates vertical scanning according to an example;

FIG. 16 is a configuration diagram of an encoding apparatus according to an embodiment;

FIG. 17 is a configuration diagram of a decoding apparatus according to an embodiment;

FIG. 18 illustrates the division of a picture in a raster-scan slice mode according to an example;

FIG. 19 illustrates the division of a picture in a rectangular slice mode according to an example;

FIG. 20 illustrates tiles and rectangular slices in a picture according to an example;

FIG. 21 illustrates the division of a picture into subpictures according to an example;

FIG. 22 illustrates code for subpicture ID mapping according to an example;

FIG. 23 illustrates signaling of slice addresses in a raster-scan slice mode according to an example;

FIG. 24 illustrates signaling of slice addresses in a rectangular slice mode according to an example;

FIG. 25 illustrates a part of the structure of a sequence parameter set according to an example;

FIG. 26 illustrates a method for deriving a subpicture ID according to an example;

FIG. 27 illustrates a scheme for utilizing a slice index for deriving the address of a CTB in a slice according to an example;

FIG. 28 illustrates a scheme for utilizing a slice index for deriving a CTB address in a target slice according to an example;

FIG. 29 illustrates a part of the structure of a picture parameter set according to an example;

FIG. 30 illustrates code for subpicture ID mapping according to an example;

FIG. 31 illustrates a change in a subpicture ID according to an example;

FIG. 32 illustrates a part of the structure of a sequence parameter set indicating whether there is syntax related to a reference picture list according to an example;

FIG. 33 illustrates a part of the structure of a picture parameter set indicating whether there are mixed NAL unit types according to an example;

FIG. 34 illustrates a slice header for configuring a reference picture list according to an embodiment;

FIG. 35 illustrates the case where the locations of subpictures in a picture are changed according to an embodiment;

FIG. 36 illustrates the case where the locations of subpictures treated as pictures are changed according to an embodiment;

FIG. 37 illustrates another part of the structure of a picture parameter set indicating whether there are mixed NAL unit types according to an example;

FIG. 38 illustrates an additional case where the locations of subpictures in a picture are changed according to an embodiment;

FIG. 39 illustrates the case where the locations of subpictures treated as pictures are changed according to an embodiment;

FIG. 40 illustrates a further part of the structure of a picture parameter set indicating whether there are mixed NAL unit types according to an example;

FIG. 41 is a flowchart of an encoding method according to an embodiment; and

FIG. 42 is a flowchart of a decoding method according to an embodiment.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

The present invention may be variously changed, and may have various embodiments, and specific embodiments will be described in detail below with reference to the attached drawings. However, it should be understood that those embodiments are not intended to limit the present invention to specific disclosure forms, and that they include all changes, equivalents or modifications included in the spirit and scope of the present invention.

Detailed descriptions of the following exemplary embodiments will be made with reference to the attached drawings illustrating specific embodiments. These embodiments are described so that those having ordinary knowledge in the technical field to which the present disclosure pertains can easily practice the embodiments. It should be noted that the various embodiments are different from each other, but do not need to be mutually exclusive of each other. For example, specific shapes, structures, and characteristics described here may be implemented as other embodiments without departing from the spirit and scope of the embodiments in relation to an embodiment. Further, it should be understood that the locations or arrangement of individual

components in each disclosed embodiment can be changed without departing from the spirit and scope of the embodiments. Therefore, the accompanying detailed description is not intended to restrict the scope of the disclosure, and the scope of the exemplary embodiments is limited only by the accompanying claims, along with equivalents thereof, as long as they are appropriately described.

In the drawings, similar reference numerals are used to designate the same or similar functions in various aspects. The shapes, sizes, etc. of components in the drawings may be exaggerated to make the description clear.

Terms such as “first” and “second” may be used to describe various components, but the components are not restricted by the terms. The terms are used only to distinguish one component from another component. For example, a first component may be named a second component without departing from the scope of the present specification. Likewise, a second component may be named a first component. The terms “and/or” may include combinations of a plurality of related described items or any of a plurality of related described items.

It will be understood that when a component is referred to as being “connected” or “coupled” to another component, the two components may be directly connected or coupled to each other, or intervening components may be present between the two components. On the other hand, it will be understood that when a component is referred to as being “directly connected or coupled”, no intervening components are present between the two components.

Also, components described in the embodiments are independently shown in order to indicate different characteristic functions, but this does not mean that each of the components is formed of a separate piece of hardware or software. That is, the components are arranged and included separately for convenience of description. For example, at least two of the components may be integrated into a single component. Conversely, one component may be divided into multiple components. An embodiment into which the components are integrated or an embodiment in which some components are separated is included in the scope of the present specification as long as it does not depart from the essence of the present specification.

Further, in the exemplary embodiments, an expression describing that a component “comprises” a specific component means that additional components may be included within the scope of the practice or the technical spirit of exemplary embodiments, but does not preclude the presence of components other than the specific component.

The terms used in the present specification are merely used to describe specific embodiments and are not intended to limit the present invention. A singular expression includes a plural expression unless a description to the contrary is specifically pointed out in context. In the present specification, it should be understood that the terms such as “include” or “have” are merely intended to indicate that features, numbers, steps, operations, components, parts, or combinations thereof are present, and are not intended to exclude the possibility that one or more other features, numbers, steps, operations, components, parts, or combinations thereof will be present or added. That is, in the present invention, an expression describing that a component “comprises” a specific component means that additional components may be included within the scope of the practice of the present invention or the technical spirit of the present invention, but does not preclude the presence of components other than the specific component.

Some components of the present invention are not essential components for performing essential functions, but may be optional components for improving only performance. The embodiments may be implemented using only essential components for implementing the essence of the embodiments. For example, a structure including only essential components, excluding optional components used only to improve performance, is also included in the scope of the embodiments.

Embodiments will be described in detail below with reference to the accompanying drawings so that those having ordinary knowledge in the technical field to which the embodiments pertain can easily practice the embodiments. In the following description of the embodiments, detailed descriptions of known functions or configurations which are deemed to make the gist of the present specification obscure will be omitted. Further, the same reference numerals are used to designate the same components throughout the drawings, and repeated descriptions of the same components will be omitted.

Hereinafter, "image" may mean a single picture constituting a video, or may mean the video itself. For example, "encoding and/or decoding of an image" may mean "encoding and/or decoding of a video", and may also mean "encoding and/or decoding of any one of images constituting the video".

Hereinafter, the terms "video" and "motion picture" may be used to have the same meaning, and may be used interchangeably with each other.

Hereinafter, a target image may be an encoding target image, which is the target to be encoded, and/or a decoding target image, which is the target to be decoded. Further, the target image may be an input image that is input to an encoding apparatus or an input image that is input to a decoding apparatus. And, a target image may be a current image, that is, the target to be currently encoded and/or decoded. For example, the terms "target image" and "current image" may be used to have the same meaning, and may be used interchangeably with each other.

Hereinafter, the terms "image", "picture", "frame", and "screen" may be used to have the same meaning and may be used interchangeably with each other.

Hereinafter, a target block may be an encoding target block, i.e. the target to be encoded and/or a decoding target block, i.e. the target to be decoded. Further, the target block may be a current block, i.e. the target to be currently encoded and/or decoded. Here, the terms "target block" and "current block" may be used to have the same meaning, and may be used interchangeably with each other. A current block may denote an encoding target block, which is the target of encoding, during encoding and/or a decoding target block, which is the target of decoding, during decoding. Also, the current block may be at least one of a coding block, a prediction block, a residual block, and a transform block.

Hereinafter, the terms "block" and "unit" may be used to have the same meaning, and may be used interchangeably with each other. Alternatively, "block" may denote a specific unit.

Hereinafter, the terms "region" and "segment" may be used interchangeably with each other.

In the following embodiments, specific information, data, a flag, an index, an element, and an attribute may have their respective values. A value of "0" corresponding to each of the information, data, flag, index, element, and attribute may indicate a false, a logical false or a first predefined value. In other words, the value of "0", a false, logical false, and a first predefined value may be used interchangeably with each

other. A value of "1" corresponding to each of the information, data, flag, index, element, and attribute may indicate a true, a logical true or a second predefined value. In other words, the value of "1", true, logical true, and a second predefined value may be used interchangeably with each other.

When a variable such as *i* or *j* is used to indicate a row, a column, or an index, the value of *i* may be an integer of 0 or more or an integer of 1 or more. In other words, in the embodiments, each of a row, a column, and an index may be counted from 0 or may be counted from 1.

In embodiments, the term "one or more" or the term "at least one" may mean the term "plural". The term "one or more" or the term "at least one" may be used interchangeably with "plural".

Below, the terms to be used in embodiments will be described.

Encoder: An encoder denotes a device for performing encoding. That is, an encoder may mean an encoding apparatus.

Decoder: A decoder denotes a device for performing decoding. That is, a decoder may mean a decoding apparatus.

Unit: A unit may denote the unit of image encoding and decoding. The terms "unit" and "block" may be used to have the same meaning, and may be used interchangeably with each other.

A unit may be an $M \times N$ array of samples. Each of *M* and *N* may be a positive integer. A unit may typically mean an array of samples in the form of two-dimensions.

In the encoding and decoding of an image, "unit" may be an area generated by the partitioning of one image. In other words, "unit" may be a region specified in one image. A single image may be partitioned into multiple units. Alternatively, one image may be partitioned into sub-parts, and the unit may denote each partitioned sub-part when encoding or decoding is performed on the partitioned sub-part.

In the encoding and decoding of an image, predefined processing may be performed on each unit depending on the type of the unit.

Depending on functions, the unit types may be classified into a macro unit, a Coding Unit (CU), a Prediction Unit (PU), a residual unit, a Transform Unit (TU), etc. Alternatively, depending on functions, the unit may denote a block, a macroblock, a coding tree unit, a coding tree block, a coding unit, a coding block, a prediction unit, a prediction block, a residual unit, a residual block, a transform unit, a transform block, etc. For example, a target unit, which is the target of encoding and/or decoding, may be at least one of a CU, a PU, a residual unit, and a TU.

The term "unit" may mean information including a luminance (luma) component block, a chrominance (chroma) component block corresponding thereto, and syntax elements for respective blocks so that the unit is designated to be distinguished from a block.

The size and shape of a unit may be variously implemented. Further, a unit may have any of various sizes and shapes. In particular, the shapes of the unit may include not only a square, but also a geometric figure that can be represented in two dimensions (2D), such as a rectangle, a trapezoid, a triangle, and a pentagon.

Further, unit information may include one or more of the type of a unit, the size of a unit, the depth of a unit, the order of encoding of a unit and the order of decoding

of a unit, etc. For example, the type of a unit may indicate one of a CU, a PU, a residual unit and a TU. One unit may be partitioned into sub-units, each having a smaller size than that of the relevant unit.

Depth: A depth may mean an extent to which the unit is partitioned. Further, the depth of the unit may indicate the level at which the corresponding unit is present when unit(s) are represented by a tree structure.

Unit partition information may include a depth indicating the depth of a unit. A depth may indicate the number of times the unit is partitioned and/or the degree to which the unit is partitioned.

In a tree structure, it may be considered that the depth of a root node is the smallest, and the depth of a leaf node is the largest. The root node may be the highest (top) node. The leaf node may be a lowest node.

A single unit may be hierarchically partitioned into multiple sub-units while having depth information based on a tree structure. In other words, the unit and sub-units, generated by partitioning the unit, may correspond to a node and child nodes of the node, respectively. Each of the partitioned sub-units may have a unit depth. Since the depth indicates the number of times the unit is partitioned and/or the degree to which the unit is partitioned, the partition information of the sub-units may include information about the sizes of the sub-units.

In a tree structure, the top node may correspond to the initial node before partitioning. The top node may be referred to as a “root node”. Further, the root node may have a minimum depth value. Here, the top node may have a depth of level ‘0’.

A node having a depth of level ‘1’ may denote a unit generated when the initial unit is partitioned once. A node having a depth of level ‘2’ may denote a unit generated when the initial unit is partitioned twice.

A leaf node having a depth of level ‘n’ may denote a unit generated when the initial unit has been partitioned n times.

The leaf node may be a bottom node, which cannot be partitioned any further. The depth of the leaf node may be the maximum level. For example, a predefined value for the maximum level may be 3.

A QT depth may denote a depth for a quad-partitioning. A BT depth may denote a depth for a binary-partitioning. A TT depth may denote a depth for a ternary-partitioning.

Sample: A sample may be a base unit constituting a block.

A sample may be represented by values from 0 to $2^{Bd}-1$ depending on the bit depth (Bd).

A sample may be a pixel or a pixel value.

Hereinafter, the terms “pixel” and “sample” may be used to have the same meaning, and may be used interchangeably with each other.

A Coding Tree Unit (CTU): A CTU may be composed of a single luma component (Y) coding tree block and two chroma component (Cb, Cr) coding tree blocks related to the luma component coding tree block. Further, a CTU may mean information including the above blocks and a syntax element for each of the blocks.

Each coding tree unit (CTU) may be partitioned using one or more partitioning methods, such as a quad tree (QT), a binary tree (BT), and a ternary tree (TT) so as to configure sub-units, such as a coding unit, a prediction unit, and a transform unit. A quad tree may mean a quaternary tree. Further, each coding tree unit may be

partitioned using a multitype tree (MTT) using one or more partitioning methods.

“CTU” may be used as a term designating a pixel block, which is a processing unit in an image-decoding and encoding process, as in the case of partitioning of an input image.

Coding Tree Block (CTB): “CTB” may be used as a term designating any one of a Y coding tree block, a Cb coding tree block, and a Cr coding tree block.

Neighbor block: A neighbor block (or neighboring block) may mean a block adjacent to a target block. A neighbor block may mean a reconstructed neighbor block.

Hereinafter, the terms “neighbor block” and “adjacent block” may be used to have the same meaning and may be used interchangeably with each other.

A neighbor block may mean a reconstructed neighbor block.

Spatial neighbor block: A spatial neighbor block may a block spatially adjacent to a target block. A neighbor block may include a spatial neighbor block.

The target block and the spatial neighbor block may be included in a target picture.

The spatial neighbor block may mean a block, the boundary of which is in contact with the target block, or a block located within a predetermined distance from the target block.

The spatial neighbor block may mean a block adjacent to the vertex of the target block. Here, the block adjacent to the vertex of the target block may mean a block vertically adjacent to a neighbor block which is horizontally adjacent to the target block or a block horizontally adjacent to a neighbor block which is vertically adjacent to the target block.

Temporal neighbor block: A temporal neighbor block may be a block temporally adjacent to a target block. A neighbor block may include a temporal neighbor block.

The temporal neighbor block may include a co-located block (col block).

The col block may be a block in a previously reconstructed co-located picture (col picture). The location of the col block in the col-picture may correspond to the location of the target block in a target picture. Alternatively, the location of the col block in the col-picture may be equal to the location of the target block in the target picture. The col picture may be a picture included in a reference picture list.

The temporal neighbor block may be a block temporally adjacent to a spatial neighbor block of a target block.

Prediction mode: The prediction mode may be information indicating the mode used for intra prediction, or the mode used for inter prediction.

Prediction unit: A prediction unit may be a base unit for prediction, such as inter prediction, intra prediction, inter compensation, intra compensation, and motion compensation.

A single prediction unit may be divided into multiple partitions having smaller sizes or sub-prediction units. The multiple partitions may also be base units in the performance of prediction or compensation. The partitions generated by dividing the prediction unit may also be prediction units.

Prediction unit partition: A prediction unit partition may be the shape into which a prediction unit is divided.

Reconstructed neighbor unit: A reconstructed neighbor unit may be a unit which has already been decoded and reconstructed neighboring a target unit.

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A reconstructed neighbor unit may be a unit that is spatially adjacent to the target unit or that is temporally adjacent to the target unit.

A reconstructed spatial neighbor unit may be a unit which is included in a target picture and which has already been reconstructed through encoding and/or decoding.

A reconstructed temporal neighbor unit may be a unit which is included in a reference image and which has already been reconstructed through encoding and/or decoding. The location of the reconstructed temporal neighbor unit in the reference image may be identical to that of the target unit in the target picture, or may correspond to the location of the target unit in the target picture. Also, a reconstructed temporal neighbor unit may be a block neighboring the corresponding block in a reference image. Here, the location of the corresponding block in the reference image may correspond to the location of the target block in the target image. Here, the fact that the locations of blocks correspond to each other may mean that the locations of the blocks are identical to each other, may mean that one block is included in another block, or may mean that one block occupies a specific location in another block.

Subpicture: A picture may be divided into one or more subpictures. A subpicture may be composed of one or more rows and one or more tile columns.

A subpicture may be a region having a rectangular shape in a picture. Further, a subpicture may include one or more CTUs.

A subpicture may be a rectangular region of one or more slices in a picture.

One subpicture may include one or more tiles, one or more bricks, and/or one or more slices. One or more slices may collectively cover a rectangular region of a picture.

Tile: A tile may be a region having a square or rectangular (i.e., a non-square rectangular) shape in a picture.

A picture may be partitioned into one or more tiles.

A tile may include one or more CTUs.

A tile may be partitioned into one or more bricks.

Brick: A brick may denote one or more CTU rows in a tile.

A tile may be partitioned into one or more bricks. Each brick may include one or more CTU rows.

A tile that is not partitioned into two parts may also denote a brick.

Slice: A slice may include one or more tiles in a picture. Alternatively, a slice may include one or more bricks in a tile.

A subpicture may contain one or more slices that collectively cover a rectangular region of a picture. Consequently, each subpicture boundary is also always a slice boundary, and each vertical subpicture boundary is always also a vertical tile boundary.

Parameter set: A parameter set may correspond to header information in the internal structure of a bitstream.

A parameter set may include at least one of a video parameter set (VPS), a sequence parameter set (SPS), a picture parameter set (PPS), an adaptation parameter set (APS), a decoding parameter set (DPS), etc.

Information signaled through each parameter set may be applied to pictures which refer to the corresponding parameter set. For example, information in a VPS may be applied to pictures which refer to the VPS. Information in an SPS may be applied to pictures which refer to the SPS. Information in a PPS may be applied to pictures which refer to the PPS.

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Each parameter set may refer to a higher parameter set. For example, a PPS may refer to an SPS. An SPS may refer to a VPS.

Further, a parameter set may include a tile group, slice header information, and tile header information. The tile group may be a group including multiple tiles. Also, the meaning of "tile group" may be identical to that of "slice".

Rate-distortion optimization: An encoding apparatus may use rate-distortion optimization so as to provide high coding efficiency by utilizing combinations of the size of a coding unit (CU), a prediction mode, the size of a prediction unit (PU), motion information, and the size of a transform unit (TU).

A rate-distortion optimization scheme may calculate rate-distortion costs of respective combinations so as to select an optimal combination from among the combinations. The rate-distortion costs may be calculated using the equation " $D+\lambda \cdot R$ ". Generally, a combination enabling the rate-distortion cost to be minimized may be selected as the optimal combination in the rate-distortion optimization scheme.

D may denote distortion. D may be the mean of squares of differences (i.e. mean square error) between original transform coefficients and reconstructed transform coefficients in a transform unit.

R may denote the rate, which may denote a bit rate using related-context information.

k denotes a Lagrangian multiplier. R may include not only coding parameter information, such as a prediction mode, motion information, and a coded block flag, but also bits generated due to the encoding of transform coefficients.

An encoding apparatus may perform procedures, such as inter prediction and/or intra prediction, transform, quantization, entropy encoding, inverse quantization (dequantization), and/or inverse transform so as to calculate precise D and R. These procedures may greatly increase the complexity of the encoding apparatus.

Bitstream: A bitstream may denote a stream of bits including encoded image information.

Parsing: Parsing may be the decision on the value of a syntax element, made by performing entropy decoding on a bitstream. Alternatively, the term "parsing" may mean such entropy decoding itself.

Symbol: A symbol may be at least one of the syntax element, the coding parameter, and the transform coefficient of an encoding target unit and/or a decoding target unit. Further, a symbol may be the target of entropy encoding or the result of entropy decoding.

Reference picture: A reference picture may be an image referred to by a unit so as to perform inter prediction or motion compensation. Alternatively, a reference picture may be an image including a reference unit referred to by a target unit so as to perform inter prediction or motion compensation.

Hereinafter, the terms "reference picture" and "reference image" may be used to have the same meaning, and may be used interchangeably with each other.

Reference picture list: A reference picture list may be a list including one or more reference images used for inter prediction or motion compensation.

The types of a reference picture list may include List Combined (LC), List 0 (L0), List 1 (L1), List 2 (L2), List 3 (L3), etc.

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For inter prediction, one or more reference picture lists may be used.

Inter-prediction indicator: An inter-prediction indicator may indicate the inter-prediction direction for a target unit. Inter prediction may be one of unidirectional prediction and bidirectional prediction. Alternatively, the inter-prediction indicator may denote the number of reference pictures used to generate a prediction unit of a target unit. Alternatively, the inter-prediction indicator may denote the number of prediction blocks used for inter prediction or motion compensation of a target unit.

Prediction list utilization flag: A prediction list utilization flag may indicate whether a prediction unit is generated using at least one reference picture in a specific reference picture list.

An inter-prediction indicator may be derived using the prediction list utilization flag. In contrast, the prediction list utilization flag may be derived using the inter-prediction indicator. For example, the case where the prediction list utilization flag indicates "0", which is a first value, may indicate that, for a target unit, a prediction block is not generated using a reference picture in a reference picture list. The case where the prediction list utilization flag indicates "1", which is a second value, may indicate that, for a target unit, a prediction unit is generated using the reference picture list.

Reference picture index: A reference picture index may be an index indicating a specific reference picture in a reference picture list.

Picture Order Count (POC): A POC value for a picture may denote an order in which the corresponding picture is displayed.

Motion vector (MV): A motion vector may be a 2D vector used for inter prediction or motion compensation. A motion vector may mean an offset between a target image and a reference image.

For example, a MV may be represented in a form such as (mv_x, mv_y) . mv_x may indicate a horizontal component, and mv_y may indicate a vertical component.

Search range: A search range may be a 2D area in which a search for a MV is performed during inter prediction.

For example, the size of the search range may be $M \times N$. M and N may be respective positive integers.

Motion vector candidate: A motion vector candidate may be a block that is a prediction candidate or the motion vector of the block that is a prediction candidate when a motion vector is predicted.

A motion vector candidate may be included in a motion vector candidate list.

Motion vector candidate list: A motion vector candidate list may be a list configured using one or more motion vector candidates.

Motion vector candidate index: A motion vector candidate index may be an indicator for indicating a motion vector candidate in the motion vector candidate list. Alternatively, a motion vector candidate index may be the index of a motion vector predictor.

Motion information: Motion information may be information including at least one of a reference picture list, a reference image, a motion vector candidate, a motion vector candidate index, a merge candidate, and a merge index, as well as a motion vector, a reference picture index, and an inter-prediction indicator.

Merge candidate list: A merge candidate list may be a list configured using one or more merge candidates.

Merge candidate: A merge candidate may be a spatial merge candidate, a temporal merge candidate, a combined

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merge candidate, a combined bi-prediction merge candidate, a candidate based on a history, a candidate based on an average of two candidates, a zero-merge candidate, etc. A merge candidate may include an inter-prediction indicator, and may include motion information such as prediction type information, a reference picture index for each list, a motion vector, a prediction list utilization flag, and an inter-prediction indicator.

Merge index: A merge index may be an indicator for indicating a merge candidate in a merge candidate list.

A merge index may indicate a reconstructed unit used to derive a merge candidate between a reconstructed unit spatially adjacent to a target unit and a reconstructed unit temporally adjacent to the target unit.

A merge index may indicate at least one of pieces of motion information of a merge candidate.

Transform unit: A transform unit may be the base unit of residual signal encoding and/or residual signal decoding, such as transform, inverse transform, quantization, dequantization, transform coefficient encoding, and transform coefficient decoding. A single transform unit may be partitioned into multiple sub-transform units having a smaller size. Here, a transform may include one or more of a primary transform and a secondary transform, and an inverse transform may include one or more of a primary inverse transform and a secondary inverse transform.

Scaling: Scaling may denote a procedure for multiplying a factor by a transform coefficient level.

As a result of scaling of the transform coefficient level, a transform coefficient may be generated. Scaling may also be referred to as "dequantization".

Quantization Parameter (QP): A quantization parameter may be a value used to generate a transform coefficient level for a transform coefficient in quantization. Alternatively, a quantization parameter may also be a value used to generate a transform coefficient by scaling the transform coefficient level in dequantization. Alternatively, a quantization parameter may be a value mapped to a quantization step size.

Delta quantization parameter: A delta quantization parameter may mean a difference value between a predicted quantization parameter and the quantization parameter of a target unit.

Scan: Scan may denote a method for aligning the order of coefficients in a unit, a block or a matrix. For example, a method for aligning a 2D array in the form of a one-dimensional (1D) array may be referred to as a "scan". Alternatively, a method for aligning a 1D array in the form of a 2D array may also be referred to as a "scan" or an "inverse scan".

Transform coefficient: A transform coefficient may be a coefficient value generated as an encoding apparatus performs a transform. Alternatively, the transform coefficient may be a coefficient value generated as a decoding apparatus performs at least one of entropy decoding and dequantization.

A quantized level or a quantized transform coefficient level generated by applying quantization to a transform coefficient or a residual signal may also be included in the meaning of the term "transform coefficient".

Quantized level: A quantized level may be a value generated as the encoding apparatus performs quantization on a transform coefficient or a residual signal. Alternatively, the quantized level may be a value that is the target of dequantization as the decoding apparatus performs dequantization.

A quantized transform coefficient level, which is the result of transform and quantization, may also be included in the meaning of a quantized level.

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Non-zero transform coefficient: A non-zero transform coefficient may be a transform coefficient having a value other than 0 or a transform coefficient level having a value other than 0. Alternatively, a non-zero transform coefficient may be a transform coefficient, the magnitude of the value of which is not 0, or a transform coefficient level, the magnitude of the value of which is not 0.

Quantization matrix: A quantization matrix may be a matrix used in a quantization procedure or a dequantization procedure so as to improve the subjective image quality or objective image quality of an image. A quantization matrix may also be referred to as a “scaling list”.

Quantization matrix coefficient: A quantization matrix coefficient may be each element in a quantization matrix. A quantization matrix coefficient may also be referred to as a “matrix coefficient”.

Default matrix: A default matrix may be a quantization matrix predefined by the encoding apparatus and the decoding apparatus.

Non-default matrix: A non-default matrix may be a quantization matrix that is not predefined by the encoding apparatus and the decoding apparatus. The non-default matrix may mean a quantization matrix to be signaled from the encoding apparatus to the decoding apparatus by a user.

Most Probable Mode (MPM): An MPM may denote an intra-prediction mode having a high probability of being used for intra prediction for a target block.

An encoding apparatus and a decoding apparatus may determine one or more MPMs based on coding parameters related to the target block and the attributes of entities related to the target block.

The encoding apparatus and the decoding apparatus may determine one or more MPMs based on the intra-prediction mode of a reference block. The reference block may include multiple reference blocks. The multiple reference blocks may include spatial neighbor blocks adjacent to the left of the target block and spatial neighbor blocks adjacent to the top of the target block. In other words, depending on which intra-prediction modes have been used for the reference blocks, one or more different MPMs may be determined.

The one or more MPMs may be determined in the same manner both in the encoding apparatus and in the decoding apparatus. That is, the encoding apparatus and the decoding apparatus may share the same MPM list including one or more MPMs.

MPM list: An MPM list may be a list including one or more MPMs. The number of the one or more MPMs in the MPM list may be defined in advance.

MPM indicator: An MPM indicator may indicate an MPM to be used for intra prediction for a target block among one or more MPMs in the MPM list. For example, the MPM indicator may be an index for the MPM list.

Since the MPM list is determined in the same manner both in the encoding apparatus and in the decoding apparatus, there may be no need to transmit the MPM list itself from the encoding apparatus to the decoding apparatus.

The MPM indicator may be signaled from the encoding apparatus to the decoding apparatus. As the MPM indicator is signaled, the decoding apparatus may determine the MPM to be used for intra prediction for the target block among the MPMs in the MPM list.

MPM use indicator: An MPM use indicator may indicate whether an MPM usage mode is to be used for prediction for a target block. The MPM usage mode may be a mode in which the MPM to be used for intra prediction for the target block is determined using the MPM list.

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The MPM use indicator may be signaled from the encoding apparatus to the decoding apparatus.

Signaling: “signaling” may denote that information is transferred from an encoding apparatus to a decoding apparatus. Alternatively, “signaling” may mean information is included in a bitstream or a recording medium by an encoding apparatus. Information signaled by an encoding apparatus may be used by a decoding apparatus.

The encoding apparatus may generate encoded information by performing encoding on information to be signaled. The encoded information may be transmitted from the encoding apparatus to the decoding apparatus. The decoding apparatus may obtain information by decoding the transmitted encoded information. Here, the encoding may be entropy encoding, and the decoding may be entropy decoding.

Selective Signaling: Information may be signaled selectively. A selective signaling FOR information may mean that an encoding apparatus selectively includes information (according to a specific condition) in a bitstream or a recording medium. Selective signaling for information may mean that a decoding apparatus selectively extracts information from a bitstream (according to a specific condition).

Omission of signaling: Signaling for information may be omitted. Omission of signaling for information on information may mean that an encoding apparatus does not include information (according to a specific condition) in a bitstream or a recording medium. Omission of signaling for information may mean that a decoding apparatus does not extract information from a bitstream (according to a specific condition).

Statistic value: A variable, a coding parameter, a constant, etc. may have values that can be calculated. The statistic value may be a value generated by performing calculations (operations) on the values of specified targets. For example, the statistic value may indicate one or more of the average, weighted average, weighted sum, minimum value, maximum value, mode, median value, and interpolated value of the values of a specific variable, a specific coding parameter, a specific constant, or the like.

FIG. 1 is a block diagram illustrating the configuration of an embodiment of an encoding apparatus to which the present disclosure is applied.

An encoding apparatus 100 may be an encoder, a video encoding apparatus or an image encoding apparatus. A video may include one or more images (pictures). The encoding apparatus 100 may sequentially encode one or more images of the video.

An encoding apparatus may generate encoded information by encoding information to be signaled. The encoded information may be transmitted from the encoding apparatus to a decoding apparatus. The decoding apparatus may acquire information by decoding the received encoded information. Here, encoding may be entropy encoding, and decoding may be entropy decoding.

Referring to FIG. 1, the encoding apparatus 100 includes an inter-prediction unit 110, an intra-prediction unit 120, a switch 115, a subtractor 125, a transform unit 130, a quantization unit 140, an entropy encoding unit 150, a dequantization (inverse quantization) unit 160, an inverse transform unit 170, an adder 175, a filter unit 180, and a reference picture buffer 190.

The encoding apparatus 100 may perform encoding on a target image using an intra mode and/or an inter mode. In other words, a prediction mode for a target block may be one of an intra mode and an inter mode.

Hereinafter, the terms “intra mode”, “intra-prediction mode”, “intra-picture mode” and “intra-picture prediction mode” may be used to have the same meaning, and may be used interchangeably with each other.

Hereinafter, the terms “inter mode”, “inter-prediction mode”, “inter-picture mode” and “inter-picture prediction mode” may be used to have the same meaning, and may be used interchangeably with each other.

Hereinafter, the term “image” may indicate only part of an image, or may indicate a block. Also, the processing of an “image” may indicate sequential processing of multiple blocks.

Further, the encoding apparatus **100** may generate a bitstream, including encoded information, via encoding on the target image, and may output and store the generated bitstream. The generated bitstream may be stored in a computer-readable storage medium and may be streamed through a wired and/or wireless transmission medium.

When the intra mode is used as a prediction mode, the switch **115** may switch to the intra mode. When the inter mode is used as a prediction mode, the switch **115** may switch to the inter mode.

The encoding apparatus **100** may generate a prediction block of a target block. Further, after the prediction block has been generated, the encoding apparatus **100** may encode a residual block for the target block using a residual between the target block and the prediction block.

When the prediction mode is the intra mode, the intra-prediction unit **120** may use pixels of previously encoded/decoded neighbor blocks adjacent to the target block as reference samples. The intra-prediction unit **120** may perform spatial prediction on the target block using the reference samples, and may generate prediction samples for the target block via spatial prediction. the prediction samples may mean samples in the prediction block.

The inter-prediction unit **110** may include a motion prediction unit and a motion compensation unit.

When the prediction mode is an inter mode, the motion prediction unit may search a reference image for the area most closely matching the target block in a motion prediction procedure, and may derive a motion vector for the target block and the found area based on the found area. Here, the motion-prediction unit may use a search range as a target area for searching.

The reference image may be stored in the reference picture buffer **190**. More specifically, an encoded and/or decoded reference image may be stored in the reference picture buffer **190** when the encoding and/or decoding of the reference image have been processed.

Since a decoded picture is stored, the reference picture buffer **190** may be a Decoded Picture Buffer (DPB).

The motion compensation unit may generate a prediction block for the target block by performing motion compensation using a motion vector. Here, the motion vector may be a two-dimensional (2D) vector used for inter-prediction. Further, the motion vector may indicate an offset between the target image and the reference image.

The motion prediction unit and the motion compensation unit may generate a prediction block by applying an interpolation filter to a partial area of a reference image when the motion vector has a value other than an integer. In order to perform inter prediction or motion compensation, it may be determined which one of a skip mode, a merge mode, an advanced motion vector prediction (AMVP) mode, and a current picture reference mode corresponds to a method for predicting the motion of a PU included in a CU, based on the

CU, and compensating for the motion, and inter prediction or motion compensation may be performed depending on the mode.

The subtractor **125** may generate a residual block, which is the differential between the target block and the prediction block. A residual block may also be referred to as a “residual signal”.

The residual signal may be the difference between an original signal and a prediction signal. Alternatively, the residual signal may be a signal generated by transforming or quantizing the difference between an original signal and a prediction signal or by transforming and quantizing the difference. A residual block may be a residual signal for a block unit.

The transform unit **130** may generate a transform coefficient by transforming the residual block, and may output the generated transform coefficient. Here, the transform coefficient may be a coefficient value generated by transforming the residual block.

The transform unit **130** may use one of multiple predefined transform methods when performing a transform.

The multiple predefined transform methods may include a Discrete Cosine Transform (DCT), a Discrete Sine Transform (DST), a Karhunen-Loeve Transform (KLT), etc.

The transform method used to transform a residual block may be determined depending on at least one of coding parameters for a target block and/or a neighbor block. For example, the transform method may be determined based on at least one of an inter-prediction mode for a PU, an intra-prediction mode for a PU, the size of a TU, and the shape of a TU. Alternatively, transformation information indicating the transform method may be signaled from the encoding apparatus **100** to the decoding apparatus **200**.

When a transform skip mode is used, the transform unit **130** may omit transforming the residual block.

By applying quantization to the transform coefficient, a quantized transform coefficient level or a quantized level may be generated. Hereinafter, in the embodiments, each of the quantized transform coefficient level and the quantized level may also be referred to as a ‘transform coefficient’.

The quantization unit **140** may generate a quantized transform coefficient level (i.e., a quantized level or a quantized coefficient) by quantizing the transform coefficient depending on quantization parameters. The quantization unit **140** may output the quantized transform coefficient level that is generated. In this case, the quantization unit **140** may quantize the transform coefficient using a quantization matrix.

The entropy encoding unit **150** may generate a bitstream by performing probability distribution-based entropy encoding based on values, calculated by the quantization unit **140**, and/or coding parameter values, calculated in the encoding procedure. The entropy encoding unit **150** may output the generated bitstream.

The entropy encoding unit **150** may perform entropy encoding on information about the pixels of the image and information required to decode the image. For example, the information required to decode the image may include syntax elements or the like.

When entropy encoding is applied, fewer bits may be assigned to more frequently occurring symbols, and more bits may be assigned to rarely occurring symbols. As symbols are represented by means of this assignment, the size of a bit string for target symbols to be encoded may be reduced. Therefore, the compression performance of video encoding may be improved through entropy encoding.

Further, for entropy encoding, the entropy encoding unit **150** may use a coding method such as exponential Golomb, Context-Adaptive Variable Length Coding (CAVLC), or Context-Adaptive Binary Arithmetic Coding (CABAC). For example, the entropy encoding unit **150** may perform entropy encoding using a Variable Length Coding/Code (VLC) table. For example, the entropy encoding unit **150** may derive a binarization method for a target symbol. Further, the entropy encoding unit **150** may derive a probability model for a target symbol/bin. The entropy encoding unit **150** may perform arithmetic coding using the derived binarization method, a probability model, and a context model.

The entropy encoding unit **150** may transform the coefficient of the form of a 2D block into the form of a 1D vector through a transform coefficient scanning method so as to encode a quantized transform coefficient level.

The coding parameters may be information required for encoding and/or decoding. The coding parameters may include information encoded by the encoding apparatus **100** and transferred from the encoding apparatus **100** to a decoding apparatus, and may also include information that may be derived in the encoding or decoding procedure. For example, information transferred to the decoding apparatus may include syntax elements.

The coding parameters may include not only information (or a flag or an index), such as a syntax element, which is encoded by the encoding apparatus and is signaled by the encoding apparatus to the decoding apparatus, but also information derived in an encoding or decoding process. Further, the coding parameters may include information required so as to encode or decode images. For example, the coding parameters may include at least one value, combinations or statistics of a size of a unit/block, a shape/form of a unit/block, a depth of a unit/block, partition information of a unit/block, a partition structure of a unit/block, information indicating whether a unit/block is partitioned in a quad-tree structure, information indicating whether a unit/block is partitioned in a binary tree structure, a partitioning direction of a binary tree structure (horizontal direction or vertical direction), a partitioning form of a binary tree structure (symmetrical partitioning or asymmetrical partitioning), information indicating whether a unit/block is partitioned in a ternary tree structure, a partitioning direction of a ternary tree structure (horizontal direction or vertical direction), a partitioning form of a ternary tree structure (symmetrical partitioning or asymmetrical partitioning, etc.), information indicating whether a unit/block is partitioned in a multi-type tree structure, a combination and a direction (horizontal direction or vertical direction, etc.) of a partitioning of the multi-type tree structure, a partitioning form of a multi-type tree structure (symmetrical partitioning or asymmetrical partitioning, etc.), a partitioning tree (a binary tree or a ternary tree) of the multi-type tree form, a type of a prediction (intra prediction or inter prediction), an intra-prediction mode/direction, an intra luma prediction mode/direction, an intra chroma prediction mode/direction, an intra partitioning information, an inter partitioning information, a coding block partitioning flag, a prediction block partitioning flag, a transform block partitioning flag, a reference sample filtering method, a reference sample filter tap, a reference sample filter coefficient, a prediction block filtering method, a prediction block filter tap, a prediction block filter coefficient, a prediction block boundary filtering method, a prediction block boundary filter tap, a prediction block boundary filter coefficient, an inter-prediction mode, motion information, a motion vector, a motion vector dif-

ference, a reference picture index, an inter-prediction direction, an inter-prediction indicator, a prediction list utilization flag, a reference picture list, a reference image, a POC, a motion vector predictor, a motion vector prediction index, a motion vector prediction candidate, a motion vector candidate list, information indicating whether a merge mode is used, a merge index, a merge candidate, a merge candidate list, information indicating whether a skip mode is used, a type of an interpolation filter, a tap of an interpolation filter, a filter coefficient of an interpolation filter, a magnitude of a motion vector, accuracy of motion vector representation, a transform type, a transform size, information indicating whether a first transform is used, information indicating whether an additional (secondary) transform is used, first transform selection information (or a first transform index), secondary transform selection information (or a secondary transform index), information indicating a presence or absence of a residual signal, a coded block pattern, a coded block flag, a quantization parameter, a residual quantization parameter, a quantization matrix, information about an intra-loop filter, information indicating whether an intra-loop filter is applied, a coefficient of an intra-loop filter, a tap of an intra-loop filter, a shape/form of an intra-loop filter, information indicating whether a deblocking filter is applied, a coefficient of a deblocking filter, a tap of a deblocking filter, deblocking filter strength, a shape/form of a deblocking filter, information indicating whether an adaptive sample offset is applied, a value of an adaptive sample offset, a category of an adaptive sample offset, a type of an adaptive sample offset, information indicating whether an adaptive in-loop filter is applied, a coefficient of an adaptive in-loop filter, a tap of an adaptive in-loop filter, a shape/form of an adaptive in-loop filter, a binarization/inverse binarization method, a context model, a context model decision method, a context model update method, information indicating whether a regular mode is performed, information whether a bypass mode is performed, a significant coefficient flag, a last significant coefficient flag, a coding flag for a coefficient group, a position of a last significant coefficient, information indicating whether a value of a coefficient is greater than 1, information indicating whether a value of a coefficient is greater than 2, information indicating whether a value of a coefficient is greater than 3, a remaining coefficient value information, a sign information, a reconstructed luma sample, a reconstructed chroma sample, a context bin, a bypass bin, a residual luma sample, a residual chroma sample, a transform coefficient, a luma transform coefficient, a chroma transform coefficient, a quantized level, a luma quantized level, a chroma quantized level, a transform coefficient level, a transform coefficient level scanning method, a size of a motion vector search region on a side of a decoding apparatus, a shape/form of a motion vector search region on a side of a decoding apparatus, the number of a motion vector search on a side of a decoding apparatus, a size of a CTU, a minimum block size, a maximum block size, a maximum block depth, a minimum block depth, an image display/output order, slice identification information, a slice type, slice partition information, tile group identification information, a tile group type, a tile group partitioning information, tile identification information, a tile type, tile partitioning information, a picture type, bit depth, input sample bit depth, reconstructed sample bit depth, residual sample bit depth, transform coefficient bit depth, quantized level bit depth, information about a luma signal, information about a chroma signal, a color space of a target block and a color space of a residual block. Further, the above-described coding parameter-related information may also be included

in the coding parameter. Information used to calculate and/or derive the above-described coding parameter may also be included in the coding parameter. Information calculated or derived using the above-described coding parameter may also be included in the coding parameter.

The first transform selection information may indicate a first transform which is applied to a target block.

The second transform selection information may indicate a second transform which is applied to a target block.

The residual signal may denote the difference between the original signal and a prediction signal. Alternatively, the residual signal may be a signal generated by transforming the difference between the original signal and the prediction signal. Alternatively, the residual signal may be a signal generated by transforming and quantizing the difference between the original signal and the prediction signal. A residual block may be the residual signal for a block.

Here, signaling information may mean that the encoding apparatus **100** includes an entropy-encoded information, generated by performing entropy encoding a flag or an index, in a bitstream, and that the decoding apparatus **200** acquires information by performing entropy decoding on the entropy-encoded information, extracted from the bitstream. Here, the information may comprises a flag, an index, etc.

A signal may mean information to be signaled. Hereinafter, information for an image and a block may be referred to as a signal. Further, hereinafter, the terms “information” and “signal” may be used to have the same meaning and may be used interchangeably with each other. For example, a specific signal may be a signal representing a specific block. An original signal may be a signal representing a target block. A prediction signal may be a signal representing a prediction block. A residual signal may be a signal representing a residual block.

A bitstream may include information based on a specific syntax. The encoding apparatus **100** may generate a bitstream including information depending on a specific syntax. The encoding apparatus **200** may acquire information from the bitstream depending on a specific syntax.

Since the encoding apparatus **100** performs encoding via inter prediction, the encoded target image may be used as a reference image for additional image(s) to be subsequently processed. Therefore, the encoding apparatus **100** may reconstruct or decode the encoded target image and store the reconstructed or decoded image as a reference image in the reference picture buffer **190**. For decoding, dequantization and inverse transform on the encoded target image may be processed.

The quantized level may be inversely quantized by the dequantization unit **160**, and may be inversely transformed by the inverse transform unit **170**. The dequantization unit **160** may generate an inversely quantized coefficient by performing inverse transform for the quantized level. The inverse transform unit **170** may generate a inversely quantized and inversely transformed coefficient by performing inverse transform for the inversely quantized coefficient.

The inversely quantized and inversely transformed coefficient may be added to the prediction block by the adder **175**. The inversely quantized and inversely transformed coefficient and the prediction block are added, and then a reconstructed block may be generated. Here, the inversely quantized and/or inversely transformed coefficient may denote a coefficient on which one or more of dequantization and inverse transform are performed, and may also denote a reconstructed residual block. Here, the reconstructed block may mean a recovered block or a decoded block.

The reconstructed block may be subjected to filtering through the filter unit **180**. The filter unit **180** may apply one or more of a deblocking filter, a Sample Adaptive Offset (SAO) filter, an Adaptive Loop Filter (ALF), and a Non Local Filter (NLF) to a reconstructed sample, the reconstructed block or a reconstructed picture. The filter unit **180** may also be referred to as an “in-loop filter”.

The deblocking filter may eliminate block distortion occurring at the boundaries between blocks in a reconstructed picture. In order to determine whether to apply the deblocking filter, the number of columns or rows which are included in a block and which include pixel(s) based on which it is determined whether to apply the deblocking filter to a target block may be decided on.

When the deblocking filter is applied to the target block, the applied filter may differ depending on the strength of the required deblocking filtering. In other words, among different filters, a filter decided on in consideration of the strength of deblocking filtering may be applied to the target block. When a deblocking filter is applied to a target block, one or more filters of a long-tap filter, a strong filter, a weak filter and Gaussian filter may be applied to the target block depending on the strength of required deblocking filtering.

Also, when vertical filtering and horizontal filtering are performed on the target block, the horizontal filtering and the vertical filtering may be processed in parallel.

The SAO may add a suitable offset to the values of pixels to compensate for coding error. The SAO may perform, for the image to which deblocking is applied, correction that uses an offset in the difference between an original image and the image to which deblocking is applied, on a pixel basis. To perform an offset correction for an image, a method for dividing the pixels included in the image into a certain number of regions, determining a region to which an offset is to be applied, among the divided regions, and applying an offset to the determined region may be used, and a method for applying an offset in consideration of edge information of each pixel may also be used.

The ALF may perform filtering based on a value obtained by comparing a reconstructed image with an original image. After pixels included in an image have been divided into a predetermined number of groups, filters to be applied to each group may be determined, and filtering may be differentially performed for respective groups. Information related to whether to apply an adaptive loop filter may be signaled for each CU. Such information may be signaled for a luma signal. The shapes and filter coefficients of ALFs to be applied to respective blocks may differ for respective blocks. Alternatively, regardless of the features of a block, an ALF having a fixed form may be applied to the block.

A non-local filter may perform filtering based on reconstructed blocks, similar to a target block. A region similar to the target block may be selected from a reconstructed picture, and filtering of the target block may be performed using the statistical properties of the selected similar region. Information about whether to apply a non-local filter may be signaled for a Coding Unit (CU). Also, the shapes and filter coefficients of the non-local filter to be applied to blocks may differ depending on the blocks.

The reconstructed block or the reconstructed image subjected to filtering through the filter unit **180** may be stored in the reference picture buffer **190** as a reference picture. The reconstructed block subjected to filtering through the filter unit **180** may be a part of a reference picture. In other words, the reference picture may be a reconstructed picture composed of reconstructed blocks subjected to filtering through

the filter unit **180**. The stored reference picture may be subsequently used for inter prediction or a motion compensation.

FIG. 2 is a block diagram illustrating the configuration of an embodiment of a decoding apparatus to which the present disclosure is applied.

A decoding apparatus **200** may be a decoder, a video decoding apparatus or an image decoding apparatus.

Referring to FIG. 2, the decoding apparatus **200** may include an entropy decoding unit **210**, a dequantization (inverse quantization) unit **220**, an inverse transform unit **230**, an intra-prediction unit **240**, an inter-prediction unit **250**, a switch **245**, an adder **255**, a filter unit **260**, and a reference picture buffer **270**.

The decoding apparatus **200** may receive a bitstream output from the encoding apparatus **100**. The decoding apparatus **200** may receive a bitstream stored in a computer-readable storage medium, and may receive a bitstream that is streamed through a wired/wireless transmission medium.

The decoding apparatus **200** may perform decoding on the bitstream in an intra mode and/or an inter mode. Further, the decoding apparatus **200** may generate a reconstructed image or a decoded image via decoding, and may output the reconstructed image or decoded image.

For example, switching to an intra mode or an inter mode based on the prediction mode used for decoding may be performed by the switch **245**. When the prediction mode used for decoding is an intra mode, the switch **245** may be operated to switch to the intra mode. When the prediction mode used for decoding is an inter mode, the switch **245** may be operated to switch to the inter mode.

The decoding apparatus **200** may acquire a reconstructed residual block by decoding the input bitstream, and may generate a prediction block. When the reconstructed residual block and the prediction block are acquired, the decoding apparatus **200** may generate a reconstructed block, which is the target to be decoded, by adding the reconstructed residual block and the prediction block.

The entropy decoding unit **210** may generate symbols by performing entropy decoding on the bitstream based on the probability distribution of a bitstream. The generated symbols may include symbols in a form of a quantized transform coefficient level (i.e., a quantized level or a quantized coefficient). Here, the entropy decoding method may be similar to the above-described entropy encoding method. That is, the entropy decoding method may be the reverse procedure of the above-described entropy encoding method.

The entropy decoding unit **210** may change a coefficient having a one-dimensional (1D) vector form to a 2D block shape through a transform coefficient scanning method in order to decode a quantized transform coefficient level.

For example, the coefficients of the block may be changed to 2D block shapes by scanning the block coefficients using up-right diagonal scanning. Alternatively, which one of up-right diagonal scanning, vertical scanning, and horizontal scanning is to be used may be determined depending on the size and/or the intra-prediction mode of the corresponding block.

The quantized coefficient may be inversely quantized by the dequantization unit **220**. The dequantization unit **220** may generate an inversely quantized coefficient by performing dequantization on the quantized coefficient. Further, the inversely quantized coefficient may be inversely transformed by the inverse transform unit **230**. The inverse transform unit **230** may generate a reconstructed residual block by performing an inverse transform on the inversely quantized coefficient. As a result of performing dequantiza-

tion and the inverse transform on the quantized coefficient, the reconstructed residual block may be generated. Here, the dequantization unit **220** may apply a quantization matrix to the quantized coefficient when generating the reconstructed residual block.

When the intra mode is used, the intra-prediction unit **240** may generate a prediction block by performing spatial prediction that uses the pixel values of previously decoded neighbor blocks adjacent to a target block for the target block.

The inter-prediction unit **250** may include a motion compensation unit. Alternatively, the inter-prediction unit **250** may be designated as a "motion compensation unit".

When the inter mode is used, the motion compensation unit may generate a prediction block by performing motion compensation that uses a motion vector and a reference image stored in the reference picture buffer **270** for the target block.

The motion compensation unit may apply an interpolation filter to a partial area of the reference image when the motion vector has a value other than an integer, and may generate a prediction block using the reference image to which the interpolation filter is applied. In order to perform motion compensation, the motion compensation unit may determine which one of a skip mode, a merge mode, an Advanced Motion Vector Prediction (AMVP) mode, and a current picture reference mode corresponds to the motion compensation method used for a PU included in a CU, based on the CU, and may perform motion compensation depending on the determined mode.

The reconstructed residual block and the prediction block may be added to each other by the adder **255**. The adder **255** may generate a reconstructed block by adding the reconstructed residual block to the prediction block.

The reconstructed block may be subjected to filtering through the filter unit **260**. The filter unit **260** may apply at least one of a deblocking filter, an SAO filter, an ALF, and a NLF to the reconstructed block or the reconstructed image. The reconstructed image may be a picture including the reconstructed block.

The filter unit may output the reconstructed image.

The reconstructed image and/or the reconstructed block subjected to filtering through the filter unit **260** may be stored as a reference picture in the reference picture buffer **270**. The reconstructed block subjected to filtering through the filter unit **260** may be a part of the reference picture. In other words, the reference picture may be an image composed of reconstructed blocks subjected to filtering through the filter unit **260**. The stored reference picture may be subsequently used for inter prediction or a motion compensation.

FIG. 3 is a diagram schematically illustrating the partition structure of an image when the image is encoded and decoded.

FIG. 3 may schematically illustrate an example in which a single unit is partitioned into multiple sub-units.

In order to efficiently partition the image, a Coding Unit (CU) may be used in encoding and decoding. The term "unit" may be used to collectively designate 1) a block including image samples and 2) a syntax element. For example, the "partitioning of a unit" may mean the "partitioning of a block corresponding to a unit".

A CU may be used as a base unit for image encoding/decoding. A CU may be used as a unit to which one mode selected from an intra mode and an inter mode in image encoding/decoding is applied. In other words, in image

encoding/decoding, which one of an intra mode and an inter mode is to be applied to each CU may be determined.

Further, a CU may be a base unit in prediction, transform, quantization, inverse transform, dequantization, and encoding/decoding of transform coefficients.

Referring to FIG. 3, an image 200 may be sequentially partitioned into units corresponding to a Largest Coding Unit (LCU), and a partition structure may be determined for each LCU. Here, the LCU may be used to have the same meaning as a Coding Tree Unit (CTU).

The partitioning of a unit may mean the partitioning of a block corresponding to the unit. Block partition information may include depth information about the depth of a unit. The depth information may indicate the number of times the unit is partitioned and/or the degree to which the unit is partitioned. A single unit may be hierarchically partitioned into a plurality of sub-units while having depth information based on a tree structure. Each of partitioned sub-units may have depth information. The depth information may be information indicating the size of a CU. The depth information may be stored for each CU.

Each CU may have depth information. When the CU is partitioned, CUs resulting from partitioning may have a depth increased from the depth of the partitioned CU by 1.

The partition structure may mean the distribution of Coding Units (CUs) to efficiently encode the image in an LCU 310. Such a distribution may be determined depending on whether a single CU is to be partitioned into multiple CUs. The number of CUs generated by partitioning may be a positive integer of 2 or more, including 2, 3, 4, 8, 16, etc.

The horizontal size and the vertical size of each of CUs generated by the partitioning may be less than the horizontal size and the vertical size of a CU before being partitioned, depending on the number of CUs generated by partitioning. For example, the horizontal size and the vertical size of each of CUs generated by the partitioning may be half of the horizontal size and the vertical size of a CU before being partitioned.

Each partitioned CU may be recursively partitioned into four CUs in the same way. Via the recursive partitioning, at least one of the horizontal size and the vertical size of each partitioned CU may be reduced compared to at least one of the horizontal size and the vertical size of the CU before being partitioned.

The partitioning of a CU may be recursively performed up to a predefined depth or a predefined size.

For example, the depth of a CU may have a value ranging from 0 to 3. The size of the CU may range from a size of 64×64 to a size of 8×8 depending on the depth of the CU.

For example, the depth of an LCU 310 may be 0, and the depth of a Smallest Coding Unit (SCU) may be a predefined maximum depth. Here, as described above, the LCU may be the CU having the maximum coding unit size, and the SCU may be the CU having the minimum coding unit size.

Partitioning may start at the LCU 310, and the depth of a CU may be increased by 1 whenever the horizontal and/or vertical sizes of the CU are reduced by partitioning.

For example, for respective depths, a CU that is not partitioned may have a size of 2N×2N. Further, in the case of a CU that is partitioned, a CU having a size of 2N×2N may be partitioned into four CUs, each having a size of N×N. The value of N may be halved whenever the depth is increased by 1.

Referring to FIG. 3, an LCU having a depth of 0 may have 64×64 pixels or 64×64 blocks. 0 may be a minimum depth. An SCU having a depth of 3 may have 8×8 pixels or 8×8 blocks. 3 may be a maximum depth. Here, a CU having

64×64 blocks, which is the LCU, may be represented by a depth of 0. A CU having 32×32 blocks may be represented by a depth of 1. A CU having 16×16 blocks may be represented by a depth of 2. A CU having 8×8 blocks, which is the SCU, may be represented by a depth of 3.

Information about whether the corresponding CU is partitioned may be represented by the partition information of the CU. The partition information may be 1-bit information. All CUs except the SCU may include partition information.

For example, the value of the partition information of a CU that is not partitioned may be a first value. The value of the partition information of a CU that is partitioned may be a second value. When the partition information indicates whether a CU is partitioned or not, the first value may be “0” and the second value may be “1”.

For example, when a single CU is partitioned into four CUs, the horizontal size and vertical size of each of four CUs generated by partitioning may be half the horizontal size and the vertical size of the CU before being partitioned. When a CU having a 32×32 size is partitioned into four CUs, the size of each of four partitioned CUs may be 16×16. When a single CU is partitioned into four CUs, it may be considered that the CU has been partitioned in a quad-tree structure. In other words, it may be considered that a quad-tree partition has been applied to a CU.

For example, when a single CU is partitioned into two CUs, the horizontal size or the vertical size of each of two CUs generated by partitioning may be half the horizontal size or the vertical size of the CU before being partitioned. When a CU having a 32×32 size is vertically partitioned into two CUs, the size of each of two partitioned CUs may be 16×32. When a CU having a 32×32 size is horizontally partitioned into two CUs, the size of each of two partitioned CUs may be 32×16. When a single CU is partitioned into two CUs, it may be considered that the CU has been partitioned in a binary-tree structure. In other words, it may be considered that a binary-tree partition has been applied to a CU.

For example, when a single CU is partitioned (or split) into three CUs, the original CU before being partitioned is partitioned so that the horizontal size or vertical size thereof is divided at a ratio of 1:2:1, thus enabling three sub-CUs to be generated. For example, when a CU having a 16×32 size is horizontally partitioned into three sub-CUs, the three sub-CUs resulting from the partitioning may have sizes of 16×8, 16×16, and 16×8, respectively, in a direction from the top to the bottom. For example, when a CU having a 32×32 size is vertically partitioned into three sub-CUs, the three sub-CUs resulting from the partitioning may have sizes of 8×32, 16×32, and 8×32, respectively, in a direction from the left to the right. When a single CU is partitioned into three CUs, it may be considered that the CU is partitioned in a ternary-tree form. In other words, it may be considered that a ternary-tree partition has been applied to the CU.

Both of quad-tree partitioning and binary-tree partitioning are applied to the LCU 310 of FIG. 3.

In the encoding apparatus 100, a Coding Tree Unit (CTU) having a size of 64×64 may be partitioned into multiple smaller CUs by a recursive quad-tree structure. A single CU may be partitioned into four CUs having the same size. Each CU may be recursively partitioned, and may have a quad-tree structure.

By the recursive partitioning of a CU, an optimal partitioning method that incurs a minimum rate-distortion cost may be selected.

The Coding Tree Unit (CTU) 320 in FIG. 3 is an example of a CTU to which all of a quad-tree partition, a binary-tree partition, and a ternary-tree partition are applied.

As described above, in order to partition a CTU, at least one of a quad-tree partition, a binary-tree partition, and a ternary-tree partition may be applied to the CTU. Partitions may be applied based on specific priority.

For example, a quad-tree partition may be preferentially applied to the CTU. A CU that cannot be partitioned in a quad-tree form any further may correspond to a leaf node of a quad-tree. A CU corresponding to the leaf node of the quad-tree may be a root node of a binary tree and/or a ternary tree. That is, the CU corresponding to the leaf node of the quad-tree may be partitioned in a binary-tree form or a ternary-tree form, or may not be partitioned any further. In this case, each CU, which is generated by applying a binary-tree partition or a ternary-tree partition to the CU corresponding to the leaf node of a quad-tree, is prevented from being subjected again to quad-tree partitioning, thus effectively performing partitioning of a block and/or signaling of block partition information.

The partition of a CU corresponding to each node of a quad-tree may be signaled using quad-partition information. Quad-partition information having a first value (e.g., "1") may indicate that the corresponding CU is partitioned in a quad-tree form. Quad-partition information having a second value (e.g., "0") may indicate that the corresponding CU is not partitioned in a quad-tree form. The quad-partition information may be a flag having a specific length (e.g., 1 bit).

Priority may not exist between a binary-tree partition and a ternary-tree partition. That is, a CU corresponding to the leaf node of a quad-tree may be partitioned in a binary-tree form or a ternary-tree form. Also, the CU generated through a binary-tree partition or a ternary-tree partition may be further partitioned in a binary-tree form or a ternary-tree form, or may not be partitioned any further.

Partitioning performed when priority does not exist between a binary-tree partition and a ternary-tree partition may be referred to as a "multi-type tree partition". That is, a CU corresponding to the leaf node of a quad-tree may be the root node of a multi-type tree. Partitioning of a CU corresponding to each node of the multi-type tree may be signaled using at least one of information indicating whether the CU is partitioned in a multi-type tree, partition direction information, and partition tree information. For partitioning of a CU corresponding to each node of a multi-type tree, information indicating whether partitioning in the multi-type tree is performed, partition direction information, and partition tree information may be sequentially signaled.

For example, information indicating whether a CU is partitioned in a multi-type tree and having a first value (e.g., "1") may indicate that the corresponding CU is partitioned in a multi-type tree form. Information indicating whether a CU is partitioned in a multi-type tree and having a second value (e.g., "0") may indicate that the corresponding CU is not partitioned in a multi-type tree form.

When a CU corresponding to each node of a multi-type tree is partitioned in a multi-type tree form, the corresponding CU may further include partition direction information.

The partition direction information may indicate the partition direction of the multi-type tree partition. Partition direction information having a first value (e.g., "1") may indicate that the corresponding CU is partitioned in a vertical direction. Partition direction information having a second value (e.g., "0") may indicate that the corresponding CU is partitioned in a horizontal direction.

When a CU corresponding to each node of a multi-type tree is partitioned in a multi-type tree form, the corresponding CU may further include partition-tree information. The partition-tree information may indicate the tree that is used for a multi-type tree partition.

For example, partition-tree information having a first value (e.g., "1") may indicate that the corresponding CU is partitioned in a binary-tree form. Partition-tree information having a second value (e.g., "0") may indicate that the corresponding CU is partitioned in a ternary-tree form.

Here, each of the above-described information indicating whether partitioning in the multi-type tree is performed, partition-tree information, and partition direction information may be a flag having a specific length (e.g., 1 bit).

At least one of the above-described quad-partition information, information indicating whether partitioning in the multi-type tree is performed, partition direction information, and partition-tree information may be entropy-encoded and/or entropy-decoded. In order to perform entropy encoding/decoding of such information, information of a neighbor CU adjacent to a target CU may be used.

For example, it may be considered that there is a high probability that the partition form of a left CU and/or an above CU (i.e., partitioning/non-partitioning, a partition tree and/or a partition direction) and the partition form of a target CU will be similar to each other. Therefore, based on the information of a neighbor CU, context information for entropy encoding and/or entropy decoding of the information of the target CU may be derived. Here, the information of the neighbor CU may include at least one of 1) quad-partition information of the neighbor CU, 2) information indicating whether the neighbor CU is partitioned in a multi-type tree, 3) partition direction information of the neighbor CU, and 4) partition-tree information of the neighbor CU.

In another embodiment, of a binary-tree partition and a ternary-tree partition, the binary-tree partition may be preferentially performed. That is, the binary-tree partition may be first applied, and then a CU corresponding to the leaf node of a binary tree may be set to the root node of a ternary tree. In this case, a quad-tree partition or a binary-tree partition may not be performed on the CU corresponding to the node of the ternary tree.

A CU, which is not partitioned any further through a quad-tree partition, a binary-tree partition, and/or a ternary-tree partition, may be the unit of encoding, prediction and/or transform. That is, the CU may not be partitioned any further for prediction and/or transform. Therefore, a partition structure for partitioning the CU into Prediction Units (PUs) and/or Transform Units (TUs), partition information thereof, etc. may not be present in a bitstream.

However, when the size of a CU, which is the unit of partitioning, is greater than the size of a maximum transform block, the CU may be recursively partitioned until the size of the CU becomes less than or equal to the size of the maximum transform block. For example, when the size of a CU is 64×64 and the size of the maximum transform block is 32×32, the CU may be partitioned into four 32×32 blocks so as to perform a transform. For example, when the size of a CU is 32×64 and the size of the maximum transform block is 32×32, the CU may be partitioned into two 32×32 blocks.

In this case, information indicating whether a CU is partitioned for a transform may not be separately signaled. Without signaling, whether a CU is partitioned may be determined via a comparison between the horizontal size (and/or vertical size) of the CU and the horizontal size (and/or vertical size) of the maximum transform block. For

example, when the horizontal size of the CU is greater than the horizontal size of the maximum transform block, the CU may be vertically bisected. Further, when the vertical size of the CU is greater than the vertical size of the maximum transform block, the CU may be horizontally bisected.

Information about the maximum size and/or minimum size of a CU and information about the maximum size and/or minimum size of a transform block may be signaled or determined at a level higher than that of the CU. For example, the higher level may be a sequence level, a picture level, a tile level, a tile group level or a slice level. For example, the minimum size of the CU may be set to 4×4. For example, the maximum size of the transform block may be set to 64×64. For example, the maximum size of the transform block may be set to 4×4.

Information about the minimum size of a CU corresponding to the leaf node of a quad-tree (i.e., the minimum size of the quad-tree) and/or information about the maximum depth of a path from the root node to the leaf node of a multi-type tree (i.e., the maximum depth of a multi-type tree) may be signaled or determined at a level higher than that of the CU. For example, the higher level may be a sequence level, a picture level, a slice level, a tile group level or a tile level. Information about the minimum size of a quad-tree and/or information about the maximum depth of a multi-type tree may be separately signaled or determined at each of an intra-slice level and an inter-slice level.

Information about the difference between the size of a CTU and the maximum size of a transform block may be signaled or determined at a level higher than that of a CU. For example, the higher level may be a sequence level, a picture level, a slice level, a tile group level or a tile level. Information about the maximum size of a CU corresponding to each node of a binary tree (i.e., the maximum size of the binary tree) may be determined based on the size and the difference information of a CTU. The maximum size of a CU corresponding to each node of a ternary tree (i.e., the maximum size of the ternary tree) may have different values depending on the type of slice. For example, the maximum size of the ternary tree at an intra-slice level may be 32×32. For example, the maximum size of the ternary tree at an inter-slice level may be 128×128. For example, the minimum size of a CU corresponding to each node of a binary tree (i.e., the minimum size of the binary tree) and/or the minimum size of a CU corresponding to each node of a ternary tree (i.e., the minimum size of the ternary tree) may be set to the minimum size of a CU.

In a further example, the maximum size of a binary tree and/or the maximum size of a ternary tree may be signaled or determined at a slice level. Also, the minimum size of a binary tree and/or the minimum size of a ternary tree may be signaled or determined at a slice level.

Based on the above-described various block sizes and depths, quad-partition information, information indicating whether partitioning in a multi-type tree is performed, partition tree information and/or partition direction information may or may not be present in a bitstream.

For example, when the size of a CU is not greater than the minimum size of a quad-tree, the CU may not include quad-partition information, and quad-partition information of the CU may be inferred as a second value.

For example, when the size of a CU corresponding to each node of a multi-type tree (horizontal size and vertical size) is greater than the maximum size of a binary tree (horizontal size and vertical size) and/or the maximum size of a ternary tree (horizontal size and vertical size), the CU may not be partitioned in a binary-tree form and/or a ternary-tree form.

By means of this determination manner, information indicating whether partitioning in a multi-type tree is performed may not be signaled, but may be inferred as a second value.

Alternatively, when the size of a CU corresponding to each node of a multi-type tree (horizontal size and vertical size) is equal to the minimum size of a binary tree (horizontal size and vertical size), or when the size of a CU (horizontal size and vertical size) is equal to twice the minimum size of a ternary tree (horizontal size and vertical size), the CU may not be partitioned in a binary tree form and/or a ternary tree form. By means of this determination manner, information indicating whether partitioning in a multi-type tree is performed may not be signaled, but may be inferred as a second value. The reason for this is that, when a CU is partitioned in a binary tree form and/or a ternary tree form, a CU smaller than the minimum size of the binary tree and/or the minimum size of the ternary tree is generated.

Alternatively, a binary-tree partition or a ternary-tree partition may be limited based on the size of a virtual pipeline data unit (i.e., the size of a pipeline buffer). For example, when a CU is partitioned into sub-CUs unsuitable for the size of a pipeline buffer through a binary-tree partition or a ternary-tree partition, a binary-tree partition or a ternary-tree partition may be limited. The size of the pipeline buffer may be equal to the maximum size of a transform block (e.g., 64×64).

For example, when the size of the pipeline buffer is 64×64, the following partitions may be limited.

Ternary-tree partition for N×M CU (where N and/or M are 128)

Horizontal binary-tree partition for 128×N CU (where N≤64)

Vertical binary-tree partition for N×128 CU (where N≤64)

Alternatively, when the depth of a CU corresponding to each node of a multi-type tree is equal to the maximum depth of the multi-type tree, the CU may not be partitioned in a binary-tree form and/or a ternary-tree form. By means of this determination manner, information indicating whether partitioning in a multi-type tree is performed may not be signaled, but may be inferred as a second value.

Alternatively, information indicating whether partitioning in a multi-type tree is performed may be signaled only when at least one of a vertical binary-tree partition, a horizontal binary-tree partition, a vertical ternary-tree partition, and a horizontal ternary-tree partition is possible for a CU corresponding to each node of a multi-type tree. Otherwise, the CU may not be partitioned in a binary-tree form and/or a ternary-tree form. By means of this determination manner, information indicating whether partitioning in a multi-type tree is performed may not be signaled, but may be inferred as a second value.

Alternatively, partition direction information may be signaled only when both a vertical binary-tree partition and a horizontal binary-tree partition are possible or only when both a vertical ternary-tree partition and a horizontal ternary-tree partition are possible, for a CU corresponding to each node of a multi-type tree. Otherwise, the partition direction information may not be signaled, but may be inferred as a value indicating the direction in which the CU can be partitioned.

Alternatively, partition tree information may be signaled only when both a vertical binary-tree partition and a vertical ternary-tree partition are possible or only when both a horizontal binary-tree partition and a horizontal ternary-tree partition are possible, for a CU corresponding to each node

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of a multi-type tree. Otherwise, the partition tree information may not be signaled, but may be inferred as a value indicating a tree that can be applied to the partition of the CU.

FIG. 4 is a diagram illustrating the form of a Prediction Unit that a Coding Unit can include.

When, among CUs partitioned from an LCU, a CU, which is not partitioned any further, may be divided into one or more Prediction Units (PUs). Such division is also referred to as "partitioning".

A PU may be a base unit for prediction. A PU may be encoded and decoded in any one of a skip mode, an inter mode, and an intra mode. A PU may be partitioned into various shapes depending on respective modes. For example, the target block, described above with reference to FIG. 1, and the target block, described above with reference to FIG. 2, may each be a PU.

A CU may not be split into PUs. When the CU is not split into PUs, the size of the CU and the size of a PU may be equal to each other.

In a skip mode, partitioning may not be present in a CU. In the skip mode, a $2N \times 2N$ mode 410, in which the sizes of a PU and a CU are identical to each other, may be supported without partitioning.

In an inter mode, 8 types of partition shapes may be present in a CU. For example, in the inter mode, the $2N \times 2N$ mode 410, a $2N \times N$ mode 415, an $N \times 2N$ mode 420, an $N \times N$ mode 425, a $2N \times nU$ mode 430, a $2N \times nD$ mode 435, an $nL \times 2N$ mode 440, and an $nR \times 2N$ mode 445 may be supported.

In an intra mode, the $2N \times 2N$ mode 410 and the $N \times N$ mode 425 may be supported.

In the $2N \times 2N$ mode 410, a PU having a size of $2N \times 2N$ may be encoded. The PU having a size of $2N \times 2N$ may mean a PU having a size identical to that of the CU. For example, the PU having a size of $2N \times 2N$ may have a size of 64×64 , 32×32 , 16×16 or 8×8 .

In the $N \times N$ mode 425, a PU having a size of $N \times N$ may be encoded.

For example, in intra prediction, when the size of a PU is 8×8 , four partitioned PUs may be encoded. The size of each partitioned PU may be 4×4 .

When a PU is encoded in an intra mode, the PU may be encoded using any one of multiple intra-prediction modes. For example, HEVC technology may provide 35 intra-prediction modes, and the PU may be encoded in any one of the 35 intra-prediction modes.

Which one of the $2N \times 2N$ mode 410 and the $N \times N$ mode 425 is to be used to encode the PU may be determined based on rate-distortion cost.

The encoding apparatus 100 may perform an encoding operation on a PU having a size of $2N \times 2N$. Here, the encoding operation may be the operation of encoding the PU in each of multiple intra-prediction modes that can be used by the encoding apparatus 100. Through the encoding operation, the optimal intra-prediction mode for a PU having a size of $2N \times 2N$ may be derived. The optimal intra-prediction mode may be an intra-prediction mode in which a minimum rate-distortion cost occurs upon encoding the PU having a size of $2N \times 2N$, among multiple intra-prediction modes that can be used by the encoding apparatus 100.

Further, the encoding apparatus 100 may sequentially perform an encoding operation on respective PUs obtained from $N \times N$ partitioning. Here, the encoding operation may be the operation of encoding a PU in each of multiple intra-prediction modes that can be used by the encoding apparatus 100. By means of the encoding operation, the optimal

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intra-prediction mode for the PU having a size of $N \times N$ may be derived. The optimal intra-prediction mode may be an intra-prediction mode in which a minimum rate-distortion cost occurs upon encoding the PU having a size of $N \times N$, among multiple intra-prediction modes that can be used by the encoding apparatus 100.

The encoding apparatus 100 may determine which of a PU having a size of $2N \times 2N$ and PUs having sizes of $N \times N$ to be encoded based on a comparison of a rate-distortion cost of the PU having a size of $2N \times 2N$ and a rate-distortion costs of the PUs having sizes of $N \times N$.

A single CU may be partitioned into one or more PUs, and a PU may be partitioned into multiple PUs.

For example, when a single PU is partitioned into four PUs, the horizontal size and vertical size of each of four PUs generated by partitioning may be half the horizontal size and the vertical size of the PU before being partitioned. When a PU having a 32×32 size is partitioned into four PUs, the size of each of four partitioned PUs may be 16×16 . When a single PU is partitioned into four PUs, it may be considered that the PU has been partitioned in a quad-tree structure.

For example, when a single PU is partitioned into two PUs, the horizontal size or the vertical size of each of two PUs generated by partitioning may be half the horizontal size or the vertical size of the PU before being partitioned. When a PU having a 32×32 size is vertically partitioned into two PUs, the size of each of two partitioned PUs may be 16×32 . When a PU having a 32×32 size is horizontally partitioned into two PUs, the size of each of two partitioned PUs may be 32×16 . When a single PU is partitioned into two PUs, it may be considered that the PU has been partitioned in a binary-tree structure.

FIG. 5 is a diagram illustrating the form of a Transform Unit that can be included in a Coding Unit.

A Transform Unit (TU) may have a base unit that is used for a procedure, such as transform, quantization, inverse transform, dequantization, entropy encoding, and entropy decoding, in a CU.

A TU may have a square shape or a rectangular shape. A shape of a TU may be determined based on a size and/or a shape of a CU.

Among CUs partitioned from the LCU, a CU which is not partitioned into CUs any further may be partitioned into one or more TUs. Here, the partition structure of a TU may be a quad-tree structure. For example, as shown in FIG. 5, a single CU 510 may be partitioned one or more times depending on the quad-tree structure. By means of this partitioning, the single CU 510 may be composed of TUs having various sizes.

It can be considered that when a single CU is split two or more times, the CU is recursively split. Through splitting, a single CU may be composed of Transform Units (TUs) having various sizes.

Alternatively, a single CU may be split into one or more TUs based on the number of vertical lines and/or horizontal lines that split the CU.

A CU may be split into symmetric TUs or asymmetric TUs. For splitting into asymmetric TUs, information about the size and/or shape of each TU may be signaled from the encoding apparatus 100 to the decoding apparatus 200. Alternatively, the size and/or shape of each TU may be derived from information about the size and/or shape of the CU.

A CU may not be split into TUs. When the CU is not split into TUs, the size of the CU and the size of a TU may be equal to each other.

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A single CU may be partitioned into one or more TUs, and a TU may be partitioned into multiple TUs.

For example, when a single TU is partitioned into four TUs, the horizontal size and vertical size of each of four TUs generated by partitioning may be half the horizontal size and the vertical size of the TU before being partitioned. When a TU having a 32×32 size is partitioned into four TUs, the size of each of four partitioned TUs may be 16×16. When a single TU is partitioned into four TUs, it may be considered that the TU has been partitioned in a quad-tree structure.

For example, when a single TU is partitioned into two TUs, the horizontal size or the vertical size of each of two TUs generated by partitioning may be half the horizontal size or the vertical size of the TU before being partitioned. When a TU having a 32×32 size is vertically partitioned into two TUs, the size of each of two partitioned TUs may be 16×32. When a TU having a 32×32 size is horizontally partitioned into two TUs, the size of each of two partitioned TUs may be 32×16. When a single TU is partitioned into two TUs, it may be considered that the TU has been partitioned in a binary-tree structure.

In a way differing from that illustrated in FIG. 5, a CU may be split.

For example, a single CU may be split into three CUs. The horizontal sizes or vertical sizes of the three CUs generated from splitting may be $\frac{1}{4}$, $\frac{1}{2}$, and $\frac{1}{4}$, respectively, of the horizontal size or vertical size of the original CU before being split.

For example, when a CU having a 32×32 size is vertically split into three CUs, the sizes of the three CUs generated from the splitting may be 8×32, 16×32, and 8×32, respectively. In this way, when a single CU is split into three CUs, it may be considered that the CU is split in the form of a ternary tree.

One of exemplary splitting forms, that is, quad-tree splitting, binary tree splitting, and ternary tree splitting, may be applied to the splitting of a CU, and multiple splitting schemes may be combined and used together for splitting of a CU. Here, the case where multiple splitting schemes are combined and used together may be referred to as “complex tree-format splitting”.

FIG. 6 illustrates the splitting of a block according to an example.

In a video encoding and/or decoding process, a target block may be split, as illustrated in FIG. 6. For example, the target block may be a CU.

For splitting of the target block, an indicator indicating split information may be signaled from the encoding apparatus 100 to the decoding apparatus 200. The split information may be information indicating how the target block is split.

The split information may be one or more of a split flag (hereinafter referred to as “split_flag”), a quad-binary flag (hereinafter referred to as “QB_flag”), a quad-tree flag (hereinafter referred to as “quadtree_flag”), a binary tree flag (hereinafter referred to as “binarytree_flag”), and a binary type flag (hereinafter referred to as “Btype_flag”).

“split_flag” may be a flag indicating whether a block is split. For example, a split_flag value of 1 may indicate that the corresponding block is split. A split_flag value of 0 may indicate that the corresponding block is not split.

“QB_flag” may be a flag indicating which one of a quad-tree form and a binary tree form corresponds to the shape in which the block is split. For example, a QB_flag value of 0 may indicate that the block is split in a quad-tree form. A QB_flag value of 1 may indicate that the block is split in a binary tree form. Alternatively, a QB_flag value of

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0 may indicate that the block is split in a binary tree form. A QB_flag value of 1 may indicate that the block is split in a quad-tree form.

“quadtree_flag” may be a flag indicating whether a block is split in a quad-tree form. For example, a quadtree_flag value of 1 may indicate that the block is split in a quad-tree form. A quadtree_flag value of 0 may indicate that the block is not split in a quad-tree form.

“binarytree_flag” may be a flag indicating whether a block is split in a binary tree form. For example, a binarytree_flag value of 1 may indicate that the block is split in a binary tree form. A binarytree_flag value of 0 may indicate that the block is not split in a binary tree form.

“Btype_flag” may be a flag indicating which one of a vertical split and a horizontal split corresponds to a split direction when a block is split in a binary tree form. For example, a Btype_flag value of 0 may indicate that the block is split in a horizontal direction. A Btype_flag value of 1 may indicate that a block is split in a vertical direction. Alternatively, a Btype_flag value of 0 may indicate that the block is split in a vertical direction. A Btype_flag value of 1 may indicate that a block is split in a horizontal direction.

For example, the split information of the block in FIG. 6 may be derived by signaling at least one of quadtree_flag, binarytree_flag, and Btype_flag, as shown in the following Table 1.

TABLE 1

quadtree_flag	binarytree_flag	Btype_flag
1		
0	1	1
	0	
	0	
1		
0	1	0
	0	
	0	
0	0	
0	0	
0	0	
	1	0
	1	1
	0	
	0	
	0	
0	0	

For example, the split information of the block in FIG. 6 may be derived by signaling at least one of split_flag, QB_flag and Btype_flag, as shown in the following Table 2.

TABLE 2

split_flag	QB_flag	Btype_flag
1		
	0	
1	1	1

TABLE 2-continued

split_flag	QB_flag	Btype_flag
0		
0		
1		
	0	
1	1	
		0
0		
0		
0		
0		
1		
	1	
1		0
		1
0		
0		
0		

The splitting method may be limited only to a quad-tree or to a binary tree depending on the size and/or shape of the block. When this limitation is applied, split_flag may be a flag indicating whether a block is split in a quad-tree form or a flag indicating whether a block is split in a binary tree form. The size and shape of a block may be derived depending on the depth information of the block, and the depth information may be signaled from the encoding apparatus 100 to the decoding apparatus 200.

When the size of a block falls within a specific range, only splitting in a quad-tree form may be possible. For example, the specific range may be defined by at least one of a maximum block size and a minimum block size at which only splitting in a quad-tree form is possible.

Information indicating the maximum block size and the minimum block size at which only splitting in a quad-tree form is possible may be signaled from the encoding apparatus 100 to the decoding apparatus 200 through a bitstream. Further, this information may be signaled for at least one of units such as a video, a sequence, a picture, a parameter, a tile group, and a slice (or a segment).

Alternatively, the maximum block size and/or the minimum block size may be fixed sizes predefined by the encoding apparatus 100 and the decoding apparatus 200. For example, when the size of a block is above 64×64 and below 256×256, only splitting in a quad-tree form may be possible. In this case, split_flag may be a flag indicating whether splitting in a quad-tree form is performed.

When the size of a block is greater than the maximum size of a transform block, only partitioning in a quad-tree form may be possible. Here, a sub-block resulting from partitioning may be at least one of a CU and a TU.

In this case, split_flag may be a flag indicating whether a CU is partitioned in a quad-tree form.

When the size of a block falls within the specific range, only splitting in a binary tree form or a ternary tree form may be possible. For example, the specific range may be defined by at least one of a maximum block size and a minimum block size at which only splitting in a binary tree form or a ternary tree form is possible.

Information indicating the maximum block size and/or the minimum block size at which only splitting in a binary tree form or splitting in a ternary tree form is possible may be signaled from the encoding apparatus 100 to the decoding apparatus 200 through a bitstream. Further, this information

may be signaled for at least one of units such as a sequence, a picture, and a slice (or a segment).

Alternatively, the maximum block size and/or the minimum block size may be fixed sizes predefined by the encoding apparatus 100 and the decoding apparatus 200. For example, when the size of a block is above 8×8 and below 16×16, only splitting in a binary tree form may be possible. In this case, split_flag may be a flag indicating whether splitting in a binary tree form or a ternary tree form is performed.

The above description of partitioning in a quad-tree form may be equally applied to a binary-tree form and/or a ternary-tree form.

The partition of a block may be limited by a previous partition. For example, when a block is partitioned in a specific binary-tree form and then multiple sub-blocks are generated from the partitioning, each sub-block may be additionally partitioned only in a specific tree form. Here, the specific tree form may be at least one of a binary-tree form, a ternary-tree form, and a quad-tree form.

When the horizontal size or vertical size of a partition block is a size that cannot be split further, the above-described indicator may not be signaled.

FIG. 7 is a diagram for explaining an embodiment of an intra-prediction process.

Arrows radially extending from the center of the graph in FIG. 7 indicate the prediction directions of intra-prediction modes. Further, numbers appearing near the arrows indicate examples of mode values assigned to intra-prediction modes or to the prediction directions of the intra-prediction modes.

In FIG. 7, A number 0 may represent a Planar mode which is a non-directional intra prediction mode. A number 1 may represent a DC mode which is a non-directional intra prediction mode.

Intra encoding and/or decoding may be performed using a reference sample of neighbor unit of a target block. The neighbor block may be a reconstructed neighbor block. The reference sample may mean a neighbor sample.

For example, intra encoding and/or decoding may be performed using the value of a reference sample which are included in a reconstructed neighbor block or the coding parameters of the reconstructed neighbor block.

The encoding apparatus 100 and/or the decoding apparatus 200 may generate a prediction block by performing intra prediction on a target block based on information about samples in a target image. When intra prediction is performed, the encoding apparatus 100 and/or the decoding apparatus 200 may generate a prediction block for the target block by performing intra prediction based on information about samples in the target image. When intra prediction is performed, the encoding apparatus 100 and/or the decoding apparatus 200 may perform directional prediction and/or non-directional prediction based on at least one reconstructed reference sample.

A prediction block may be a block generated as a result of performing intra prediction. A prediction block may correspond to at least one of a CU, a PU, and a TU.

The unit of a prediction block may have a size corresponding to at least one of a CU, a PU, and a TU. The prediction block may have a square shape having a size of 2N×2N or N×N. The size of N×N may include sizes of 4×4, 8×8, 16×16, 32×32, 64×64, or the like.

Alternatively, a prediction block may have a square block having a size of 2×2, 4×4, 8×8, 16×16, 32×32, 64×64 or the like or a rectangular block having a size of 2×8, 4×8, 2×16, 4×16, 8×16, or the like.

Intra prediction may be performed in consideration of the intra-prediction mode for the target block. The number of intra-prediction modes that the target block can have may be a predefined fixed value, and may be a value determined differently depending on the attributes of a prediction block. For example, the attributes of the prediction block may include the size of the prediction block, the type of prediction block, etc. Further, the attribute of a prediction block may indicate a coding parameter for the prediction block.

For example, the number of intra-prediction modes may be fixed at N regardless of the size of a prediction block. Alternatively, the number of intra-prediction modes may be, for example, 3, 5, 9, 17, 34, 35, 36, 65, 67 or 95.

The intra-prediction modes may be non-directional modes or directional modes.

For example, the intra-prediction modes may include two non-directional modes and 65 directional modes corresponding to numbers 0 to 66 illustrated in FIG. 7.

For example, the intra-prediction modes may include two non-directional modes and 93 directional modes corresponding to numbers -14 to 80 illustrated in FIG. 7 in a case that a specific intra prediction method is used.

The two non-directional modes may include a DC mode and a planar mode.

A directional mode may be a prediction mode having a specific direction or a specific angle. The directional mode may also be referred to as an “angular mode”.

An intra-prediction mode may be represented by at least one of a mode number, a mode value, a mode angle, and a mode direction. In other words, the terms “(mode) number of the intra-prediction mode”, “(mode) value of the intra-prediction mode”, “(mode) angle of the intra-prediction mode”, and “(mode) direction of the intra-prediction mode” may be used to have the same meaning, and may be used interchangeably with each other.

The number of intra-prediction modes may be M. The value of M may be 1 or more. In other words, the number of intra-prediction modes may be M, which includes the number of non-directional modes and the number of directional modes.

The number of intra-prediction modes may be fixed to M regardless of the size and/or the color component of a block. For example, the number of intra-prediction modes may be fixed at any one of 35 and 67 regardless of the size of a block.

Alternatively, the number of intra-prediction modes may differ depending on the shape, the size and/or the type of the color component of a block.

For example, in FIG. 7, directional prediction modes illustrated as dashed lines may be applied only for a prediction for a non-square block.

For example, the larger the size of the block, the greater the number of intra-prediction modes. Alternatively, the larger the size of the block, the smaller the number of intra-prediction modes. When the size of the block is 4x4 or 8x8, the number of intra-prediction modes may be 67. When the size of the block is 16x16, the number of intra-prediction modes may be 35. When the size of the block is 32x32, the number of intra-prediction modes may be 19. When the size of a block is 64x64, the number of intra-prediction modes may be 7.

For example, the number of intra prediction modes may differ depending on whether a color component is a luma signal or a chroma signal. Alternatively, the number of intra-prediction modes corresponding to a luma component block may be greater than the number of intra-prediction modes corresponding to a chroma component block.

For example, in a vertical mode having a mode value of 50, prediction may be performed in a vertical direction based on the pixel value of a reference sample. For example, in a horizontal mode having a mode value of 18, prediction may be performed in a horizontal direction based on the pixel value of a reference sample.

Even in directional modes other than the above-described mode, the encoding apparatus 100 and the decoding apparatus 200 may perform intra prediction on a target unit using reference samples depending on angles corresponding to the directional modes.

Intra-prediction modes located on a right side with respect to the vertical mode may be referred to as ‘vertical-right modes’. Intra-prediction modes located below the horizontal mode may be referred to as ‘horizontal-below modes’. For example, in FIG. 7, the intra-prediction modes in which a mode value is one of 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, and 66 may be vertical-right modes. Intra-prediction modes in which a mode value is one of 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, and 17 may be horizontal-below modes.

The non-directional mode may include a DC mode and a planar mode. For example, a value of the DC mode may be 1. A value of the planar mode may be 0.

The directional mode may include an angular mode. Among the plurality of the intra prediction modes, remaining modes except for the DC mode and the planar mode may be directional modes.

When the intra-prediction mode is a DC mode, a prediction block may be generated based on the average of pixel values of a plurality of reference pixels. For example, a value of a pixel of a prediction block may be determined based on the average of pixel values of a plurality of reference pixels.

The number of above-described intra-prediction modes and the mode values of respective intra-prediction modes are merely exemplary. The number of above-described intra-prediction modes and the mode values of respective intra-prediction modes may be defined differently depending on the embodiments, implementation and/or requirements.

In order to perform intra prediction on a target block, the step of checking whether samples included in a reconstructed neighbor block can be used as reference samples of a target block may be performed. When a sample that cannot be used as a reference sample of the target block is present among samples in the neighbor block, a value generated via copying and/or interpolation that uses at least one sample value, among the samples included in the reconstructed neighbor block, may replace the sample value of the sample that cannot be used as the reference sample. When the value generated via copying and/or interpolation replaces the sample value of the existing sample, the sample may be used as the reference sample of the target block.

When intra prediction is used, a filter may be applied to at least one of a reference sample and a prediction sample based on at least one of the intra-prediction mode and the size of the target block.

The type of filter to be applied to at least one of a reference sample and a prediction sample may differ depending on at least one of the intra-prediction mode of a target block, the size of the target block, and the shape of the target block. The types of filters may be classified depending on one or more of the length of filter tap, the value of a filter coefficient, and filter strength. The length of filter tap may mean the number of filter taps. Also, the number of filter tap may mean the length of the filter.

When the intra-prediction mode is a planar mode, a sample value of a prediction target block may be generated using a weighted sum of an above reference sample of the target block, a left reference sample of the target block, an above-right reference sample of the target block, and a below-left reference sample of the target block depending on the location of the prediction target sample in the prediction block when the prediction block of the target block is generated.

When the intra-prediction mode is a DC mode, the average of reference samples above the target block and the reference samples to the left of the target block may be used when the prediction block of the target block is generated. Also, filtering using the values of reference samples may be performed on specific rows or specific columns in the target block. The specific rows may be one or more upper rows adjacent to the reference sample. The specific columns may be one or more left columns adjacent to the reference sample.

When the intra-prediction mode is a directional mode, a prediction block may be generated using the above reference samples, left reference samples, above-right reference sample and/or below-left reference sample of the target block.

In order to generate the above-described prediction sample, real-number-based interpolation may be performed.

The intra-prediction mode of the target block may be predicted from intra prediction mode of a neighbor block adjacent to the target block, and the information used for prediction may be entropy-encoded/decoded.

For example, when the intra-prediction modes of the target block and the neighbor block are identical to each other, it may be signaled, using a predefined flag, that the intra-prediction modes of the target block and the neighbor block are identical.

For example, an indicator for indicating an intra-prediction mode identical to that of the target block, among intra-prediction modes of multiple neighbor blocks, may be signaled.

When the intra-prediction modes of the target block and a neighbor block are different from each other, information about the intra-prediction mode of the target block may be encoded and/or decoded using entropy encoding and/or decoding.

FIG. 8 is a diagram illustrating reference samples used in an intra-prediction procedure.

Reconstructed reference samples used for intra prediction of the target block may include below-left reference samples, left reference samples, an above-left corner reference sample, above reference samples, and above-right reference samples.

For example, the left reference samples may mean reconstructed reference pixels adjacent to the left side of the target block. The above reference samples may mean reconstructed reference pixels adjacent to the top of the target block. The above-left corner reference sample may mean a reconstructed reference pixel located at the above-left corner of the target block. The below-left reference samples may mean reference samples located below a left sample line composed of the left reference samples, among samples located on the same line as the left sample line. The above-right reference samples may mean reference samples located to the right of an above sample line composed of the above reference samples, among samples located on the same line as the above sample line.

When the size of a target block is $N \times N$, the numbers of the below-left reference samples, the left reference samples, the above reference samples, and the above-right reference samples may each be N .

By performing intra prediction on the target block, a prediction block may be generated. The generation of the prediction block may include the determination of the values of pixels in the prediction block. The sizes of the target block and the prediction block may be equal.

The reference samples used for intra prediction of the target block may vary depending on the intra-prediction mode of the target block. The direction of the intra-prediction mode may represent a dependence relationship between the reference samples and the pixels of the prediction block.

For example, the value of a specified reference sample may be used as the values of one or more specified pixels in the prediction block. In this case, the specified reference sample and the one or more specified pixels in the prediction block may be the sample and pixels which are positioned in a straight line in the direction of an intra-prediction mode. In other words, the value of the specified reference sample may be copied as the value of a pixel located in a direction reverse to the direction of the intra-prediction mode. Alternatively, the value of a pixel in the prediction block may be the value of a reference sample located in the direction of the intra-prediction mode with respect to the location of the pixel.

In an example, when the intra-prediction mode of a target block is a vertical mode, the above reference samples may be used for intra prediction. When the intra-prediction mode is the vertical mode, the value of a pixel in the prediction block may be the value of a reference sample vertically located above the location of the pixel. Therefore, the above reference samples adjacent to the top of the target block may be used for intra prediction. Furthermore, the values of pixels in one row of the prediction block may be identical to those of the above reference samples.

In an example, when the intra-prediction mode of a target block is a horizontal mode, the left reference samples may be used for intra prediction. When the intra-prediction mode is the horizontal mode, the value of a pixel in the prediction block may be the value of a reference sample horizontally located left to the location of the pixel. Therefore, the left reference samples adjacent to the left of the target block may be used for intra prediction. Furthermore, the values of pixels in one column of the prediction block may be identical to those of the left reference samples.

In an example, when the mode value of the intra-prediction mode of the current block is 34, at least some of the left reference samples, the above-left corner reference sample, and at least some of the above reference samples may be used for intra prediction. When the mode value of the intra-prediction mode is 34, the value of a pixel in the prediction block may be the value of a reference sample diagonally located at the above-left corner of the pixel.

Further, At least a part of the above-right reference samples may be used for intra prediction in a case that an intra prediction mode of which a mode value is a value ranging from 52 to 66.

Further, At least a part of the below-left reference samples may be used for intra prediction in a case that an intra prediction mode of which a mode value is a value ranging from 2 to 17.

Further, the above-left corner reference sample may be used for intra prediction in a case that an intra prediction mode of which a mode value is a value ranging from 19 to 49.

The number of reference samples used to determine the pixel value of one pixel in the prediction block may be either 1, or 2 or more.

As described above, the pixel value of a pixel in the prediction block may be determined depending on the location of the pixel and the location of a reference sample indicated by the direction of the intra-prediction mode. When the location of the pixel and the location of the reference sample indicated by the direction of the intra-prediction mode are integer positions, the value of one reference sample indicated by an integer position may be used to determine the pixel value of the pixel in the prediction block.

When the location of the pixel and the location of the reference sample indicated by the direction of the intra-prediction mode are not integer positions, an interpolated reference sample based on two reference samples closest to the location of the reference sample may be generated. The value of the interpolated reference sample may be used to determine the pixel value of the pixel in the prediction block. In other words, when the location of the pixel in the prediction block and the location of the reference sample indicated by the direction of the intra-prediction mode indicate the location between two reference samples, an interpolated value based on the values of the two samples may be generated.

The prediction block generated via prediction may not be identical to an original target block. In other words, there may be a prediction error which is the difference between the target block and the prediction block, and there may also be a prediction error between the pixel of the target block and the pixel of the prediction block.

Hereinafter, the terms “difference”, “error”, and “residual” may be used to have the same meaning, and may be used interchangeably with each other.

For example, in the case of directional intra prediction, the longer the distance between the pixel of the prediction block and the reference sample, the greater the prediction error that may occur. Such a prediction error may result in discontinuity between the generated prediction block and neighbor blocks.

In order to reduce the prediction error, filtering for the prediction block may be used. Filtering may be configured to adaptively apply a filter to an area, regarded as having a large prediction error, in the prediction block. For example, the area regarded as having a large prediction error may be the boundary of the prediction block. Further, an area regarded as having a large prediction error in the prediction block may differ depending on the intra-prediction mode, and the characteristics of filters may also differ depending thereon.

As illustrated in FIG. 8, for intra prediction of a target block, at least one of reference line 0 to reference line 3 may be used.

Each reference line in FIG. 8 may indicate a reference sample line comprising one or more reference samples. As the number of the reference line is lower, a line of reference samples closer to a target block may be indicated.

Samples in segment A and segment F may be acquired through padding that uses samples closest to the target block in segment B and segment E instead of being acquired from reconstructed neighbor blocks.

Index information indicating a reference sample line to be used for intra-prediction of the target block may be signaled. The index information may indicate a reference sample line to be used for intra-prediction of the target block, among

multiple reference sample lines. For example, the index information may have a value corresponding to any one of 0 to 3.

When the top boundary of the target block is the boundary of a CTU, only reference sample line 0 may be available. Therefore, in this case, index information may not be signaled. When an additional reference sample line other than reference sample line 0 is used, filtering of a prediction block, which will be described later, may not be performed.

In the case of inter-color intra prediction, a prediction block for a target block of a second color component may be generated based on the corresponding reconstructed block of a first color component.

For example, the first color component may be a luma component, and the second color component may be a chroma component.

In order to perform inter-color intra prediction, parameters for a linear model between the first color component and the second color component may be derived based on a template.

The template may include reference samples above the target block (above reference samples) and/or reference samples to the left of the target block (left reference samples), and may include above reference samples and/or left reference samples of a reconstructed block of the first color component, which correspond to the reference samples.

For example, parameters for a linear model may be derived using 1) the value of the sample of a first color component having the maximum value, among the samples in the template, 2) the value of the sample of a second color component corresponding to the sample of the first color component, 3) the value of the sample of a first color component having the minimum value, among the samples in the template, and 4) the value of the sample of a second color component corresponding to the sample of the first color component.

When the parameters for the linear model are derived, a prediction block for the target block may be generated by applying the corresponding reconstructed block to the linear model.

Depending on the image format, sub-sampling may be performed on samples neighbor the reconstructed block of the first color component and the corresponding reconstructed block of the first color component. For example, when one sample of the second color component corresponds to four samples of the first color component, one corresponding sample may be calculated by performing sub-sampling on the four samples of the first color component. When sub-sampling is performed, derivation of the parameters for the linear model and inter-color intra prediction may be performed based on the sub-sampled corresponding sample.

Information about whether inter-color intra prediction is performed and/or the range of the template may be signaled in an intra-prediction mode.

The target block may be partitioned into two or four sub-blocks in a horizontal direction and/or a vertical direction.

The sub-blocks resulting from the partitioning may be sequentially reconstructed. That is, as intra-prediction is performed on each sub-block, a sub-prediction block for the sub-block may be generated. Also, as dequantization (inverse quantization) and/or an inverse transform are performed on each sub-block, a sub-residual block for the corresponding sub-block may be generated. A reconstructed sub-block may be generated by adding the sub-prediction

block to the sub-residual block. The reconstructed sub-block may be used as a reference sample for intra prediction of the sub-block having the next priority.

A sub-block may be a block including a specific number (e.g., 16) of samples or more. For example, when the target block is an 8×4 block or a 4×8 block, the target block may be partitioned into two sub-blocks. Also, when the target block is a 4×4 block, the target block cannot be partitioned into sub-blocks. When the target block has another size, the target block may be partitioned into four sub-blocks.

Information about whether intra prediction based on such sub-blocks is performed and/or information about a partition direction (horizontal direction or vertical direction) may be signaled.

Such sub-block-based intra prediction may be limited such that it is performed only when reference sample line 0 is used. When sub-block-based intra-prediction is performed, filtering of a prediction block, which will be described below, may not be performed.

A final prediction block may be generated by performing filtering on the prediction block generated via intra prediction.

Filtering may be performed by applying specific weights to a filtering target sample, which is the target to be filtered, a left reference sample, an above reference sample, and/or an above-left reference sample.

The weights and/or reference samples (e.g., the range of reference samples, the locations of the reference samples, etc.) used for filtering may be determined based on at least one of a block size, an intra-prediction mode, and the location of the filtering target sample in a prediction block.

For example, filtering may be performed only in a specific intra-prediction mode (e.g., DC mode, planar mode, vertical mode, horizontal mode, diagonal mode and/or adjacent diagonal mode).

The adjacent diagonal mode may be a mode having a number obtained by adding k to the number of the diagonal mode, and may be a mode having a number obtained by subtracting k from the number of the diagonal mode. In other words, the number of the adjacent diagonal mode may be the sum of the number of the diagonal mode and k , or may be the difference between the number of the diagonal mode and k . For example, k may be a positive integer of 8 or less.

The intra-prediction mode of the target block may be derived using the intra-prediction mode of a neighbor block present near the target block, and such a derived intra-prediction mode may be entropy-encoded and/or entropy-decoded.

For example, when the intra-prediction mode of the target block is identical to the intra-prediction mode of the neighbor block, information indicating that the intra-prediction mode of the target block is identical to the intra-prediction mode of the neighbor block may be signaled using specific flag information.

Further, for example, indicator information for a neighbor block having an intra-prediction mode identical to the intra-prediction mode of the target block, among intra-prediction modes of multiple neighbor blocks, may be signaled.

For example, when the intra-prediction mode of the target block is different from the intra-prediction mode of the neighbor block, entropy encoding and/or entropy decoding may be performed on information about the intra-prediction mode of the target block by performing entropy encoding and/or entropy decoding based on the intra-prediction mode of the neighbor block.

FIG. 9 is a diagram for explaining an embodiment of an inter prediction procedure.

The rectangles shown in FIG. 9 may represent images (or pictures). Further, in FIG. 9, arrows may represent prediction directions. An arrow pointing from a first picture to a second picture means that the second picture refers to the first picture. That is, each image may be encoded and/or decoded depending on the prediction direction.

Images may be classified into an Intra Picture (I picture), a Uni-prediction Picture or Predictive Coded Picture (P picture), and a Bi-prediction Picture or Bi-predictive Coded Picture (B picture) depending on the encoding type. Each picture may be encoded and/or decoded depending on the encoding type thereof.

When a target image that is the target to be encoded is an I picture, the target image may be encoded using data contained in the image itself without inter prediction that refers to other images. For example, an I picture may be encoded only via intra prediction.

When a target image is a P picture, the target image may be encoded via inter prediction, which uses reference pictures existing in one direction. Here, the one direction may be a forward direction or a backward direction.

When a target image is a B picture, the image may be encoded via inter prediction that uses reference pictures existing in two directions, or may be encoded via inter prediction that uses reference pictures existing in one of a forward direction and a backward direction. Here, the two directions may be the forward direction and the backward direction.

A P picture and a B picture that are encoded and/or decoded using reference pictures may be regarded as images in which inter prediction is used.

Below, inter prediction in an inter mode according to an embodiment will be described in detail.

Inter prediction or a motion compensation may be performed using a reference image and motion information.

In an inter mode, the encoding apparatus 100 may perform inter prediction and/or motion compensation on a target block. The decoding apparatus 200 may perform inter prediction and/or motion compensation, corresponding to inter prediction and/or motion compensation performed by the encoding apparatus 100, on a target block.

Motion information of the target block may be individually derived by the encoding apparatus 100 and the decoding apparatus 200 during the inter prediction. The motion information may be derived using motion information of a reconstructed neighbor block, motion information of a col block, and/or motion information of a block adjacent to the col block.

For example, the encoding apparatus 100 or the decoding apparatus 200 may perform prediction and/or motion compensation by using motion information of a spatial candidate and/or a temporal candidate as motion information of the target block. The target block may mean a PU and/or a PU partition.

A spatial candidate may be a reconstructed block which is spatially adjacent to the target block.

A temporal candidate may be a reconstructed block corresponding to the target block in a previously reconstructed co-located picture (col picture).

In inter prediction, the encoding apparatus 100 and the decoding apparatus 200 may improve encoding efficiency and decoding efficiency by utilizing the motion information of a spatial candidate and/or a temporal candidate. The motion information of a spatial candidate may be referred to

as 'spatial motion information'. The motion information of a temporal candidate may be referred to as 'temporal motion information'.

Below, the motion information of a spatial candidate may be the motion information of a PU including the spatial candidate. The motion information of a temporal candidate may be the motion information of a PU including the temporal candidate. The motion information of a candidate block may be the motion information of a PU including the candidate block.

Inter prediction may be performed using a reference picture.

The reference picture may be at least one of a picture previous to a target picture and a picture subsequent to the target picture. The reference picture may be an image used for the prediction of the target block.

In inter prediction, a region in the reference picture may be specified by utilizing a reference picture index (or refIdx) for indicating a reference picture, a motion vector, which will be described later, etc. Here, the region specified in the reference picture may indicate a reference block.

Inter prediction may select a reference picture, and may also select a reference block corresponding to the target block from the reference picture. Further, inter prediction may generate a prediction block for the target block using the selected reference block.

The motion information may be derived during inter prediction by each of the encoding apparatus **100** and the decoding apparatus **200**.

A spatial candidate may be a block 1) which is present in a target picture, 2) which has been previously reconstructed via encoding and/or decoding, and 3) which is adjacent to the target block or is located at the corner of the target block. Here, the "block located at the corner of the target block" may be either a block vertically adjacent to a neighbor block that is horizontally adjacent to the target block, or a block horizontally adjacent to a neighbor block that is vertically adjacent to the target block. Further, "block located at the corner of the target block" may have the same meaning as "block adjacent to the corner of the target block". The meaning of "block located at the corner of the target block" may be included in the meaning of "block adjacent to the target block".

For example, a spatial candidate may be a reconstructed block located to the left of the target block, a reconstructed block located above the target block, a reconstructed block located at the below-left corner of the target block, a reconstructed block located at the above-right corner of the target block, or a reconstructed block located at the above-left corner of the target block.

Each of the encoding apparatus **100** and the decoding apparatus **200** may identify a block present at the location spatially corresponding to the target block in a col picture. The location of the target block in the target picture and the location of the identified block in the col picture may correspond to each other.

Each of the encoding apparatus **100** and the decoding apparatus **200** may determine a col block present at the predefined relative location for the identified block to be a temporal candidate. The predefined relative location may be a location present inside and/or outside the identified block.

For example, the col block may include a first col block and a second col block. When the coordinates of the identified block are (xP, yP) and the size of the identified block is represented by (nPSW, nPSH), the first col block may be a block located at coordinates (xP+nPSW, yP+nPSH). The second col block may be a block located at

coordinates (xP+(nPSW>>1), yP+(nPSH>>1)). The second col block may be selectively used when the first col block is unavailable.

The motion vector of the target block may be determined based on the motion vector of the col block. Each of the encoding apparatus **100** and the decoding apparatus **200** may scale the motion vector of the col block. The scaled motion vector of the col block may be used as the motion vector of the target block. Further, a motion vector for the motion information of a temporal candidate stored in a list may be a scaled motion vector.

The ratio of the motion vector of the target block to the motion vector of the col block may be identical to the ratio of a first temporal distance to a second temporal distance. The first temporal distance may be the distance between the reference picture and the target picture of the target block. The second temporal distance may be the distance between the reference picture and the col picture of the col block.

The scheme for deriving motion information may change depending on the inter-prediction mode of a target block. For example, as inter-prediction modes applied for inter prediction, an Advanced Motion Vector Predictor (AMVP) mode, a merge mode, a skip mode, a merge mode with a motion vector difference, a sub block merge mode, a triangle partition mode, an inter-intra combined prediction mode, an affine inter mode, a current picture reference mode, etc. may be present. The merge mode may also be referred to as a "motion merge mode". Individual modes will be described in detail below.

1) AMVP Mode

When an AMVP mode is used, the encoding apparatus **100** may search a neighbor region of a target block for a similar block. The encoding apparatus **100** may acquire a prediction block by performing prediction on the target block using motion information of the found similar block. The encoding apparatus **100** may encode a residual block, which is the difference between the target block and the prediction block.

1-1) Creation of List of Prediction Motion Vector Candidates

When an AMVP mode is used as the prediction mode, each of the encoding apparatus **100** and the decoding apparatus **200** may create a list of prediction motion vector candidates using the motion vector of a spatial candidate, the motion vector of a temporal candidate, and a zero vector. The prediction motion vector candidate list may include one or more prediction motion vector candidates. At least one of the motion vector of a spatial candidate, the motion vector of a temporal candidate, and a zero vector may be determined and used as a prediction motion vector candidate.

Hereinafter, the terms "prediction motion vector (candidate)" and "motion vector (candidate)" may be used to have the same meaning, and may be used interchangeably with each other.

Hereinafter, the terms "prediction motion vector candidate" and "AMVP candidate" may be used to have the same meaning, and may be used interchangeably with each other.

Hereinafter, the terms "prediction motion vector candidate list" and "AMVP candidate list" may be used to have the same meaning, and may be used interchangeably with each other.

Spatial candidates may include a reconstructed spatial neighbor block. In other words, the motion vector of the reconstructed neighbor block may be referred to as a "spatial prediction motion vector candidate".

Temporal candidates may include a col block and a block adjacent to the col block. In other words, the motion vector

of the col block or the motion vector of the block adjacent to the col block may be referred to as a “temporal prediction motion vector candidate”.

The zero vector may be a (0, 0) motion vector.

The prediction motion vector candidates may be motion vector predictors for predicting a motion vector. Also, in the encoding apparatus 100, each prediction motion vector candidate may be an initial search location for a motion vector.

1-2) Search for Motion Vectors that Use List of Prediction Motion Vector Candidates

The encoding apparatus 100 may determine the motion vector to be used to encode a target block within a search range using a list of prediction motion vector candidates. Further, the encoding apparatus 100 may determine a prediction motion vector candidate to be used as the prediction motion vector of the target block, among prediction motion vector candidates present in the prediction motion vector candidate list.

The motion vector to be used to encode the target block may be a motion vector that can be encoded at minimum cost.

Further, the encoding apparatus 100 may determine whether to use the AMVP mode to encode the target block.

1-3) Transmission of Inter-Prediction Information

The encoding apparatus 100 may generate a bitstream including inter-prediction information required for inter prediction. The decoding apparatus 200 may perform inter prediction on the target block using the inter-prediction information of the bitstream.

The inter-prediction information may contain 1) mode information indicating whether an AMVP mode is used, 2) a prediction motion vector index, 3) a Motion Vector Difference (MVD), 4) a reference direction, and 5) a reference picture index.

Hereinafter, the terms “prediction motion vector index” and “AMVP index” may be used to have the same meaning, and may be used interchangeably with each other.

Further, the inter-prediction information may contain a residual signal.

The decoding apparatus 200 may acquire a prediction motion vector index, an MVD, a reference direction, and a reference picture index from the bitstream through entropy decoding when mode information indicates that the AMVP mode is used.

The prediction motion vector index may indicate a prediction motion vector candidate to be used for the prediction of a target block, among prediction motion vector candidates included in the prediction motion vector candidate list.

1-4) Inter Prediction in AMVP Mode that Uses Inter-Prediction Information

The decoding apparatus 200 may derive prediction motion vector candidates using a prediction motion vector candidate list, and may determine the motion information of a target block based on the derived prediction motion vector candidates.

The decoding apparatus 200 may determine a motion vector candidate for the target block, among the prediction motion vector candidates included in the prediction motion vector candidate list, using a prediction motion vector index. The decoding apparatus 200 may select a prediction motion vector candidate, indicated by the prediction motion vector index, from among prediction motion vector candidates included in the prediction motion vector candidate list, as the prediction motion vector of the target block.

The encoding apparatus 100 may generate an entropy-encoded prediction motion vector index by applying entropy

encoding to a prediction motion vector index, and may generate a bitstream including the entropy-encoded prediction motion vector index. The entropy-encoded prediction motion vector index may be signaled from the encoding apparatus 100 to the decoding apparatus 200 through a bitstream. The decoding apparatus 200 may extract the entropy-encoded prediction motion vector index from the bitstream, and may acquire the prediction motion vector index by applying entropy decoding to the entropy-encoded prediction motion vector index.

The motion vector to be actually used for inter prediction of the target block may not match the prediction motion vector. In order to indicate the difference between the motion vector to be actually used for inter prediction of the target block and the prediction motion vector, an MVD may be used. The encoding apparatus 100 may derive a prediction motion vector similar to the motion vector to be actually used for inter prediction of the target block so as to use an MVD that is as small as possible.

A Motion Vector Difference (MVD) may be the difference between the motion vector of the target block and the prediction motion vector. The encoding apparatus 100 may calculate the MVD, and may generate an entropy-encoded MVD by applying entropy encoding to the MVD. The encoding apparatus 100 may generate a bitstream including the entropy-encoded MVD.

The MVD may be transmitted from the encoding apparatus 100 to the decoding apparatus 200 through the bitstream. The decoding apparatus 200 may extract the entropy-encoded MVD from the bitstream, and may acquire the MVD by applying entropy decoding to the entropy-encoded MVD.

The decoding apparatus 200 may derive the motion vector of the target block by summing the MVD and the prediction motion vector. In other words, the motion vector of the target block derived by the decoding apparatus 200 may be the sum of the MVD and the motion vector candidate.

Also, the encoding apparatus 100 may generate entropy-encoded MVD resolution information by applying entropy encoding to calculated MVD resolution information, and may generate a bitstream including the entropy-encoded MVD resolution information. The decoding apparatus 200 may extract the entropy-encoded MVD resolution information from the bitstream, and may acquire MVD resolution information by applying entropy decoding to the entropy-encoded MVD resolution information. The decoding apparatus 200 may adjust the resolution of the MVD using the MVD resolution information.

Meanwhile, the encoding apparatus 100 may calculate an MVD based on an affine model. The decoding apparatus 200 may derive the affine control motion vector of the target block through the sum of the MVD and an affine control motion vector candidate, and may derive the motion vector of a sub-block using the affine control motion vector.

The reference direction may indicate a list of reference pictures to be used for prediction of the target block. For example, the reference direction may indicate one of a reference picture list L0 and a reference picture list L1.

The reference direction merely indicates the reference picture list to be used for prediction of the target block, and may not mean that the directions of reference pictures are limited to a forward direction or a backward direction. In other words, each of the reference picture list L0 and the reference picture list L1 may include pictures in a forward direction and/or a backward direction.

That the reference direction is unidirectional may mean that a single reference picture list is used. That the reference

direction is bidirectional may mean that two reference picture lists are used. In other words, the reference direction may indicate one of the case where only the reference picture list L0 is used, the case where only the reference picture list L1 is used, and the case where two reference picture lists are used.

The reference picture index may indicate a reference picture that is used for prediction of the target block, among reference pictures present in a reference picture list. The encoding apparatus 100 may generate an entropy-encoded reference picture index by applying entropy encoding to the reference picture index, and may generate a bitstream including the entropy-encoded reference picture index. The entropy-encoded reference picture index may be signaled from the encoding apparatus 100 to the decoding apparatus 200 through the bitstream. The decoding apparatus 200 may extract the entropy-encoded reference picture index from the bitstream, and may acquire the reference picture index by applying entropy decoding to the entropy-encoded reference picture index.

When two reference picture lists are used to predict the target block, a single reference picture index and a single motion vector may be used for each of the reference picture lists. Further, when two reference picture lists are used to predict the target block, two prediction blocks may be specified for the target block. For example, the (final) prediction block of the target block may be generated using the average or weighted sum of the two prediction blocks for the target block.

The motion vector of the target block may be derived by the prediction motion vector index, the MVD, the reference direction, and the reference picture index.

The decoding apparatus 200 may generate a prediction block for the target block based on the derived motion vector and the reference picture index. For example, the prediction block may be a reference block, indicated by the derived motion vector, in the reference picture indicated by the reference picture index.

Since the prediction motion vector index and the MVD are encoded without the motion vector itself of the target block being encoded, the number of bits transmitted from the encoding apparatus 100 to the decoding apparatus 200 may be decreased, and encoding efficiency may be improved.

For the target block, the motion information of reconstructed neighbor blocks may be used. In a specific inter-prediction mode, the encoding apparatus 100 may not separately encode the actual motion information of the target block. The motion information of the target block is not encoded, and additional information that enables the motion information of the target block to be derived using the motion information of reconstructed neighbor blocks may be encoded instead. As the additional information is encoded, the number of bits transmitted to the decoding apparatus 200 may be decreased, and encoding efficiency may be improved.

For example, as inter-prediction modes in which the motion information of the target block is not directly encoded, there may be a skip mode and/or a merge mode. Here, each of the encoding apparatus 100 and the decoding apparatus 200 may use an identifier and/or an index that indicates a unit, the motion information of which is to be used as the motion information of the target unit, among reconstructed neighbor units.

2) Merge Mode

As a scheme for deriving the motion information of a target block, there is merging. The term "merging" may

mean the merging of the motion of multiple blocks. "Merging" may mean that the motion information of one block is also applied to other blocks. In other words, a merge mode may be a mode in which the motion information of the target block is derived from the motion information of a neighbor block.

When a merge mode is used, the encoding apparatus 100 may predict the motion information of a target block using the motion information of a spatial candidate and/or the motion information of a temporal candidate. The spatial candidate may include a reconstructed spatial neighbor block that is spatially adjacent to the target block. The spatial neighbor block may include a left neighbor block and an above neighbor block. The temporal candidate may include a col block. The terms "spatial candidate" and "spatial merge candidate" may be used to have the same meaning, and may be used interchangeably with each other. The terms "temporal candidate" and "temporal merge candidate" may be used to have the same meaning, and may be used interchangeably with each other.

The encoding apparatus 100 may acquire a prediction block via prediction. The encoding apparatus 100 may encode a residual block, which is the difference between the target block and the prediction block.

2-1) Creation of Merge Candidate List

When the merge mode is used, each of the encoding apparatus 100 and the decoding apparatus 200 may create a merge candidate list using the motion information of a spatial candidate and/or the motion information of a temporal candidate. The motion information may include 1) a motion vector, 2) a reference picture index, and 3) a reference direction. The reference direction may be unidirectional or bidirectional. The reference direction may mean an inter prediction indicator.

The merge candidate list may include merge candidates. The merge candidates may be motion information. In other words, the merge candidate list may be a list in which pieces of motion information are stored.

The merge candidates may be pieces of motion information of temporal candidates and/or spatial candidates. In other words, the merge candidates list may comprise motion information of a temporal candidates and/or spatial candidates, etc.

Further, the merge candidate list may include new merge candidates generated by a combination of merge candidates that are already present in the merge candidate list. In other words, the merge candidate list may include new motion information generated by a combination of pieces of motion information previously present in the merge candidate list.

Also, a merge candidate list may include history-based merge candidates. The history-based merge candidates may be the motion information of a block which is encoded and/or decoded prior to a target block.

Also, a merge candidate list may include a merge candidate based on an average of two merge candidates.

The merge candidates may be specific modes deriving inter prediction information. The merge candidate may be information indicating a specific mode deriving inter prediction information. Inter prediction information of a target block may be derived according to a specific mode which the merge candidate indicates. Furthermore, the specific mode may include a process of deriving a series of inter prediction information. This specific mode may be an inter prediction information derivation mode or a motion information derivation mode.

The inter prediction information of the target block may be derived according to the mode indicated by the merge

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candidate selected by the merge index among the merge candidates in the merge candidate list.

For example, the motion information derivation modes in the merge candidate list may be at least one of 1) motion information derivation mode for a sub-block unit and 2) an affine motion information derivation mode.

Furthermore, the merge candidate list may include motion information of a zero vector. The zero vector may also be referred to as a “zero-merge candidate”.

In other words, pieces of motion information in the merge candidate list may be at least one of 1) motion information of a spatial candidate, 2) motion information of a temporal candidate, 3) motion information generated by a combination of pieces of motion information previously present in the merge candidate list, and 4) a zero vector.

Motion information may include 1) a motion vector, 2) a reference picture index, and 3) a reference direction. The reference direction may also be referred to as an “inter-prediction indicator”. The reference direction may be unidirectional or bidirectional. The unidirectional reference direction may indicate L0 prediction or L1 prediction.

The merge candidate list may be created before prediction in the merge mode is performed.

The number of merge candidates in the merge candidate list may be predefined. Each of the encoding apparatus 100 and the decoding apparatus 200 may add merge candidates to the merge candidate list depending on the predefined scheme and predefined priorities so that the merge candidate list has a predefined number of merge candidates. The merge candidate list of the encoding apparatus 100 and the merge candidate list of the decoding apparatus 200 may be made identical to each other using the predefined scheme and the predefined priorities.

Merging may be applied on a CU basis or a PU basis. When merging is performed on a CU basis or a PU basis, the encoding apparatus 100 may transmit a bitstream including predefined information to the decoding apparatus 200. For example, the predefined information may contain 1) information indicating whether to perform merging for individual block partitions, and 2) information about a block with which merging is to be performed, among blocks that are spatial candidates and/or temporal candidates for the target block.

2-2) Search for Motion Vector that Uses Merge Candidate List

The encoding apparatus 100 may determine merge candidates to be used to encode a target block. For example, the encoding apparatus 100 may perform prediction on the target block using merge candidates in the merge candidate list, and may generate residual blocks for the merge candidates. The encoding apparatus 100 may use a merge candidate that incurs the minimum cost in prediction and in the encoding of residual blocks to encode the target block.

Further, the encoding apparatus 100 may determine whether to use a merge mode to encode the target block.

2-3) Transmission of Inter-Prediction Information

The encoding apparatus 100 may generate a bitstream that includes inter-prediction information required for inter prediction. The encoding apparatus 100 may generate entropy-encoded inter-prediction information by performing entropy encoding on inter-prediction information, and may transmit a bitstream including the entropy-encoded inter-prediction information to the decoding apparatus 200. Through the bitstream, the entropy-encoded inter-prediction information may be signaled to the decoding apparatus 200 by the encoding apparatus 100. The decoding apparatus 200 may extract entropy-encoded inter-prediction information from

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the bitstream, and may acquire inter-prediction information by applying entropy decoding to the entropy-encoded inter-prediction information.

The decoding apparatus 200 may perform inter prediction on the target block using the inter-prediction information of the bitstream.

The inter-prediction information may contain 1) mode information indicating whether a merge mode is used, 2) a merge index and 3) correction information.

Further, the inter-prediction information may contain a residual signal.

The decoding apparatus 200 may acquire the merge index from the bitstream only when the mode information indicates that the merge mode is used.

The mode information may be a merge flag. The unit of the mode information may be a block. Information about the block may include mode information, and the mode information may indicate whether a merge mode is applied to the block.

The merge index may indicate a merge candidate to be used for the prediction of the target block, among merge candidates included in the merge candidate list. Alternatively, the merge index may indicate a block with which the target block is to be merged, among neighbor blocks spatially or temporally adjacent to the target block.

The encoding apparatus 100 may select a merge candidate having the highest encoding performance among the merge candidates included in the merge candidate list and set a value of the merge index to indicate the selected merge candidate.

Correction information may be information used to correct a motion vector. The encoding apparatus 100 may generate correction information. The decoding apparatus 200 may correct the motion vector of a merge candidate selected by a merge index based on the correction information.

The correction information may include at least one of information indicating whether correction is to be performed, correction direction information, and correction size information. A prediction mode in which the motion vector is corrected based on the signaled correction information may be referred to as a “merge mode having a motion vector difference”.

2-4) Inter Prediction of Merge Mode that Uses Inter-Prediction Information

The decoding apparatus 200 may perform prediction on the target block using the merge candidate indicated by the merge index, among merge candidates included in the merge candidate list.

The motion vector of the target block may be specified by the motion vector, reference picture index, and reference direction of the merge candidate indicated by the merge index.

3) Skip Mode

A skip mode may be a mode in which the motion information of a spatial candidate or the motion information of a temporal candidate is applied to the target block without change. Also, the skip mode may be a mode in which a residual signal is not used. In other words, when the skip mode is used, a reconstructed block may be the same as a prediction block.

The difference between the merge mode and the skip mode lies in whether or not a residual signal is transmitted or used. That is, the skip mode may be similar to the merge mode except that a residual signal is not transmitted or used.

When the skip mode is used, the encoding apparatus 100 may transmit information about a block, the motion infor-

mation of which is to be used as the motion information of the target block, among blocks that are spatial candidates or temporal candidates, to the decoding apparatus 200 through a bitstream. The encoding apparatus 100 may generate entropy-encoded information by performing entropy encoding on the information, and may signal the entropy-encoded information to the decoding apparatus 200 through a bitstream. The decoding apparatus 200 may extract entropy-encoded information from the bitstream, and may acquire information by applying entropy decoding to the entropy-encoded information.

Further, when the skip mode is used, the encoding apparatus 100 may not transmit other syntax information, such as an MVD, to the decoding apparatus 200. For example, when the skip mode is used, the encoding apparatus 100 may not signal a syntax element related to at least one of an MVD, a coded block flag, and a transform coefficient level to the decoding apparatus 200.

3-1) Creation of Merge Candidate List

The skip mode may also use a merge candidate list. In other words, a merge candidate list may be used both in the merge mode and in the skip mode. In this aspect, the merge candidate list may also be referred to as a “skip candidate list” or a “merge/skip candidate list”.

Alternatively, the skip mode may use an additional candidate list different from that of the merge mode. In this case, in the following description, a merge candidate list and a merge candidate may be replaced with a skip candidate list and a skip candidate, respectively.

The merge candidate list may be created before prediction in the skip mode is performed.

3-2) Search for Motion Vector that Uses Merge Candidate List

The encoding apparatus 100 may determine the merge candidates to be used to encode a target block. For example, the encoding apparatus 100 may perform prediction on the target block using the merge candidates in a merge candidate list. The encoding apparatus 100 may use a merge candidate that incurs the minimum cost in prediction to encode the target block.

Further, the encoding apparatus 100 may determine whether to use a skip mode to encode the target block.

3-3) Transmission of Inter-Prediction Information

The encoding apparatus 100 may generate a bitstream that includes inter-prediction information required for inter prediction. The decoding apparatus 200 may perform inter prediction on the target block using the inter-prediction information of the bitstream.

The inter-prediction information may include 1) mode information indicating whether a skip mode is used, and 2) a skip index.

The skip index may be identical to the above-described merge index.

When the skip mode is used, the target block may be encoded without using a residual signal. The inter-prediction information may not contain a residual signal. Alternatively, the bitstream may not include a residual signal.

The decoding apparatus 200 may acquire a skip index from the bitstream only when the mode information indicates that the skip mode is used. As described above, a merge index and a skip index may be identical to each other. The decoding apparatus 200 may acquire the skip index from the bitstream only when the mode information indicates that the merge mode or the skip mode is used.

The skip index may indicate the merge candidate to be used for the prediction of the target block, among the merge candidates included in the merge candidate list.

3-4) Inter Prediction in Skip Mode that Uses Inter-Prediction Information

The decoding apparatus 200 may perform prediction on the target block using a merge candidate indicated by a skip index, among the merge candidates included in a merge candidate list.

The motion vector of the target block may be specified by the motion vector, reference picture index, and reference direction of the merge candidate indicated by the skip index.

4) Current Picture Reference Mode

The current picture reference mode may denote a prediction mode that uses a previously reconstructed region in a target picture to which a target block belongs.

A motion vector for specifying the previously reconstructed region may be used. Whether the target block has been encoded in the current picture reference mode may be determined using the reference picture index of the target block.

A flag or index indicating whether the target block is a block encoded in the current picture reference mode may be signaled by the encoding apparatus 100 to the decoding apparatus 200. Alternatively, whether the target block is a block encoded in the current picture reference mode may be inferred through the reference picture index of the target block.

When the target block is encoded in the current picture reference mode, the target picture may exist at a fixed location or an arbitrary location in a reference picture list for the target block.

For example, the fixed location may be either a location where a value of the reference picture index is 0 or the last location.

When the target picture exists at an arbitrary location in the reference picture list, an additional reference picture index indicating such an arbitrary location may be signaled by the encoding apparatus 100 to the decoding apparatus 200.

5) Sub-Block Merge Mode

A sub-block merge mode may be a mode in which motion information is derived from the sub-block of a CU.

When the sub-block merge mode is applied, a sub-block merge candidate list may be generated using the motion information of a co-located sub-block (col-sub-block) of a target sub-block (i.e., a sub-block-based temporal merge candidate) in a reference image and/or an affine control point motion vector merge candidate.

6) Triangle Partition Mode

In a triangle partition mode, a target block may be partitioned in a diagonal direction, and sub-target blocks resulting from partitioning may be generated. For each sub-target block, motion information of the corresponding sub-target block may be derived, and a prediction sample for each sub-target block may be derived using the derived motion information. A prediction sample for the target block may be derived through a weighted sum of the prediction samples for the sub-target blocks resulting from the partitioning.

7) Combination Inter-Intra Prediction Mode

The combination inter-intra prediction mode may be a mode in which a prediction sample for a target block is derived using a weighted sum of a prediction sample generated via inter-prediction and a prediction sample generated via intra-prediction.

In the above-described modes, the decoding apparatus 200 may autonomously correct derived motion information. For example, the decoding apparatus 200 may search a specific area for motion information having the minimum

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sum of Absolute Differences (SAD) based on a reference block indicated by the derived motion information, and may derive the found motion information as corrected motion information.

In the above-described modes, the decoding apparatus 200 may compensate for the prediction sample derived via inter prediction using an optical flow.

In the above-described AMVP mode, merge mode, skip mode, etc., motion information to be used for prediction of the target block may be specified among pieces of motion information in a list using the index information of the list.

In order to improve encoding efficiency, the encoding apparatus 100 may signal only the index of an element that incurs the minimum cost in inter prediction of the target block, among elements in the list. The encoding apparatus 100 may encode the index, and may signal the encoded index.

Therefore, the above-described lists (i.e. the prediction motion vector candidate list and the merge candidate list) must be able to be derived by the encoding apparatus 100 and the decoding apparatus 200 using the same scheme based on the same data. Here, the same data may include a reconstructed picture and a reconstructed block. Further, in order to specify an element using an index, the order of the elements in the list must be fixed.

FIG. 10 illustrates spatial candidates according to an embodiment.

In FIG. 10, the locations of spatial candidates are illustrated.

The large block in the center of the drawing may denote a target block. Five small blocks may denote spatial candidates.

The coordinates of the target block may be (xP, yP), and the size of the target block may be represented by (nPSW, nPSH).

Spatial candidate A₀ may be a block adjacent to the below-left corner of the target block. A₀ may be a block that occupies pixels located at coordinates (xP-1, yP+nPSH+1).

Spatial candidate A₁ may be a block adjacent to the left of the target block. A₁ may be a lowermost block, among blocks adjacent to the left of the target block. Alternatively, A₁ may be a block adjacent to the top of A₀. A₁ may be a block that occupies pixels located at coordinates (xP-1, yP+nPSH).

Spatial candidate B₀ may be a block adjacent to the above-right corner of the target block. B₀ may be a block that occupies pixels located at coordinates (xP+nPSW+1, yP-1).

Spatial candidate B₁ may be a block adjacent to the top of the target block. B₁ may be a rightmost block, among blocks adjacent to the top of the target block. Alternatively, B₁ may be a block adjacent to the left of B₀. B₁ may be a block that occupies pixels located at coordinates (xP+nPSW, yP-1).

Spatial candidate B₂ may be a block adjacent to the above-left corner of the target block. B₂ may be a block that occupies pixels located at coordinates (xP-1, yP-1).

Determination of Availability of Spatial Candidate and Temporal Candidate

In order to include the motion information of a spatial candidate or the motion information of a temporal candidate in a list, it must be determined whether the motion information of the spatial candidate or the motion information of the temporal candidate is available.

Hereinafter, a candidate block may include a spatial candidate and a temporal candidate.

For example, the determination may be performed by sequentially applying the following steps 1) to 4).

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Step 1) When a PU including a candidate block is out of the boundary of a picture, the availability of the candidate block may be set to "false". The expression "availability is set to false" may have the same meaning as "set to be unavailable".

Step 2) When a PU including a candidate block is out of the boundary of a slice, the availability of the candidate block may be set to "false". When the target block and the candidate block are located in different slices, the availability of the candidate block may be set to "false".

Step 3) When a PU including a candidate block is out of the boundary of a tile, the availability of the candidate block may be set to "false". When the target block and the candidate block are located in different tiles, the availability of the candidate block may be set to "false".

Step 4) When the prediction mode of a PU including a candidate block is an intra-prediction mode, the availability of the candidate block may be set to "false". When a PU including a candidate block does not use inter prediction, the availability of the candidate block may be set to "false".

FIG. 11 illustrates the order of addition of motion information of spatial candidates to a merge list according to an embodiment.

As shown in FIG. 11, when pieces of motion information of spatial candidates are added to a merge list, the order of A₁, B₁, B₀, A₀, and B₂ may be used. That is, pieces of motion information of available spatial candidates may be added to the merge list in the order of A₁, B₁, B₀, A₀, and B₂.

Method for Deriving Merge List in Merge Mode and Skip Mode

As described above, the maximum number of merge candidates in the merge list may be set. The set maximum number is indicated by "N". The set number may be transmitted from the encoding apparatus 100 to the decoding apparatus 200. The slice header of a slice may include N. In other words, the maximum number of merge candidates in the merge list for the target block of the slice may be set by the slice header. For example, the value of N may be basically 5.

Pieces of motion information (i.e., merge candidates) may be added to the merge list in the order of the following steps 1) to 4).

Step 1) Among spatial candidates, available spatial candidates may be added to the merge list. Pieces of motion information of the available spatial candidates may be added to the merge list in the order illustrated in FIG. 10. Here, when the motion information of an available spatial candidate overlaps other motion information already present in the merge list, the motion information may not be added to the merge list. The operation of checking whether the corresponding motion information overlaps other motion information present in the list may be referred to in brief as an "overlap check".

The maximum number of pieces of motion information that are added may be N.

Step 2) When the number of pieces of motion information in the merge list is less than N and a temporal candidate is available, the motion information of the temporal candidate may be added to the merge list. Here, when the motion information of the available temporal candidate overlaps other motion information already present in the merge list, the motion information may not be added to the merge list.

Step 3) When the number of pieces of motion information in the merge list is less than N and the type of a target slice is "B", combined motion information generated by combined bidirectional prediction (bi-prediction) may be added to the merge list.

The target slice may be a slice including a target block.

The combined motion information may be a combination of L0 motion information and L1 motion information. L0 motion information may be motion information that refers only to a reference picture list L0. L1 motion information may be motion information that refers only to a reference picture list L1.

In the merge list, one or more pieces of L0 motion information may be present. Further, in the merge list, one or more pieces of L1 motion information may be present.

The combined motion information may include one or more pieces of combined motion information. When the combined motion information is generated, L0 motion information and L1 motion information, which are to be used for generation, among the one or more pieces of L0 motion information and the one or more pieces of L1 motion information, may be predefined. One or more pieces of combined motion information may be generated in a predefined order via combined bidirectional prediction, which uses a pair of different pieces of motion information in the merge list. One of the pair of different pieces of motion information may be L0 motion information and the other of the pair may be L1 motion information.

For example, combined motion information that is added with the highest priority may be a combination of L0 motion information having a merge index of 0 and L1 motion information having a merge index of 1. When motion information having a merge index of 0 is not L0 motion information or when motion information having a merge index of 1 is not L1 motion information, the combined motion information may be neither generated nor added. Next, the combined motion information that is added with the next priority may be a combination of L0 motion information, having a merge index of 1, and L1 motion information, having a merge index of 0. Subsequent detailed combinations may conform to other combinations of video encoding/decoding fields.

Here, when the combined motion information overlaps other motion information already present in the merge list, the combined motion information may not be added to the merge list.

Step 4) When the number of pieces of motion information in the merge list is less than N, motion information of a zero vector may be added to the merge list.

The zero-vector motion information may be motion information for which the motion vector is a zero vector.

The number of pieces of zero-vector motion information may be one or more. The reference picture indices of one or more pieces of zero-vector motion information may be different from each other. For example, the value of the reference picture index of first zero-vector motion information may be 0. The value of the reference picture index of second zero-vector motion information may be 1.

The number of pieces of zero-vector motion information may be identical to the number of reference pictures in the reference picture list.

The reference direction of zero-vector motion information may be bidirectional. Both of the motion vectors may be zero vectors. The number of pieces of zero-vector motion information may be the smaller one of the number of reference pictures in the reference picture list L0 and the number of reference pictures in the reference picture list L1. Alternatively, when the number of reference pictures in the reference picture list L0 and the number of reference pictures in the reference picture list L1 are different from each

other, a reference direction that is unidirectional may be used for a reference picture index that may be applied only to a single reference picture list.

The encoding apparatus 100 and/or the decoding apparatus 200 may sequentially add the zero-vector motion information to the merge list while changing the reference picture index.

When zero-vector motion information overlaps other motion information already present in the merge list, the zero-vector motion information may not be added to the merge list.

The order of the above-described steps 1) to 4) is merely exemplary, and may be changed. Further, some of the above steps may be omitted depending on predefined conditions. Method for Deriving Prediction Motion Vector Candidate List in AMVP Mode

The maximum number of prediction motion vector candidates in a prediction motion vector candidate list may be predefined. The predefined maximum number is indicated by N. For example, the predefined maximum number may be 2.

Pieces of motion information (i.e. prediction motion vector candidates) may be added to the prediction motion vector candidate list in the order of the following steps 1) to 3).

Step 1) Available spatial candidates, among spatial candidates, may be added to the prediction motion vector candidate list. The spatial candidates may include a first spatial candidate and a second spatial candidate.

The first spatial candidate may be one of A_0 , A_1 , scaled A_0 , and scaled A_1 . The second spatial candidate may be one of B_0 , B_1 , B_2 , scaled B_0 , scaled B_1 , and scaled B_2 .

Pieces of motion information of available spatial candidates may be added to the prediction motion vector candidate list in the order of the first spatial candidate and the second spatial candidate. In this case, when the motion information of an available spatial candidate overlaps other motion information already present in the prediction motion vector candidate list, the motion information may not be added to the prediction motion vector candidate list. In other words, when the value of N is 2, if the motion information of a second spatial candidate is identical to the motion information of a first spatial candidate, the motion information of the second spatial candidate may not be added to the prediction motion vector candidate list.

The maximum number of pieces of motion information that are added may be N.

Step 2) When the number of pieces of motion information in the prediction motion vector candidate list is less than N and a temporal candidate is available, the motion information of the temporal candidate may be added to the prediction motion vector candidate list. In this case, when the motion information of the available temporal candidate overlaps other motion information already present in the prediction motion vector candidate list, the motion information may not be added to the prediction motion vector candidate list.

Step 3) When the number of pieces of motion information in the prediction motion vector candidate list is less than N, zero-vector motion information may be added to the prediction motion vector candidate list.

The zero-vector motion information may include one or more pieces of zero-vector motion information. The reference picture indices of the one or more pieces of zero-vector motion information may be different from each other.

The encoding apparatus 100 and/or the decoding apparatus 200 may sequentially add pieces of zero-vector motion

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information to the prediction motion vector candidate list while changing the reference picture index.

When zero-vector motion information overlaps other motion information already present in the prediction motion vector candidate list, the zero-vector motion information may not be added to the prediction motion vector candidate list.

The description of the zero-vector motion information, made above in connection with the merge list, may also be applied to zero-vector motion information. A repeated description thereof will be omitted.

The order of the above-described steps 1) to 3) is merely exemplary, and may be changed. Further, some of the steps may be omitted depending on predefined conditions.

FIG. 12 illustrates a transform and quantization process according to an example.

As illustrated in FIG. 12, quantized levels may be generated by performing a transform and/or quantization process on a residual signal.

A residual signal may be generated as the difference between an original block and a prediction block. Here, the prediction block may be a block generated via intra prediction or inter prediction.

The residual signal may be transformed into a signal in a frequency domain through a transform procedure that is a part of a quantization procedure.

A transform kernel used for a transform may include various DCT kernels, such as Discrete Cosine Transform (DCT) type 2 (DCT-II) and Discrete Sine Transform (DST) kernels.

These transform kernels may perform a separable transform or a two-dimensional (2D) non-separable transform on the residual signal. The separable transform may be a transform indicating that a one-dimensional (1D) transform is performed on the residual signal in each of a horizontal direction and a vertical direction.

The DCT type and the DST type, which are adaptively used for a 1D transform, may include DCT-V, DCT-VIII, DST-I, and DST-VII in addition to DCT-II, as shown in each of the following Table 3 and the following table 4.

TABLE 3

Transform set	Transform candidates
0	DST-VII, DCT-VIII
1	DST-VII, DST-I
2	DST-VII, DCT-V

TABLE 4

Transform set	Transform candidates
0	DST-VII, DCT-VIII, DST-I
1	DST-VII, DST-I, DCT-VIII
2	DST-VII, DCT-V, DST-I

As shown in Table 3 and Table 4, when a DCT type or a DST type to be used for a transform is derived, transform sets may be used. Each transform set may include multiple transform candidates. Each transform candidate may be a DCT type or a DST type.

The following Table 5 shows examples of a transform set to be applied to a horizontal direction and a transform set to be applied to a vertical direction depending on intra-prediction modes.

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TABLE 5

	Intra-prediction mode									
	0	1	2	3	4	5	6	7	8	9
Vertical transform set	2	1	0	1	0	1	0	1	0	1
Horizontal transform set	2	1	0	1	0	1	0	1	0	1
	Intra-prediction mode									
	10	11	12	13	14	15	16	17	18	19
Vertical transform set	0	1	0	1	0	0	0	0	0	0
Horizontal transform set	0	1	0	1	2	2	2	2	2	2
	Intra-prediction mode									
	20	21	22	23	24	25	26	27	28	29
Vertical transform set	0	0	0	1	0	1	0	1	0	1
Horizontal transform set	2	2	2	1	0	1	0	1	0	1
	Intra-prediction mode									
	30	31	32	33	34	35	36	37	38	39
Vertical transform set	0	1	0	1	0	1	0	1	0	1
Horizontal transform set	0	1	0	1	0	1	0	1	0	1
	Intra-prediction mode									
	40	41	42	43	44	45	46	47	48	49
Vertical transform set	0	1	0	1	0	1	2	2	2	2
Horizontal transform set	0	1	0	1	0	1	0	0	0	0
	Intra-prediction mode									
	50	51	52	53	54	55	56	57	58	59
Vertical transform set	2	2	2	2	2	1	0	1	0	1
Horizontal transform set	0	0	0	0	0	1	0	1	0	1
	Intra-prediction mode									
	60	61	62	63	64	65	66			
Vertical transform set	0	1	0	1	0	1	0			
Horizontal transform set	0	1	0	1	0	1	0			

In Table 5, numbers of vertical transform sets and horizontal transform sets that are to be applied to the horizontal direction of a residual signal depending on the intra-prediction modes of the target block are indicated.

As exemplified in FIGS. 4 and 5, transform sets to be applied to the horizontal direction and the vertical direction may be predefined depending on the intra-prediction mode of the target block. The encoding apparatus 100 may perform a transform and an inverse transform on the residual signal using a transform included in the transform set corresponding to the intra-prediction mode of the target block. Further, the decoding apparatus 200 may perform an inverse transform on the residual signal using a transform

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included in the transform set corresponding to the intra-prediction mode of the target block.

In the transform and inverse transform, transform sets to be applied to the residual signal may be determined, as exemplified in Tables 3, 4, and 5, and may not be signaled. Transform indication information may be signaled from the encoding apparatus **100** to the decoding apparatus **200**. The transform indication information may be information indicating which one of multiple transform candidates included in the transform set to be applied to the residual signal is used.

For example, when the size of the target block is 64×64 or less, transform sets, each having three transforms, may be configured depending on the intra-prediction modes. An optimal transform method may be selected from among a total of nine multiple transform methods resulting from combinations of three transforms in a horizontal direction and three transforms in a vertical direction. Through such an optimal transform method, the residual signal may be encoded and/or decoded, and thus coding efficiency may be improved.

Here, information indicating which one of transforms belonging to each transform set has been used for at least one of a vertical transform and a horizontal transform may be entropy-encoded and/or -decoded. Here, truncated unary binarization may be used to encode and/or decode such information.

As described above, methods using various transforms may be applied to a residual signal generated via intra prediction or inter prediction.

The transform may include at least one of a first transform and a secondary transform. A transform coefficient may be generated by performing the first transform on the residual signal, and a secondary transform coefficient may be generated by performing the secondary transform on the transform coefficient.

The first transform may be referred to as a “primary transform”. Further, the first transform may also be referred to as an “Adaptive Multiple Transform (AMT) scheme”. AMT may mean that, as described above, different transforms are applied to respective 1D directions (i.e. a vertical direction and a horizontal direction).

A secondary transform may be a transform for improving energy concentration on a transform coefficient generated by the first transform. Similar to the first transform, the secondary transform may be a separable transform or a non-separable transform. Such a non-separable transform may be a Non-Separable Secondary Transform (NSST).

The first transform may be performed using at least one of predefined multiple transform methods. For example, the predefined multiple transform methods may include a Discrete Cosine Transform (DCT), a Discrete Sine Transform (DST), a Karhunen-Loeve Transform (KLT), etc.

Further, a first transform may be a transform having various transform types depending on a kernel function that defines a Discrete Cosine Transform (DCT) or a Discrete Sine Transform (DST).

For example, the transform type may be determined based at least one of 1) a prediction mode of a target block (for example, one of an intra prediction and an inter prediction), 2) a size of a target block, 3) a shape of a target block, 4) an intra prediction mode of a target block, 5) a component of a target block (for example, one of a luma component and a chroma component), and 6) a partitioning type applied to a target block (for example, one of a Quad Tree, a Binary Tree and a Ternary Tree).

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For example, the first transform may include transforms, such as DCT-2, DCT-5, DCT-7, DST-7, DST-1, DST-8, and DCT-8 depending on the transform kernel presented in the following Table 6. In the following Table 6, various transform types and transform kernel functions for Multiple Transform Selection (MTS) are exemplified.

MTS may refer to the selection of combinations of one or more DCT and/or DST kernels so as to transform a residual signal in a horizontal and/or vertical direction.

TABLE 6

Transform type	Transform kernel function $T_i(j)$
DCT-2	$T_i(j) = \omega_0 \cdot \sqrt{\frac{2}{N}} \cdot \cos\left(\frac{\pi \cdot i \cdot (2j+1)}{2N}\right) \text{ where}$ $\omega_0 = \sqrt{\frac{2}{N}} \text{ (} i=0 \text{) or } 1 \text{ (otherwise)}$
DST-7	$T_i(j) = \sqrt{\frac{4}{2N+1}} \cdot \sin\left(\frac{\pi \cdot (2j+1) \cdot (j+1)}{2N+1}\right)$
DCT-5	$T_i(j) = \omega_0 \cdot \omega_1 \cdot \sqrt{\frac{2}{2N-1}} \cdot \cos\left(\frac{2\pi \cdot i \cdot j}{2N+1}\right) \text{ where}$ $\omega_{0/1} = \sqrt{\frac{2}{N}} \text{ (} i \text{ or } j=0 \text{) or } 1 \text{ (otherwise)}$
DCT-8	$T_i(j) = \sqrt{\frac{4}{2N+1}} \cdot \cos\left(\frac{\pi \cdot (2j+1) \cdot (2j+1)}{4N+2}\right)$
DST-1	$T_i(j) = \sqrt{\frac{2}{N+1}} \cdot \sin\left(\frac{\pi \cdot (i+1) \cdot (j+1)}{N+1}\right)$

In Table 6, i and j may be integer values that are equal to or greater than 0 and are less than or equal to $N-1$.

The secondary transform may be performed on the transform coefficient generated by performing the first transform.

As in the first transform, transform sets may also be defined in a secondary transform. The methods for deriving and/or determining the above-described transform sets may be applied not only to the first transform but also to the secondary transform.

The first transform and the secondary transform may be determined for a specific target.

For example, a first transform and a secondary transform may be applied to signal components corresponding to one or more of a luminance (luma) component and a chrominance (chroma) component. Whether to apply the first transform and/or the secondary transform may be determined depending on at least one of coding parameters for a target block and/or a neighbor block. For example, whether to apply the first transform and/or the secondary transform may be determined depending on the size and/or shape of the target block.

In the encoding apparatus **100** and the decoding apparatus **200**, transform information indicating the transform method to be used for the target may be derived by utilizing specified information.

For example, the transform information may include a transform index to be used for a primary transform and/or a secondary transform. Alternatively, the transform information may indicate that a primary transform and/or a secondary transform are not used.

For example, when the target of a primary transform and a secondary transform is a target block, the transform

method(s) to be applied to the primary transform and/or the secondary transform indicated by the transform information may be determined depending on at least one of coding parameters for the target block and/or blocks neighbor the target block.

Alternatively, transform information indicating a transform method for a specific target may be signaled from the encoding apparatus 100 to the decoding apparatus 200.

For example, for a single CU, whether to use a primary transform, an index indicating the primary transform, whether to use a secondary transform, and an index indicating the secondary transform may be derived as the transform information by the decoding apparatus 200. Alternatively, for a single CU, the transform information, which indicates whether to use a primary transform, an index indicating the primary transform, whether to use a secondary transform, and an index indicating the secondary transform, may be signaled.

The quantized transform coefficient (i.e. the quantized levels) may be generated by performing quantization on the result, generated by performing the first transform and/or the secondary transform, or on the residual signal.

FIG. 13 illustrates diagonal scanning according to an example.

FIG. 14 illustrates horizontal scanning according to an example.

FIG. 15 illustrates vertical scanning according to an example.

Quantized transform coefficients may be scanned via at least one of (up-right) diagonal scanning, vertical scanning, and horizontal scanning depending on at least one of an intra-prediction mode, a block size, and a block shape. The block may be a Transform Unit (TU).

Each scanning may be initiated at a specific start point, and may be terminated at a specific end point.

For example, quantized transform coefficients may be changed to 1D vector forms by scanning the coefficients of a block using diagonal scanning of FIG. 13. Alternatively, horizontal scanning of FIG. 14 or vertical scanning of FIG. 15, instead of diagonal scanning, may be used depending on the size and/or intra-prediction mode of a block.

Vertical scanning may be the operation of scanning 2D block-type coefficients in a column direction. Horizontal scanning may be the operation of scanning 2D block-type coefficients in a row direction.

In other words, which one of diagonal scanning, vertical scanning, and horizontal scanning is to be used may be determined depending on the size and/or inter-prediction mode of the block.

As illustrated in FIGS. 13, 14, and 15, the quantized transform coefficients may be scanned along a diagonal direction, a horizontal direction or a vertical direction.

The quantized transform coefficients may be represented by block shapes. Each block may include multiple sub-blocks. Each sub-block may be defined depending on a minimum block size or a minimum block shape.

In scanning, a scanning sequence depending on the type or direction of scanning may be primarily applied to sub-blocks. Further, a scanning sequence depending on the direction of scanning may be applied to quantized transform coefficients in each sub-block.

For example, as illustrated in FIGS. 13, 14, and 15, when the size of a target block is 8x8, quantized transform coefficients may be generated through a first transform, a secondary transform, and quantization on the residual signal of the target block. Therefore, one of three types of scanning sequences may be applied to four 4x4 sub-blocks, and

quantized transform coefficients may also be scanned for each 4x4 sub-block depending on the scanning sequence.

The encoding apparatus 100 may generate entropy-encoded quantized transform coefficients by performing entropy encoding on scanned quantized transform coefficients, and may generate a bitstream including the entropy-encoded quantized transform coefficients.

The decoding apparatus 200 may extract the entropy-encoded quantized transform coefficients from the bitstream, and may generate quantized transform coefficients by performing entropy decoding on the entropy-encoded quantized transform coefficients. The quantized transform coefficients may be aligned in the form of a 2D block via inverse scanning. Here, as the method of inverse scanning, at least one of up-right diagonal scanning, vertical scanning, and horizontal scanning may be performed.

In the decoding apparatus 200, dequantization may be performed on the quantized transform coefficients. A secondary inverse transform may be performed on the result generated by performing dequantization depending on whether to perform the secondary inverse transform. Further, a first inverse transform may be performed on the result generated by performing the secondary inverse transform depending on whether the first inverse transform is to be performed. A reconstructed residual signal may be generated by performing the first inverse transform on the result generated by performing the secondary inverse transform.

For a luma component which is reconstructed via intra prediction or inter prediction, inverse mapping having a dynamic range may be performed before in-loop filtering.

The dynamic range may be divided into 16 equal pieces, and mapping functions for respective pieces may be signaled. Such a mapping function may be signaled at a slice level or a tile group level.

An inverse mapping function for performing inverse mapping may be derived based on the mapping function.

In-loop filtering, the storage of a reference picture, and motion compensation may be performed in an inverse mapping area.

A prediction block generated via inter prediction may be changed to a mapped area through mapping using a mapping function, and the changed prediction block may be used to generate a reconstructed block. However, since intra prediction is performed in the mapped area, a prediction block generated via intra prediction may be used to generate a reconstructed block without requiring mapping and/or inverse mapping.

For example, when the target block is a residual block of a chroma component, the residual block may be changed to an inversely mapped area by scaling the chroma component of the mapped area.

Whether scaling is available may be signaled at a slice level or a tile group level.

For example, scaling may be applied only to the case where mapping is available for a luma component and where the partitioning of the luma component and the partitioning of the chroma component follow the same tree structure.

Scaling may be performed based on the average of the values of samples in a luma prediction block, which corresponds to a chroma prediction block. Here, when the target block uses inter prediction, the luma prediction block may mean a mapped luma prediction block.

A value required for scaling may be derived by referring to a look-up table using the index of a piece to which the average of sample values of the luma prediction block belongs.

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The residual block may be changed to an inversely mapped area by scaling the residual block using a finally derived value. Thereafter, for the block of a chroma component, reconstruction, intra prediction, inter prediction, in-loop filtering, and the storage of a reference picture may be performed in the inversely mapped area.

For example, information indicating whether the mapping and/or inverse mapping of a luma component and a chroma component are available may be signaled through a sequence parameter set.

A prediction block for the target block may be generated based on a block vector. The block vector may indicate displacement between the target block and a reference block. The reference block may be a block in a target image.

In this way, a prediction mode in which the prediction block is generated by referring to the target image may be referred to as an "Intra-Block Copy (IBC) mode".

An IBC mode may be applied to a CU having a specific size. For example, the IBC mode may be applied to an M×N CU. Here, M and N may be less than or equal to 64.

The IBC mode may include a skip mode, a merge mode, an AMVP mode, etc. In the case of the skip mode or the merge mode, a merge candidate list may be configured, and a merge index is signaled, and thus a single merge candidate may be specified among merge candidates present in the merge candidate list. The block vector of the specified merge candidate may be used as the block vector of the target block.

In the case of the AMVP mode, a differential block vector may be signaled. Also, a prediction block vector may be derived from the left neighbor block and the above neighbor block of the target block. Further, an index indicating which neighbor block is to be used may be signaled.

A prediction block in the IBC mode may be included in a target CTU or a left CTU, and may be limited to a block within a previously reconstructed area. For example, the value of a block vector may be limited so that a prediction block for a target block is located in a specific area. The specific area may be an area defined by three 64×64 blocks that are encoded and/or decoded prior to a 64×64 block including the target block. The value of the block vector is limited in this way, and thus memory consumption and device complexity caused by the implementation of the IBC mode may be decreased.

FIG. 16 is a configuration diagram of an encoding apparatus according to an embodiment.

An encoding apparatus 1600 may correspond to the above-described encoding apparatus 100.

The encoding apparatus 1600 may include a processing unit 1610, memory 1630, a user interface (UI) input device 1650, a UI output device 1660, and storage 1640, which communicate with each other through a bus 1690. The encoding apparatus 1600 may further include a communication unit 1620 coupled to a network 1699.

The processing unit 1610 may be a Central Processing Unit (CPU) or a semiconductor device for executing processing instructions stored in the memory 1630 or the storage 1640. The processing unit 1610 may be at least one hardware processor.

The processing unit 1610 may generate and process signals, data or information that are input to the encoding apparatus 1600, are output from the encoding apparatus 1600, or are used in the encoding apparatus 1600, and may perform examination, comparison, determination, etc. related to the signals, data or information. In other words, in embodiments, the generation and processing of data or

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information and examination, comparison and determination related to data or information may be performed by the processing unit 1610.

The processing unit 1610 may include an inter-prediction unit 110, an intra-prediction unit 120, a switch 115, a subtractor 125, a transform unit 130, a quantization unit 140, an entropy encoding unit 150, a dequantization unit 160, an inverse transform unit 170, an adder 175, a filter unit 180, and a reference picture buffer 190.

At least some of the inter-prediction unit 110, the intra-prediction unit 120, the switch 115, the subtractor 125, the transform unit 130, the quantization unit 140, the entropy encoding unit 150, the dequantization unit 160, the inverse transform unit 170, the adder 175, the filter unit 180, and the reference picture buffer 190 may be program modules, and may communicate with an external device or system. The program modules may be included in the encoding apparatus 1600 in the form of an operating system, an application program module, or other program modules.

The program modules may be physically stored in various types of well-known storage devices. Further, at least some of the program modules may also be stored in a remote storage device that is capable of communicating with the encoding apparatus 1200.

The program modules may include, but are not limited to, a routine, a subroutine, a program, an object, a component, and a data structure for performing functions or operations according to an embodiment or for implementing abstract data types according to an embodiment.

The program modules may be implemented using instructions or code executed by at least one processor of the encoding apparatus 1600.

The processing unit 1610 may execute instructions or code in the inter-prediction unit 110, the intra-prediction unit 120, the switch 115, the subtractor 125, the transform unit 130, the quantization unit 140, the entropy encoding unit 150, the dequantization unit 160, the inverse transform unit 170, the adder 175, the filter unit 180, and the reference picture buffer 190.

A storage unit may denote the memory 1630 and/or the storage 1640. Each of the memory 1630 and the storage 1640 may be any of various types of volatile or nonvolatile storage media. For example, the memory 1630 may include at least one of Read-Only Memory (ROM) 1631 and Random Access Memory (RAM) 1632.

The storage unit may store data or information used for the operation of the encoding apparatus 1600. In an embodiment, the data or information of the encoding apparatus 1600 may be stored in the storage unit.

For example, the storage unit may store pictures, blocks, lists, motion information, inter-prediction information, bitstreams, etc.

The encoding apparatus 1600 may be implemented in a computer system including a computer-readable storage medium.

The storage medium may store at least one module required for the operation of the encoding apparatus 1600. The memory 1630 may store at least one module, and may be configured such that the at least one module is executed by the processing unit 1610.

Functions related to communication of the data or information of the encoding apparatus 1600 may be performed through the communication unit 1620.

For example, the communication unit 1620 may transmit a bitstream to a decoding apparatus 1600, which will be described later.

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FIG. 17 is a configuration diagram of a decoding apparatus according to an embodiment.

The decoding apparatus 1700 may correspond to the above-described decoding apparatus 200.

The decoding apparatus 1700 may include a processing unit 1710, memory 1730, a user interface (UI) input device 1750, a UI output device 1760, and storage 1740, which communicate with each other through a bus 1790. The decoding apparatus 1700 may further include a communication unit 1720 coupled to a network 1799.

The processing unit 1710 may be a Central Processing Unit (CPU) or a semiconductor device for executing processing instructions stored in the memory 1730 or the storage 1740. The processing unit 1710 may be at least one hardware processor.

The processing unit 1710 may generate and process signals, data or information that are input to the decoding apparatus 1700, are output from the decoding apparatus 1700, or are used in the decoding apparatus 1700, and may perform examination, comparison, determination, etc. related to the signals, data or information. In other words, in embodiments, the generation and processing of data or information and examination, comparison and determination related to data or information may be performed by the processing unit 1710.

The processing unit 1710 may include an entropy decoding unit 210, a dequantization unit 220, an inverse transform unit 230, an intra-prediction unit 240, an inter-prediction unit 250, a switch 245, an adder 255, a filter unit 260, and a reference picture buffer 270.

At least some of the entropy decoding unit 210, the dequantization unit 220, the inverse transform unit 230, the intra-prediction unit 240, the inter-prediction unit 250, the adder 255, the switch 245, the filter unit 260, and the reference picture buffer 270 of the decoding apparatus 200 may be program modules, and may communicate with an external device or system. The program modules may be included in the decoding apparatus 1700 in the form of an operating system, an application program module, or other program modules.

The program modules may be physically stored in various types of well-known storage devices. Further, at least some of the program modules may also be stored in a remote storage device that is capable of communicating with the decoding apparatus 1700.

The program modules may include, but are not limited to, a routine, a subroutine, a program, an object, a component, and a data structure for performing functions or operations according to an embodiment or for implementing abstract data types according to an embodiment.

The program modules may be implemented using instructions or code executed by at least one processor of the decoding apparatus 1700.

The processing unit 1710 may execute instructions or code in the entropy decoding unit 210, the dequantization unit 220, the inverse transform unit 230, the intra-prediction unit 240, the inter-prediction unit 250, the switch 245, the adder 255, the filter unit 260, and the reference picture buffer 270.

A storage unit may denote the memory 1730 and/or the storage 1740. Each of the memory 1730 and the storage 1740 may be any of various types of volatile or nonvolatile storage media. For example, the memory 1730 may include at least one of ROM 1731 and RAM 1732.

The storage unit may store data or information used for the operation of the decoding apparatus 1700. In an embodi-

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ment, the data or information of the decoding apparatus 1700 may be stored in the storage unit.

For example, the storage unit may store pictures, blocks, lists, motion information, inter-prediction information, bit-streams, etc.

The decoding apparatus 1700 may be implemented in a computer system including a computer-readable storage medium.

The storage medium may store at least one module required for the operation of the decoding apparatus 1700. The memory 1730 may store at least one module, and may be configured such that the at least one module is executed by the processing unit 1710.

Functions related to communication of the data or information of the decoding apparatus 1700 may be performed through the communication unit 1720.

For example, the communication unit 1720 may receive a bitstream from the encoding apparatus 1700.

Concept of Picture, Slice, Tile, and Brick, and Configuration of Slice

FIG. 18 illustrates the division of a picture in a raster-scan slice mode according to an example.

One picture may be divided into one or more regions. A region may be a tile.

For example, each region may have the shape of a rectangle. In other words, one or more rectangular regions may constitute one picture.

Alternatively, each region may have a shape in which more than one or more rectangles are combined with each other. For example, the more than one or more combined rectangles may be consecutive units following a raster-scan order.

In an embodiment, a rectangle may be limited to a square.

The unit of division may be any of a slice, a tile, and a coding tree unit (CTU).

In FIG. 18, each slice is indicated by a bold solid line. Each tile is indicated by a thin solid line. Each CTU is indicated by a dotted line.

A picture may be divided into slices and tiles.

The picture may be divided into one or more slices. Alternatively, the picture may include one or more slices.

Information for division into one or more slices may be signaled through a network abstraction layer (NAL) unit. A NAL unit may include a slice header and slice data.

In FIG. 18, the picture is divided into three slices.

The picture may be divided into one or more tiles. In the division of the picture into one or more tiles, the number of rows and the number of columns corresponding to one or more tiles may be used. In other words, the picture may be divided into one or more tiles depending on n_1 and m_1 . n_1 denotes the number of one or more rows corresponding to one or more tiles. m_1 denotes the number of one or more columns corresponding to one or more tiles.

In FIG. 18, the picture is divided into 12 tiles. A first slice of the picture includes two tiles. A second slice of the picture includes five tiles. A third slice of the picture includes five tiles.

The picture may be divided into one or more Coding Tree Units (CTUs).

Each tile may include one or more CTUs. Alternatively, each tile may be divided into one or more CTUs. A tile may cover a region of a rectangular in a picture. A tile may be composed of a sequence of CTUs.

Each tile may be composed of one or more CTUs corresponding to n_2 rows and m_2 columns. n_2 may be an integer of 1 or more, and m_2 may be an integer of 1 or more. Further,

the picture may be composed of a set of one or more CTUs included in the rectangular region of each tile.

Each CTU may occupy a rectangular region in a picture. Processing of one or more CTUs in a tile may be performed using a raster-scan mode (raster-scan slice mode). Here, processing may include encoding, decoding, writing, and reading.

In FIG. 18, each tile includes 18 CTUs. The 18 CTUs may be configured in three rows and six columns.

At least a part of encoding of a slice and/or a tile and at least a part of decoding thereof may be independently performed on an additional slice and/or an additional tile. For example, encoding may include prediction, transform, quantization, entropy coding, dequantization, inverse transform, and reconstruction. Decoding may include dequantization, inverse transform, prediction, and reconstruction.

By utilizing these features, a slice and a tile may be used for parallel processing, which becomes necessary as the encoding apparatus 1600 and the decoding apparatus 1700 become more complicated, and may be used to provide a Region Of Interest (ROI) in the picture.

Each slice may be composed of one or more tiles of a picture. Alternatively, each slice may be composed of a sequence of one or more CTUs comprising a tile. For example, each slice may include one or more integer number of tiles of the picture. Alternatively, each slice may include one or more integer number of CTUs of the picture.

Each vertical slice boundary is always also a vertical tile boundary. In other words, a vertical boundary of the slice can be constrained to coincide with a boundary of the tile.

It is possible that a horizontal boundary of a slice is not a tile boundary. A horizontal boundary of a slice may be composed of horizontal CTU boundaries within a tile.

When a tile is partitioned into a plurality of rectangular slices, each slice of the plurality of rectangular slices may be composed of an integer number of consecutive complete CTU rows within the tile.

Each slice in the picture may be defined in two types of modes. A first mode may be a raster-scan slice mode. A second mode may be a rectangular slice mode.

In the raster-scan slice mode, one slice may include the sequence of complete tiles in a tile raster scan of the picture.

In the rectangular slice mode, one slice may include 1) multiple complete tiles which collectively form a rectangular region of the picture, or 2) multiple consecutive complete CTU rows of one tile. The multiple consecutive complete CTU rows of one tile may collectively form the rectangular region of the picture.

One or more tiles in a rectangular slice may be processed in a raster-scanning order.

Each sub-picture may include one or more slices which collectively cover a rectangular region of a picture. The boundary of each sub-picture may always be the boundary of a slice. Further, the boundary of each vertical sub-picture may always be the boundary of a vertical tile.

FIG. 19 illustrates the division of a picture in a rectangular slice mode according to an example.

In FIG. 19, each slice is indicated by a bold solid line. Each tile is indicated by a thin solid line. Each CTU is indicated by a dotted line.

In FIG. 19, the picture is divided into 24 tiles and nine rectangular slices. The 24 tiles are configured in six columns and four rows.

As illustrated in FIGS. 18 and 19, each slice may include multiple tiles.

FIG. 20 illustrates tiles and rectangular slices in a picture according to an example.

In FIG. 20, each slice is indicated by a bold solid line. Each tile is indicated by a thin solid line. Each CTU is indicated by a dotted line.

The picture may be divided into one or more tiles and one or more rectangular slices.

In FIG. 20, the picture is divided into four tiles. Further, the picture is divided into four rectangular slices.

The picture may be encoded using three different color planes. A color plane identifier for each slice may indicate the color plane of the corresponding slice. Here, the slice may include only a CTU in a color corresponding to the color plane identifier of the slice. Each color array may be composed of slices having the same color plane identifier.

FIG. 21 illustrates the division of a picture into sub-pictures according to an example.

In FIG. 21, each sub-picture is indicated by a bold dotted line. Each slice is indicated by a bold solid line. Each tile is indicated by a thin solid line. Each CTU is indicated by a thin dotted line.

In FIG. 21, the picture may include 15 tiles. Each of the tiles may include 4x4 CTUs. Also, the picture may be divided into 24 slices and 24 sub-pictures. As described above with reference to FIGS. 18 to 20, each slice may be composed of one or more tiles, and may be composed of one or more CTUs in the tiles.

Hereinafter, a target picture denotes a picture that is the target of current encoding/decoding. A target sub-picture denotes a sub-picture that is the target of current encoding/decoding. A target slice denotes a slice that is the target of current encoding/decoding. A target block may be a block that is the target of current encoding/decoding. A target picture may include a target sub-picture. A target sub-picture may include a target slice. A target slice may include a target block.

Configuration of Subpicture

As described above, a picture may be composed of one or more subpictures.

In order to define one or more subpictures in the picture, a layout and identifiers (ID) of the one or more subpictures may be required.

1) The layout and subpicture IDs may be signaled through a Sequence Parameter Set (SPS).

The layout may be information for specifying the locations and areas of one or more subpictures in the picture.

The widths and heights of respective subpictures may be signaled through the SPS. The unit of width and height may be a Coding Tree Unit (CTU). The width and height of each subpicture may be signaled in units of a CTU (or a CTU size).

The subpicture IDs of respective subpictures may be signaled through the SPS.

1) The layout and subpicture IDs may be signaled through a Picture Parameter Set (PPS).

The subpicture IDs of respective subpictures may be signaled through the PPS.

The configuration of rectangular slices may be signaled through the PPS.

The configuration of rectangular slices may indicate 1) the number of one or more rectangular slices and 2) the area occupied by each of the one or more rectangular slices.

Determination of Subpicture ID of Subpicture

FIG. 22 illustrates code for subpicture ID mapping according to an example.

Subpicture ID mapping may be configured to determine the subpicture ID of each of one or more subpictures in a picture.

When the PPS is parsed, the syntax element “SubpicIdVal[i]” may be derived based on the code illustrated in FIG. 22.

SubpicIdVal[i] may be the subpicture ID of an i-th subpicture, among the one or more subpictures in the picture. “i” may be equal to or greater than ‘0’ and less than or equal to ‘sps_num_subpics_minus1’.

The syntax element “sps_num_subpics_minus1+1” may indicate the number of one or more subpictures in each picture. sps_num_subpics_minus1 may be signaled through the SPS.

The syntax element “subpic_id_mapping_explicitly_signalled_flag” may be information explicitly indicating whether subpicture ID mapping is to be performed in the SPS or the PPS.

The syntax element “subpic_id_mapping_in_pps_flag” may be information (or a flag) indicating whether information about subpicture ID mapping is to be signaled through the PPS. When the information about subpicture ID mapping is not signaled through the PPS, the information about subpicture ID mapping may be signaled through the SPS.

The syntax element “pps_subpic_id[i]” may specify the subpicture ID of the i-th subpicture, among the one or more subpictures in the picture.

The syntax element “pps_subpic_id[i]” may be signaled through the PPS.

The syntax element “sps_subpic_id[i]” may specify the subpicture ID of the i-th subpicture, among the one or more subpictures in the picture.

The syntax element “sps_subpic_id[i]” may be signaled through the SPS.

Each picture may include one or more slices.

The slice header of each of the one or more slices may include slice_subpic_id (or sh_subpic_id). The slice_subpic_id of each slice may be signaled through the slice header.

slice_subpic_id may indicate the subpicture ID of a subpicture including a slice.

If there is slice_subpic_id, the value of the variable “CurrSubpicIdx” may be derived such that SubpicIdVal[CurrSubpicIdx] is identical to sh_subpic_id. If there is no slice_subpic_id, the value of CurrSubpicIdx may be derived as ‘0’.

The variable “CurrSubpicIdx” may be used to derive the location of a target subpicture.

FIG. 23 illustrates signaling of slice addresses in a raster-scan slice mode according to an example.

A PPS may include the syntax element “rect_slice_flag”. The syntax element “rect_slice_flag” may specify the slice mode used for each picture that refers to the PPS.

When the value of pps_rect_slice_flag is ‘0’, a raster-scan slice mode may be used for each picture that refers to the PPS, and a layout of slices may not be signaled in the PPS.

When the value of pps_rect_slice_flag is ‘1’, a rectangular slice mode may be used for each picture that refers to the PPS, and the layout of slices may be signaled in the PPS.

Depending on the slice mode, a scheme for determining slice addresses may differ.

In FIG. 23, signaling of slice addresses performed when the value of rect_slice_flag is ‘0’ is illustrated.

When the value of rect_slice_flag is ‘0’, the slice address of each slice may be the raster-scan tile index of the first tile of the slice. That is, the slice address may be determined by units of tiles.

The slice addresses of one or more slices of the subpicture may range from ‘0’ to ‘NumTilesInPic-1’.

In the example of FIG. 23, the value of rect_slice may be ‘0’.

Further, NumofSlice (i.e., the number of one or more slices) may be 2.

Also, slice address [1] (i.e., the address of a first slice) may be ‘1’. Also, slice address [2] (i.e., the address of a second slice) may be ‘4’.

Further, the range K of the slice address [i] may be defined as the range of integers from ‘0’ to ‘NumTileInPic-1’. NumTileInPic may be the number of one or more tiles in a picture.

FIG. 24 illustrates signaling of slice addresses in a rectangular slice mode according to an example.

In FIG. 24, signaling of slice addresses performed when the value of rect_slice_flag is ‘1’ is illustrated.

When the value of rect_slice_flag is ‘1’, the slice address of each slice may be a slice index. Here, the slice address may be determined at the level of a subpicture. That is, different slice addresses may be respectively assigned to one or more slices in a subpicture.

Since the slice addresses are determined at the level of the subpicture, the range of slice addresses may be defined as the range of integers from ‘0’ to ‘NumSliceInSubPic[CurrSubpicIdx]-1’. CurrSubpicIdx may be the subpicture index of a target subpicture. NumSliceInSubPic[i] may be the number of one or more slices in an i-th subpicture.

In the example of FIG. 24, one or more slices may be present in a subpicture having a subpicture ID of 2.

Further, rect_slice may be ‘1’.

Furthermore, the syntax element “NumSlicesInSubpic[CurrSubpicIdx]” (i.e., the number of one or more slices in the target subpicture) may be ‘2’.

Also, slice address [1] (the address of a first slice) may be ‘1’. Also, slice address [2] (the address of a second slice) may be ‘4’.

Further, the range K of the slice address[i] may be defined as the range of integers from ‘0’ to ‘NumSlicesInSubpic[CurrSubpicIdx]-1’. NumSlicesInSubpic[CurrSubpicIdx] may be the number of one or more slices in the target subpicture.

FIG. 25 illustrates a part of the structure of a sequence parameter set according to an example.

The code of FIG. 25 shows a method for signaling a subpicture ID through an SPS.

As the method for signaling the subpicture ID, 1) a method for signaling a subpicture ID through an SPS and 2) a method for signaling a subpicture ID through a PPS may be used.

When the SPS is used to signal a subpicture ID, 1) the subpicture ID may be directly signaled through the SPS. 2) Alternatively, the subpicture ID of the subpicture may be derived through the subpicture index of the subpicture.

Here, the subpicture index may be equal to or greater than ‘0’ and less than or equal to ‘sps_num_subpics_minus1’.

Depending on a specific scan type, respective subpicture indexes may be sequentially assigned to one or more subpictures. The scan type may be a raster scan, a vertical scan, a horizontal scan, a z scan, a diagonal scan or the like.

When the SPS is not used to signal the subpicture ID of the subpicture, a PPS may be used to signal the subpicture ID of the subpicture.

Depending on the layout of one or more subpictures defined in the SPS, subpicture indices of the one or more subpictures may be the same for respective pictures. In contrast, subpicture IDs of one or more subpictures may be different for each picture.

The subpicture IDs of multiple subpictures constituting one picture may be different from each other. Further, the subpicture ID of at least one of the multiple subpictures

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constituting one picture may be identical to that of other subpictures. Alternatively, all subpicture IDs of multiple subpictures may be identical to each other.

As illustrated in FIG. 25, when the value of `subpic_id_mapping_explicitly_signalled_flag` is '1', each subpicture ID may be signaled through an SPS.

Further, when the value of `subpic_id_mapping_explicitly_signalled_flag` is '0', each subpicture ID may be derived using code, which will be described later with reference to FIG. 26.

Furthermore, even if the value of `subpic_id_mapping_explicitly_signalled_flag` is '1', some subpicture IDs may not be explicitly signaled, and subpicture IDs may be derived as a specific value depending on specific schemes and specific conditions. Here, derivation may be performed based on at least one of the locations of subpictures, the number of subpictures, and previously allocated subpicture IDs. Alternatively, the specific value may be determined based on at least one of the locations of subpictures, the number of subpictures, and previously allocated subpicture IDs.

The syntax element "`subpic_id_mapping_in_sps_flag`" may be information indicating whether subpicture ID mapping is to be signaled in an SPS when the value of `subpic_id_mapping_explicitly_signalled_flag` is '1'.

When the value of `subpic_id_mapping_explicitly_signalled_flag` is '1', if the value of `subpic_id_mapping_in_sps_flag` is '1', subpicture ID mapping may be signaled in the SPS.

As illustrated in FIG. 25, when the value of `subpic_id_mapping_explicitly_signalled_flag` is '1', if the value of `subpic_id_mapping_in_sps_flag` is '1', `sps_num_subpic_minus1+1` subpicture IDs (`sps_subpic_id[i]`) ranging from '0' to '`sps_num_subpic_minus1`' may be signaled through the 'for clause'.

When the value of `subpic_id_mapping_explicitly_signalled_flag` is '1', if the value of `subpic_id_mapping_in_sps_flag` is '0', subpicture ID mapping may be signaled in a PPS. The PPS may be referred to by the coded pictures of a Coded Layer Video Sequence (CLVS).

FIG. 26 illustrates a method for deriving a subpicture ID according to an example.

The code of FIG. 26 may show a method of deriving the value of `sps_subpic_id` when the value of `subpic_id_mapping_explicitly_signalled_flag` is '0' and the value of `subpic_info_present_flag` is '1'.

As illustrated in FIG. 26, when the value of `subpic_id_mapping_explicitly_signalled_flag` is '0' and the value of `subpic_info_present_flag` is '1', the subpicture ID of a subpicture having a subpicture index of `i` may be '`i`'.

The subpicture may be divided into one or more slices. Therefore, when the slice address of each slice is derived, the location of a subpicture corresponding to the subpicture ID may be derived using the derived slice address.

FIG. 27 illustrates a scheme for utilizing a slice index for deriving the address of a CTB in a slice according to an example.

The function "`AddCtbToSlice`" of FIG. 27 may add Coding Tree Blocks (CTB) to the slices.

In `AddCtbToSlice`, the CTB address of a corresponding CTB in a target slice may be derived using the slice index `sliceIdx`.

`CtbAddrInSlice[i][j]` may indicate the address of a `j`-th CTB in an `i`-th slice.

`PicWidthInCtbs` may be the width of a picture represented by units of a CTB.

`NumCtusInSlice[i]` may indicate the number of CTBs in the `i`-th slice.

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FIG. 28 illustrates a scheme for utilizing a slice index for deriving the address of a CTB in a target slice according to an example.

In the code of FIG. 28, CTB addresses of CTBs in the target slice may be derived using a slice address. Here, the CTB address of each of the CTBs in the target slice may be derived using the variable "`CtbAddrInSlice`".

`slice_address` may indicate the slice index of the target slice (at the level of a subpicture).

`CurrSubpicIdx` may indicate the subpicture index of a target subpicture.

`picLevelSliceIdx` may be determined to indicate the slice index of the target slice at the picture level using `slice_address` and `NumSlicesInSubpic[j]`. In other words, `picLevelSliceIdx` may be initialized to have the value of `slice_address`, and a number of slices in subpictures preceding the target subpicture may be added to `picLevelSliceIdx`.

`NumCtusInCurrSlice` may represent the number of CTBs in the slice indicated by `picLevelSliceIdx`.

Through the 'for clause' using `NumCtusInCurrSlice`, `CtbAddrInSlice[picLevelSliceIdx][i]` may be copied to `CtbAddrInCurrSlice[i]`. Through this copying, CTB addresses of one or more CTBs in the target slice may be set.

Referring back to FIG. 25, the code of FIG. 25 may represent a method in which the subpicture ID of each subpicture is directly signaled through the SPS. Alternatively, the subpicture ID of each subpicture may be derived through the index of the subpicture.

FIG. 29 illustrates a part of the structure of a Picture Parameter Set (PPS) according to an example.

The code of FIG. 29 may indicate a method for signaling a subpicture ID through a PPS.

As the method for signaling the subpicture ID, 1) a method for signaling a subpicture ID through an SPS and 2) a method for signaling a subpicture ID through a PPS may be used.

As illustrated in FIG. 29, when the value of `subpic_id_mapping_explicitly_signalled_pps_flag` is '1', the subpicture ID may be signaled through the PPS.

Further, when the value of `subpic_id_mapping_explicitly_signalled_pps_flag` is '0', the subpicture ID may be derived using the code, described above with reference to FIG. 26.

The syntax element "`subpic_id_mapping_in_pps_flag`" may be information indicating whether subpicture ID mapping is to be signaled in the PPS when the value of `subpic_id_mapping_explicitly_signalled_pps_flag` is '1'.

When the value of `subpic_id_mapping_explicitly_signalled_pps_flag` is '1', if the value of `subpic_id_mapping_in_pps_flag` is '1', subpicture ID mapping may be signaled in the PPS.

As illustrated in FIG. 29, when the value of `subpic_id_mapping_explicitly_signalled_pps_flag` is '1', if the value of `subpic_id_mapping_in_pps_flag` is 1, `pps_num_subpic_minus1+1` PPS subpicture IDs (`pps_subpic_id[i]`) ranging from '0' to '`pps_num_subpic_minus1`' may be signaled through the 'for clause'.

FIG. 30 illustrates code for subpicture ID mapping according to an example.

For an SPS and a PPS, when the value of `subpic_id_mapping_explicitly_signalled_flag` is '1' or when the value of `subpic_id_mapping_explicitly_signalled_pps_flag` is '1', subpicture IDs may be signaled through the SPS or PPS. When the value of `subpic_id_mapping_explicitly_signalled_pps_flag` is '1', subpicture IDs may be signaled

through the PPS. When the value of `subpic_id_mapping_explicitly_signalled_pps_flag` is '0', subpicture IDs may be signaled through the SPS.

When the value of `subpic_id_mapping_explicitly_signalled_flag` is '0' and the value of `subpic_id_mapping_explicitly_signalled_pps_flag` is '0', the subpicture ID of each subpicture may have subpicture index values falling within a range from '0' to '`sps_num_subpics_minus1`'.

FIG. 31 illustrates a change in a subpicture ID according to an example.

When a specific subpicture is treated as a picture in an SPS, the subpicture ID of the specific subpicture in a PPS may be changed.

As illustrated in FIG. 31, the location of a subpicture in reference picture N-1 and the location of the subpicture in reference picture N (i.e., the target picture) may not be identical to each other.

Due to the difference in location, when motion compensation using the reference picture of a target subpicture is performed in inter-prediction, the locations of subpictures of the reference picture shall be derived.

In an embodiment, in order to provide the locations of the subpictures of the reference picture, the locations of the subpictures of the reference picture may be signaled at the level of at least one of a slice header, a PPS, and a picture header (PH) of the reference picture.

When bidirectional prediction is used for the target subpicture, the locations of all subpictures both for reference picture list L0 and in reference picture list L1 may be provided.

1) When a specific subpicture is treated and processed in the same way as a picture, or 2) when one subpicture is encoded and/or decoded independently of other subpictures, the target subpicture may be forced to refer to only a reference subpicture having a subpicture ID identical to that of the target subpicture. Independent encoding and/or decoding may include at least one of intra-prediction, inter-prediction, and in-loop filtering.

If the target subpicture refers to a reference subpicture having a subpicture ID different from that of the target subpicture, a procedure for determining whether such reference is to be permitted may be additionally performed. For this determination, a flag indicating whether reference is to be permitted may be signaled. Such a flag may be signaled at the level of at least one of an SPS, a PPS, and a SH. Method for Determining NAL Unit Type of Rearranged Subpicture

For one or more pictures in a sequence, when the subpicture ID of subpicture *i* of a picture is determined through an SPS, the subpicture ID of the subpicture *i* may be `sps_subpic[i]`. Here, subpicture *i* may indicate a subpicture having a subpicture index of *i*.

When the target picture is not a first picture in the sequence, the subpicture ID of the subpicture *i* of the target picture may be determined through a PPS. Here, the subpicture ID of the subpicture *i* of the target picture (i.e., `SubpicIdVal[i]` of the subpicture *i* of the target picture) may be different from the subpicture ID of the subpicture *i* of a previous picture (i.e., `SubpicIdVal[i]` of the subpicture *i* of the previous picture). The previous picture may be a picture that is encoded and/or decoded prior to the target picture in the order of encoding and/or decoding.

The rearrangement of the subpicture may mean that the subpicture ID of the subpicture *i* of the target picture has changed from the subpicture ID of the subpicture *i* of the previous picture.

When the rearrangement of the subpicture is applied, the NAL unit type of the slice of the rearranged subpicture in the target picture may be one of values ranging from 'Instantaneous Decoding Refresh (IDR)_W_Random Access Decodable Leading (RADL)' to 'CRA_NUT'. When the rearrangement of the subpicture is applied, the NAL unit type of the slice of the rearranged subpicture in the target picture may be `IDR_W_RADL`, `IDR_N_LP`, or `CRA_NUT`.

A NAL unit having the NAL unit type of `IDR_W_RADL` or `IDR_N_LP` may indicate a coded slice for an Instantaneous Decoding Refresh (IDR) picture or an IDR subpicture.

A NAL unit having a NAL unit type of `CRA_NUT` may be a coded slice of an instantaneous Clean Random Access (CRA) picture or subpicture.

In other words, when rearrangement of a subpicture occurs in the target picture in the sequence, the slice in the subpicture of the target picture may be a slice in an Intra-Random Access Point (IRAP) subpicture. The IRAP subpicture may be a subpicture for which all Video Coding Layer (VCL) NAL units have the value of the same NAL unit type and for which the value of the NAL unit type falls within a range from 'IDR_W_RADL' to 'CRA_NUT'.

When subpictures in one picture have values of different NAL unit types, the values of different NAL unit types for the subpictures in one picture may be signaled through the syntax element "`mixed_nalu_types_in_pic_flag`" in the PPS. `mixed_nalu_types_in_pic_flag` may be information indicating whether mixed NAL units are to be used for the picture.

When the value of `mixed_nalu_types_in_pic_flag` in the PPS is '0', the values of NAL unit types must be identical to each other for all VCL NAL units of the picture.

When the value of `mixed_nalu_types_in_pic_flag` in the PPS is '1', the picture may have at least two subpictures, and the VCL NAL units of the picture may have two or more different NAL unit type values.

Picture in which Mixed NAL Units are Used

A slice header may access information in a PPS through a PPS ID. A picture header may include the PPS ID.

A slice header may include the syntax element "`picture_header_in_slice_header_flag`". Here, `picture_header_in_slice_header_flag` may be information indicating whether a picture header is present in a slice header.

When the value of `picture_header_in_slice_header_flag` in the slice header is '1', the slice header may include the syntax structure of the picture header.

When the value of `picture_header_in_slice_header_flag` in the slice header is '0', the syntax structure of the picture header may not be present in the slice header, and a picture unit for the target picture shall have the NAL unit of the picture header. Further, in this case, the values of `picture_header_in_slice_header_flag` of all slices in the target picture may be '0'.

Therefore, in order to determine the value of `mixed_nalu_types_in_pic_flag`, the values of `mixed_nalu_types_in_pic_flag` in the PPS must be identified through the PPS ID of the picture header.

When the value of `mixed_nalu_types_in_pic_flag` for the picture is 1, the following conditions 1) and 2) may be applied.

1) A picture may include slices having a NAL unit type of IDR, and one or more slices of the picture may have NAL unit types corresponding to other values. Here, the following items 1-1) to 1-4) may be applied.

1-1) A certain slice may belong to a certain subpicture, in which the value of `subpic_treated_as_pic_flag[i]` is '1', among the subpictures of the target picture.

1-2) A certain slice may not be included in a subpicture which includes a Variable Length Coding/Code (VLC) NAL unit for which a NAL unit type is not IDR, among the subpictures of the target picture.

1-3) When the NAL unit type of the certain slice of the target picture is CRA, for all subsequent picture units following the target picture in a decoding order and an output order, reference picture list L0 (e.g., `RefPicList[0]`) and reference picture list L1 (e.g., `RefPicList[1]`) for slices of a certain subpicture may include none of pictures preceding the target picture in a decoding order in an activation list.

1-4) When the NAL unit type of a certain slice is `IDR_W_RADL` or `IDR_N_LP`, for all subsequent picture units following a target picture (e.g., `picA`) in a decoding order, reference picture list L0 (e.g., `RefPicList[0]`) and reference picture list L1 (e.g., `RefPicList[1]`) for slices of a target subpicture (e.g., `subpicA`) in the corresponding picture unit may include none of pictures preceding the target picture (e.g., `picA`) in a decoding order in an activation list.

2) The case where the value of `mixed_nalu_types_in_pic_flag` for a picture is '1' may mean that other pictures that refer to the PPS include slices having other NAL unit types. For example, these pictures may be pictures encoded by combining subpicture bitstreams. When this combination is performed, the encoding apparatus 100 may perform matching between the structure of the combined bitstream and the parameters of original bitstreams.

FIG. 32 illustrates a part of the structure of a sequence parameter set indicating whether there is syntax related to a reference picture list according to an example.

An SPS may include the syntax element "`sps_idr_rpl_present_flag`". The syntax element "`sps_idr_rpl_present_flag`" may be information indicating whether reference picture list (RPL) syntax elements may be present in the slice headers of slices for which a NAL unit type is `IDR_N_LP` or `IDR_W_RADL`.

When the value of `sps_idr_rpl_present_flag` is '1', the RPL syntax elements may be present in the slice headers of the slices for which a NAL unit type is `IDR_N_LP` or `IDR_W_RADL`.

When the value of `sps_idr_rpl_present_flag` is '0', the RPL syntax elements may not be present in the slice headers of slices for which a NAL unit type is `IDR_N_LP` or `IDR_W_RADL`.

FIG. 33 illustrates a part of the structure of a picture parameter set indicating whether there are mixed NAL unit types according to an example.

A PPS may include the syntax element "`mixed_nalu_types_in_pic_flag`".

When the value of `mixed_nalu_types_in_pic_flag` is '1', each picture that refers to the PPS may include one or more VCL NAL units, wherein the NAL unit type values of the one or more VCL NAL units may not be identical to each other. Also, each picture that refers to the PPS may not be an IRAP picture.

When the value of `mixed_nalu_types_in_pic_flag` is '0', each picture that refers to the PPS may include one or more VCL NAL units, wherein the NAL unit type values of the one or more VCL NAL units may be identical to each other.

When the NAL unit type of the slice is an IDR type, a reference picture list may be configured based on the setting of the syntax elements "`sps_idr_rpl_present_flag`" and "`mixed_nalu_types_in_pic_flag`".

FIG. 34 illustrates a slice header configuring a reference picture list according to an embodiment.

In an embodiment, according to the slice header of FIG. 34, the following schemes may be applied to the configuration of the reference picture list.

1) A reference picture list may be restrictively configured for a subpicture only when mixed NAL units are used for a picture. As an example of this restrictive configuration, as illustrated in FIG. 34, a reference picture list may be configured for the subpicture when the value of `sps_idr_rpl_present_flag` in an SPS is '1' and the value of `mixed_nalu_types_in_pic_flag` in a PPS is '1'.

2) Alternatively, in another example, even if the value of `mixed_nalu_types_in_pic_flag` is '0', a reference picture list may be configured for a subpicture when the value of `sps_idr_rpl_present_flag` is '1'.

3) Alternatively, in a further example, even if the value of `sps_idr_rpl_present_flag` is '0', a reference picture list may be configured for a subpicture when the value of `mixed_nalu_types_in_pic_flag` is '1'.

4) When the values of both `sps_idr_rpl_present_flag` and `mixed_nalu_types_in_pic_flag` are '0', a reference picture list for a subpicture may not be configured.

As described above with reference to the above-described embodiments, the configuration of a reference picture list may be adaptively performed based on at least one of the syntax elements `sps_idr_rpl_present_flag`, `rpl_info_in_ph_flag`, and NAL unit type-related information (e.g., `mixed_nalu_types_in_pic_flag`).

For the purpose of such adaptive performance, the syntax element "`rpl_info_in_ph_flag`" may be used. "`rpl_info_in_ph_flag`" may be information indicating whether information or syntax related to a reference picture list is present in a picture header.

Whether a reference picture list is to be configured in the slice header may be determined based on at least one of 1) the value of `rpl_info_in_ph_flag`, 2) NAL unit type, 3) the value of `sps_idr_rpl_present_flag`, and 4) the value of `mixed_nalu_types_in_pic_flag`.

For example, the configuration of a reference picture list in the slice header may be performed only when the value of `rpl_info_in_ph_flag` is '0' in the slice header.

For example, the configuration of a reference picture list in the slice header may be performed when the NAL unit type is none of specific types. The specific types may be `IDR_W_RADL` and `IDR_N_LP`.

For example, the configuration of a reference picture list in the slice header may be performed when the value of `sps_idr_rpl_present_flag` is '1' and the value of `mixed_nalu_types_in_pic_flag` is '1'.

As illustrated in FIG. 34, the above-described conditions may be combined with each other. The configuration of a reference picture list in the slice header may be performed when condition 1 is true and condition 2 is true. Condition 1 may be a condition that the value of `rpl_info_in_ph_flag` in the slice header be '0'. Condition 2 may be a condition that 1) a NAL unit type be none of specific types, or a condition that 2) the values of both `sps_idr_rpl_present_flag` and `mixed_nalu_types_in_pic_flag` be '1'.

Next, whether an override is to be performed on the number of activated (or active) reference indexes in the slice header may be determined.

Whether an override is to be performed may be determined based on at least one of 1) the value of `rpl_info_in_ph_flag`, 2) the NAL unit type, 3) the value of `sps_idr_rpl_present_flag`, 4) the value of `mixed_nalu`

_types_in_pic_flag, 5) the slice type of each slice, and 6) the number of entries in a reference picture list.

For example, an override may be performed when the value of rpl_info_in_ph_flag is '1'.

For example, an override may be performed when the NAL unit type is none of the specific types. The specific types may be IDR_W_RADL and IDR_N_LP.

For example, an override may be performed when the value of sps_idr_rpl_present_flag is '1' and the value of mixed_nalu_types_in_pic_flag is '1'.

For example, an override may be performed when the type of a slice is not 'I' and the value of num_ref_entries[0][RplIdx[0]] is greater than '1'. Alternatively, an override may be performed when the type of a slice is 'B' and the value of num_ref_entries[1][RplIdx[1]] is greater than '1'.

num_ref_entries[listIdx][rplIdx] may specify the number of entries in a ref_pic_list_struct(listIdx, rplIdx) syntax structure.

An override may include signaling of num_ref_idx_active_override_flag.

The syntax element "num_ref_idx_active_override_flag" may be information indicating whether an override is to be performed on the number of activated reference indexes for the reference picture list.

The syntax element "num_ref_idx_active_override_flag" may be information indicating whether, for a P slice and a B slice, the syntax element "num_ref_idx_active_minus1[0]" is provided, and for the B slice, the syntax element "num_ref_idx_active_minus1[1]" is provided.

When the value of num_ref_idx_active_override_flag is '0', num_ref_idx_active_minus1[0] and num_ref_idx_active_minus1[1] may not be present.

The syntax elements "num_ref_idx_active_minus1[0]" and "num_ref_idx_active_minus1[1]" may be information indicating the number of activated reference indexes in a specific reference picture list.

The syntax element "num_ref_idx_active_minus1[0]" may indicate the number of activated reference indexes-1 for reference picture list L0.

The syntax element "num_ref_idx_active_minus1[1]" may indicate the number of activated reference indexes-1 for reference picture list L1.

If the value of num_ref_idx_active_override_flag is '1', the syntax element "num_ref_idx_active_minus1[0]" may be present both for the P slice and the B slice and the syntax element "num_ref_idx_active_minus1[1]" may be present for the B slice when the value of num_ref_entries[0][RplIdx[0]] is greater than '1'.

If the value of num_ref_idx_active_override_flag is '1', signaling of num_ref_idx_active_minus1 may be performed when the value of num_ref_entries[i][RplIdx[i]] is greater than '1'. Here, when the type of a slice is 'B', signaling of num_ref_idx_active_minus1[0] and num_ref_idx_active_minus1[1] may be performed. When the type of a slice is not 'B' (i.e., when the type of the slice is 'P'), signaling of num_ref_idx_active_minus1[0] may be performed. The above-described methods for configuring reference picture lists may be applied equally or similarly to other embodiments, which will be described later. Hereinafter, repeated descriptions may be omitted.

In an embodiment, in the slice header, a reference picture list may be configured regardless of (or independently of) sps_idr_rpl_present_flag. A reference picture list may be configured regardless of (or independently of) rpl_info_in_ph_flag and the NAL unit type-related information.

The above-described methods for configuring reference picture lists may be applied equally or similarly to other

embodiments, which will be described later. Hereinafter, repeated descriptions may be omitted.

Constraints for reference picture lists configured for the slice of the subpicture of the target picture may be identical to those of reference picture lists configured for the target picture.

For example, if the value of sps_idr_rpl_present_flag is '1', when the picture is an Instantaneous Decoding Refresh (IDR) picture, or even when a subpicture is an IDR subpicture, a reference picture list may be configured for the slice.

For example, a reference picture list may be configured even in the case in which a subpicture is rearranged and is then changed to an IRAP subpicture.

For example, when a target subpicture is a Clean Random Access (CRA) picture, reference picture list L0 (e.g., RefPicList[0]) and reference picture list L1 (e.g., RefPicList[1]) may include a picture including an IRAP subpicture that precedes the target subpicture in a decoding order or an output order, and may not include a picture that precedes the picture including the IRAP subpicture in a decoding order or an output order.

For example, when the target subpicture is a Random Access Decodable Leading (RADL) picture, a specific picture for the RADL picture shall not be an active entry in reference picture list L0 or reference picture list L1. Here, the specific picture may be 1) a picture including a Random Access Skipped Leading (RASL) subpicture or 2) a picture that precedes a picture including an associated IRAP subpicture in a decoding order.

In order to use a cross-component adaptive loop filter, signaling may be performed through the syntax element "sps_ccalf_enabled_flag" defined in an SPS.

The syntax element "sps_ccalf_enabled_flag" may be information indicating whether the cross-component adaptive loop filter is to be used. When the value of sps_ccalf_enabled_flag is '0', a cross-component adaptive loop filter may not be used. When the value of sps_ccalf_enabled_flag is '1', a cross-component adaptive loop filter may be used. When sps_ccalf_enabled_flag is not signaled in the SPS or when sps_ccalf_enabled_flag is not present in the SPS, the value of sps_ccalf_enabled_flag may be regarded as '0'.

FIG. 35 illustrates the case where locations of subpictures in a picture are changed according to an embodiment.

In FIG. 35, an N-th picture, an N+1-th picture, and an N+2-th picture are illustrated.

Each of the pictures may include subpicture 1, subpicture 2, subpicture 3, and subpicture 4. For the illustrated "subpicture n", "n" may denote the subpicture ID of the corresponding subpicture.

In FIG. 35, arrows are present between corresponding subpictures n of the two pictures. An arrow from the subpicture of a first picture (i.e., N-th picture) to the subpicture of a second picture (i.e., N+1-th picture) may mean that the subpicture at the start point of the arrow is referred to by the subpicture at the end point of the arrow.

In FIG. 35, compared to the N-th picture, the locations of subpicture 4 and subpicture 2 are exchanged with each other in the N+1-th picture. In other words, rearrangement of the subpictures may be applied to the N+1-th picture.

When the locations of subpicture 4 and subpicture 2 are exchanged with each other in the N+1-th picture, the value of mixed_nalu_types_in_pic_flag in a PPS for the N+2-th picture may be set to '1'. Further, the slice type of subpicture 4 of the N+2-th picture may be one of IDR_W_RADL, IDR_N_LP, and CRA_NUT, and the slice type of subpicture 2 may be one of IDR_W_RADL, IDR_N_LP, and CRA_NUT.

In the case where a NAL unit type of a slice in subpicture 4 of the N+2-th picture is IDR, a reference picture list may be configured when the value of `sps_idr_rpl_present_flag` is '1' and the value of `mixed_nalu_types_in_pic_flag` is '1'.

After a first slice header of the target picture has been parsed, reference picture marking may be performed.

When a NAL unit type of a slice in the last subpicture (or the subpicture) of the corresponding picture is IDR, a reference picture list may be configured.

Here, when the value of `mixed_nalu_types_in_pic_flag` is set to '1', a reference picture list may be configured even if the NAL unit type of a certain slice in the picture is set to IDR.

After decoding performed on all slices in the picture has been completed, a decoded target picture may be marked with "used for short-term reference".

FIG. 36 illustrates the case where locations of subpictures treated as pictures are changed according to an embodiment.

In FIG. 36, an N-th picture, an N+1-th picture, and an N+2-th picture are illustrated.

Each of the pictures may include subpicture 1, subpicture 2, subpicture 3, and subpicture 4. For the illustrated "subpicture n", "n" may denote the subpicture ID of the corresponding subpicture.

In FIG. 36, arrows are present between corresponding subpictures n of the two pictures. An arrow from the subpicture of a first picture (i.e., N-th picture) to the subpicture of a second picture (i.e., N+1-th picture) may mean that the subpicture at the start point of the arrow is referred to by the subpicture at the end point of the arrow.

In FIG. 36, compared to the N-th picture, the locations of subpicture 1 and subpicture 3 are exchanged with each other in the N+1-th picture. In other words, rearrangement of the subpictures may be applied to the N+1-th picture.

When the value of `subpic_treated_as_pic_flag` for the subpicture is set to '1' in an SPS, the subpicture may be treated as a picture.

When the value of `mixed_nalu_types_in_pic_flag` in a PPS is '1', the NAL unit of at least one subpicture of the picture may have the value of a NAL unit type equal to STSA_NUT, RADL_NUT, RASL_NUT, IDR_W_RADL, IDR_N_LP, or CRA_NUT. In contrast, the NAL units all of other subpictures of the picture may have additional specific values of a NAL unit type. The additional specific values may be TRAIL_NUT, RADL_NUT, and RASL_NUT.

A NAL unit having a NAL unit type equal to STSA_NUT may indicate a Step-wise Temporal Sublayer Access (STSA) picture or a STSA subpicture. The STSA picture may be a coded picture for which each VCL NAL unit has a NAL unit type equal to STSA_NUT. The STSA subpicture may be a subpicture for which each VCL NAL unit has a NAL unit type equal to STSA_NUT.

A NAL unit having a NAL unit type equal to RADL_NUT may indicate a Random Access Decodable Leading (RADL) picture or an RADL subpicture. The RADL picture may be a coded picture for which each VCL NAL unit has a NAL unit type equal to RADL_NUT. The RADL subpicture may be a subpicture for which each VCL NAL unit has a NAL unit type equal to RADL_NUT.

A NAL unit having a NAL unit type equal to RASL_NUT may indicate a Random Access Skipped Leading (RASL) picture or an RASL subpicture. The RASL picture may be a coded picture for which at least one VCL NAL unit has a NAL unit type equal to RASL_NUT and other VCL NAL units all have a NAL unit type equal to RASL_NUT or

RADL_NUT. The RASL subpicture may be a subpicture for which each VCL NAL unit has a NAL unit type equal to RASL_NUT.

A NAL unit having a NAL unit type equal to IDR_W_RADL or IDR_N_LP may indicate an Instantaneous Decoding Refresh (IDR) picture or an IDR subpicture. The IDR picture may be an IRAP picture for which each NAL unit has a NAL unit type equal to IDR_W_RADL or IDR_N_LP. The IDR subpicture may be an IRAP subpicture for which each NAL unit has a NAL unit type equal to IDR_W_RADL or IDR_N_LP.

A NAL unit having a NAL unit type equal to CRA_NUT may indicate a Clean Random Access (CRA) picture or a CRA subpicture. The CRA picture may be an IRAP picture for which each NAL unit has a NAL unit type equal to CRA_NUT. The CRA subpicture may be an IRAP subpicture for which each NAL unit has a NAL unit type equal to CRA_NUT.

A NAL unit having a NAL unit type equal to TRAIL_NUT may be a trailing picture or a trailing subpicture. The trailing picture may be a picture for which each NAL unit has a NAL unit type equal to TRAIL_NUT. The trailing subpicture may be a subpicture for which each NAL unit has a NAL unit type equal to TRAIL_NUT.

As illustrated in FIG. 36, in the case where the value of `subpic_treated_as_pic_flag` is set to '1' for all subpictures in a sequence and the locations of subpictures are changed during encoding and/or decoding of each subpicture of the target picture, a collocated subpicture in another picture may not be used as a reference picture. When an area of a first subpicture of a first picture, located in the first picture, is identical to an area of a second subpicture of a second picture, located in the second picture, the first picture and the second picture may be subpictures collocated with each other.

When the collocated subpictures cannot be used as reference pictures, the subpicture of the target picture may be processed as an IRAP picture.

When rearrangement of subpictures is performed, subpictures to which rearrangement is applied may be processed as IRAP pictures. An IRAP picture may be a picture for which each VCL NAL unit has a NAL unit type equal to any one of IDR_W_RADL, IDR_N_LP, and CRA_NUT. In this case, since the subpicture is a part of the picture, the value of `mixed_nalu_types_in_pic_flag` in the PPS for the picture shall be '1'.

Calculation of a picture order count (POC) for the target picture may be performed on a first slice of the target picture.

As illustrated in FIG. 36, the locations of subpictures may be changed in the N+2-th picture. Due to this change, a subpicture at a first location may be changed from subpicture 1 to subpicture 3. Subpicture 3 changed to the first location may be processed as an IRAP picture.

Picture types of the subpictures of the N+2-th picture may include both IRAP and non-IRAP types. In other words, in the picture types of the subpictures of the N+2-th picture, both IRAP and non-IRAP types may be mixed with each other. Therefore, for decoding of non-IRAP subpictures, a reference picture list may be required. When a reference picture list may be configured, the reference picture list shall be configured for each slice in a non-IRAP picture.

Here, when the value of `sps_idr_rpl_present_flag` in the SPS is '1' and the value of `mixed_nalu_type_in_pic_flag` is '1', a reference picture list may be configured even if the NAL unit type of a slice in the subpicture is an IDR type.

For a first slice in the N+2-th picture, reference picture marking may be performed on decoded pictures in a Decoded Picture Buffer (DPB).

After reference picture marking, decoding on each slice may be performed.

After decoding performed on all slices in the target picture has been completed, the decoded target picture may be marked with “used for short-term reference”. Further, an Inter-Layer Reference Picture (ILRP) in reference picture list 0 or reference picture list 1 may be marked with “used for short-term reference”.

FIG. 37 illustrates another part of the structure of a picture parameter set indicating whether there are mixed NAL unit types according to an example.

A description made with reference to FIG. 34 may also be applied to an embodiment which will be described with reference to FIG. 37. Repeated descriptions thereof may be omitted.

As illustrated in FIG. 37, when the value of `mixed_nalu_types_in_pic_flag` in a PPS is 1, a reference picture list may be configured for a subpicture. Here, the configuration of the reference picture list may be irrelevant to the syntax element “`sps_idr_rpl_present_flag`” in an SPS. In other words, `sps_idr_rpl_present_flag` may not be present in the SPS, and a reference picture list may be configured for the subpicture when the value of `mixed_nalu_types_in_pic_flag` in the PPS is ‘1’ even if `sps_idr_rpl_present_flag` is not present in the SPS.

Unlike in FIG. 34, in FIG. 37, `sps_idr_rpl_present_flag` was deleted from two “if clauses”.

Therefore, in embodiments, 1) a determination of whether a reference picture list is to be configured in a slice header and 2) a determination of whether an override is to be performed on the number of activated reference indexes in the slice header may be performed regardless of the presence or value of `sps_idr_rpl_present_flag`. In other words, 1) a determination of whether a reference picture list is to be configured in the slice header and 2) a determination of whether an override is to be performed on the number of activated reference indexes in the slice header may be performed independently of `sps_idr_rpl_present_flag`.

Whether a reference picture list is to be configured in the slice header may be determined based on at least one of 1) the value of `rpl_info_in_ph_flag`, 2) the NAL unit type, and 3) the value of `mixed_nalu_types_in_pic_flag`.

For example, the configuration of a reference picture list in the slice header may be performed only when the value of `rpl_info_in_ph_flag` is ‘0’ in the slice header.

For example, the configuration of a reference picture list in the slice header may be performed when the NAL unit type is none of specific types. The specific types may be IDR_W_RADL and IDR_N_LP.

For example, the configuration of a reference picture list in the slice header may be performed when the value of `mixed_nalu_types_in_pic_flag` is ‘1’, regardless of the value of `sps_idr_rpl_present_flag` is ‘1’.

As illustrated in FIG. 36, the above-described conditions may be combined with each other. The configuration of a reference picture list in the slice header may be performed when condition 1 is true and condition 2 is true. Condition 1 may be a condition that the value of `rpl_info_in_ph_flag` in the slice header be ‘0’. Condition 2 may be a condition that 1) a NAL unit type be none of specific types, or a condition that 2) the value of `mixed_nalu_types_in_pic_flag` be ‘1’.

Next, whether an override is to be performed on the number of activated reference indexes in the slice header may be determined.

Whether an override is to be performed may be determined based on at least one of 1) the value of `rpl_info_in_ph_flag`, 2) the NAL unit type, 3) the value of `mixed_nalu_types_in_pic_flag`, 4) the slice type of each slice, and 5) the number of entries in a reference picture list.

For example, an override may be performed when the value of `rpl_info_in_ph_flag` is ‘1’.

For example, an override may be performed when the NAL unit type is none of the specific types. The specific types may be IDR_W_RADL and IDR_N_LP.

For example, an override may be performed when the value of `mixed_nalu_types_in_pic_flag` is ‘1’, regardless of the value of `sps_idr_rpl_present_flag` is ‘1’.

For example, an override may be performed when the type of a slice is not ‘I’ and the value of `num_ref_entries[0][RplIdx[0]]` is greater than ‘1’. Alternatively, an override may be performed when the type of a slice is ‘B’ and the value of `num_ref_entries[1][RplIdx[1]]` is greater than ‘1’.

`num_ref_entries[listIdx][rplIdx]` may specify the number of entries in a `ref_pic_list_struct(listIdx, rplIdx)` syntax structure.

An override may include signaling of `num_ref_idx_active_override_flag`.

FIG. 38 illustrates an additional case where locations of subpictures in a picture are changed according to an embodiment.

Descriptions of pictures, subpictures, arrows, and rearrangement of subpictures, made above with reference to FIG. 35, may also be applied to the embodiment which will be described below with reference to FIG. 38. Repeated descriptions thereof may be omitted.

When the locations of subpicture 4 and subpicture 2 are exchanged with each other in the N+1-th picture, the value of `mixed_nalu_types_in_pic_flag` in a PPS for the N+2-th picture may be set to ‘1’. Further, the slice type of subpicture 4 of the N+2-th picture may be one of IDR_W_RADL, IDR_N_LP, and CRA_NUT, and the slice type of subpicture 2 may be one of IDR_W_RADL, IDR_N_LP, and CRA_NUT.

In the case where the NAL unit type of a slice in subpicture 4 of the N+2-th picture is IDR, a reference picture list may be configured when the value of `mixed_nalu_types_in_pic_flag` is ‘1’. Here, the configuration of the reference picture list may be irrelevant to the syntax element “`sps_idr_rpl_present_flag`” in an SPS. In other words, `sps_idr_rpl_present_flag` may not be present in the SPS, and a reference picture list may be configured for the subpicture when the value of `mixed_nalu_types_in_pic_flag` in the PPS is ‘1’ even if `sps_idr_rpl_present_flag` is not present in the SPS.

After a first slice header of the target picture has been parsed, reference picture marking may be performed.

When a NAL unit type of a slice in the last subpicture (or the subpicture) of the corresponding picture is IDR, a reference picture list may be configured.

Here, when the value of `mixed_nalu_types_in_pic_flag` is set to ‘1’, a reference picture list may be configured even if the NAL unit type of a certain slice in the picture is set to IDR.

After decoding performed on all slices in the picture has been completed, a decoded target picture may be marked with “used for short-term reference”.

FIG. 39 illustrates the case where locations of subpictures treated as pictures are changed according to an embodiment.

Descriptions of pictures, subpictures, arrows, and rearrangement of subpictures, made above with reference to FIG. 36 may also be applied to the embodiment which will be described below with reference to FIG. 39. Repeated descriptions thereof may be omitted.

When the value of `subpic_treated_as_pic_flag` for a subpicture is set to '1' in an SPS, the subpicture may be treated as a picture.

When the value of `mixed_nalu_types_in_pic_flag` in a PPS is '1', a subpicture of a picture may have a NAL unit having a NAL unit type equal to STSA_NUT, RADL_NUT, RASL_NUT, IDR_W_RADL, IDR_N_LP, or CRA_NUT. In contrast, all of other subpictures of the picture shall have an additional specific value of the NAL unit type. The additional specific value may be TRAIL_NUT, RADL_NUT or RASL_NUT.

Descriptions of STSA_NUT, RADL_NUT, RASL_NUT, IDR_W_RADL, IDR_N_LP, and CRA_NUT, made above with reference to FIG. 36, may be applied to the embodiment which is described with reference to FIG. 39. Repeated descriptions thereof may be omitted.

As illustrated in FIG. 39, in the case where the value of `subpic_treated_as_pic_flag` is set to '1' for all subpictures in a sequence and the locations of subpictures are changed during encoding and/or decoding of each subpicture of the target picture, a collocated subpicture in another picture may not be used as a reference picture. When an area of a first subpicture of a first picture, located in the first picture, is identical to an area of a second subpicture of a second picture, located in the second picture, the first picture and the second picture may be subpictures collocated with each other.

When the collocated subpictures cannot be used as reference pictures, the subpicture of the target picture may be processed as an IRAP picture.

When rearrangement of subpictures is performed, subpictures to which rearrangement is applied may be processed as IRAP pictures. An IRAP picture may be a picture for which each VCL NAL unit has a NAL unit type equal to any one of IDR_W_RADL, IDR_N_LP, and CRA_NUT. In this case, since the subpicture is a part of the picture, the value of `mixed_nalu_types_in_pic_flag` in the PPS for the picture shall be '1'.

Calculation of a picture order count (POC) for the target picture may be performed on a first slice of the target picture.

As illustrated in FIG. 39, the locations of subpictures may be changed in the N+2-th picture. Due to this change, a subpicture at a first location may be changed from subpicture 1 to subpicture 3. Subpicture 3 changed to the first location may be processed as an IRAP picture.

Picture types of the subpictures of the N+2-th picture may include both IRAP and non-IRAP types. In other words, in the picture types of the subpictures of the N+2-th picture, both IRAP and non-IRAP types may be mixed with each other. Therefore, for decoding of non-IRAP subpictures, a reference picture list may be required. When a reference picture list may be configured, the reference picture list shall be configured for each slice in a non-IRAP picture.

When the value of `mixed_nalu_type_in_pic_flag` is '1', a reference picture list may be configured even if the NAL unit type of the slice in the subpicture is IDR. Here, the configuration of the reference picture list may be irrelevant to the syntax element "`sps_idr_rpl_present_flag`" in an SPS. In other words, `sps_idr_rpl_present_flag` may not be present in the SPS, and a reference picture list may be configured for

the subpicture when the value of `mixed_nalu_types_in_pic_flag` in the PPS is '1' even if `sps_idr_rpl_present_flag` is not present in the SPS.

For a first slice in the N+2-th picture, reference picture marking may be performed on decoded pictures in a Decoded Picture Buffer (DPB).

After reference picture marking, decoding on each slice may be performed.

After decoding performed on all slices in the target picture has been completed, the decoded target picture may be marked with "used for short-term reference". Further, an Inter-Layer Reference Picture (ILRP) in reference picture list 0 or reference picture list 1 may be marked with "used for short-term reference".

FIG. 40 illustrates a further part of the structure of a picture parameter set indicating whether there are mixed NAL unit types according to an example.

A description made with reference to FIG. 34 may also be applied to an embodiment which will be described below with reference to FIG. 40. Repeated descriptions thereof may be omitted.

When IDR is applied to a single picture other than a subpicture, a reference picture list may be configured if the value of `sps_idr_rplpresent_flag` in an SPS is '1'. Further, even if the value of `sps_idr_rplpresent_flag` is '0', when the picture is changed to an IDR picture through rearrangement of subpictures, the value of `mixed_nalu_types_in_pic_flag` is set to '1' in the PPS, and thus a reference picture list may be configured for a slice in the subpicture. In other words, when the value of `sps_idr_rpl_present_flag` is '1' or when the value of `mixed_nalu_types_in_pic_flag` is '1', a reference picture list may be configured.

Therefore, in the embodiments, when the value of `mixed_nalu_types_in_pic_flag` is '1', 1) a determination of whether a reference picture list is to be configured in a slice header and 2) a determination of whether an override is to be performed on the number of activated reference indexes in the slice header may be performed regardless of the presence or value of `sps_idr_rpl_present_flag`. In other words, when the value of `mixed_nalu_types_in_pic_flag` is '1', 1) a determination of whether a reference picture list is to be configured in the slice header and 2) a determination of whether an override is to be performed on the number of activated reference indexes in the slice header may be independent of `sps_idr_rpl_present_flag`.

FIG. 41 is a flowchart of an encoding method according to an embodiment.

At step 4110, the processing unit 1610 of the encoding apparatus 1600 may generate information about a target picture by performing encoding on the target picture.

The information about the target picture may include information described in connection with the foregoing embodiments and information included in syntax elements illustrated in FIGS. 22 to 40.

For example, the information about the target picture may include information for a determination of a reference picture list for the target picture and information for prediction based on the reference picture list.

For example, the information about the target picture may be information used to decode the target picture by the decoding apparatus 1700. Decoding on the target picture may be performed by utilizing the information about the target picture.

For example, the information about the target picture may include the above-described picture information, information about subpictures, information about slices, information about an SPS, information about a PPS, information about a

slice header, etc. Further, the information about the target picture may include information that is the target to be signaled.

In an embodiment, encoding of specific information may mean that the specific information is included in or added to a bitstream. Alternatively, encoding of the specific information may mean signaling of the specific information.

At step 4120, the processing unit 1610 of the encoding apparatus 1600 may generate a bitstream including the information about the target picture.

At step 4130, the communication unit 1620 of the encoding apparatus 1600 may transmit the bitstream including the information about the target picture to the decoding apparatus 1700.

Alternatively, the storage unit 1630 of the encoding apparatus 1600 may store the bitstream including the information about the target picture.

FIG. 42 is a flowchart of a decoding method according to an embodiment.

At step 4210, the processing unit 1710 of the decoding apparatus 1700 may receive a bitstream including information about a target picture from the encoding apparatus 1600.

Alternatively, the processing unit 1710 of the decoding apparatus 1700 may acquire the bitstream including information about the target picture from the storage unit 1730.

Alternatively, the processing unit 1710 of the decoding apparatus 1700 may acquire a bitstream including information about the target picture from a computer-readable storage medium. The computer-readable storage medium may include the bitstream including the information about the target picture.

The processing unit 1710 of the decoding apparatus 1700 may acquire the information about the target picture from the bitstream.

At step 4220, the processing unit 1700 of the encoding apparatus 1710 may perform decoding on the target picture using the information about the target picture.

For example, the decoding may include a determination of a reference picture list for the target picture and prediction based on the reference picture list. The information about the target picture may include information described in connection with the foregoing embodiments and information included in the syntax elements illustrated in FIGS. 22 to 40.

For example, the information about the target picture may include the above-described picture information, information about subpictures, information about slices, information about an SPS, information about a PPS, information about a slice header, etc. Further, the information about the target picture may include information that is the target to be signaled.

The processing unit 1710 of the decoding apparatus 1700 may extract the information about the target picture from the bitstream.

The information about the target picture may be decoded depending on the priority described in the embodiments described above with reference to FIGS. 22 to 40.

In an embodiment, decoding of specific information may mean that the specific information is extracted from the bitstream. Alternatively, decoding of the specific information may mean signaling of the specific information.

Provided are a method, an apparatus and a storage medium that perform encoding/decoding on an image using a subpicture.

Provided are a method, an apparatus and a storage medium that configure a reference picture list for motion compensation in the case where the location of a reference

subpicture to be referred to for motion compensation is different from the location of a target subpicture.

Provided are a method, an apparatus and a storage medium that configure a reference picture list and provide an override for the number of activated reference indexes.

The embodiments may be performed using the same method by the encoding apparatus 1600 and by the decoding apparatus 1700. Also, the image may be encoded/decoded using at least one of the embodiments or at least one combination thereof.

The order of application of the embodiments may be different from each other by the encoding apparatus 1600 and the decoding apparatus 1700, and the order of application of the embodiments may be (at least partially) identical to each other by the encoding apparatus 1600 and the decoding apparatus 1700.

The embodiments may be performed for each of a luma signal and a chroma signal, and may be equally performed for the luma signal and the chroma signal.

The form of a block to which the embodiments of the present disclosure are applied may have a square or non-square shape.

The embodiments of the present disclosure may be applied according to the size of at least one of a target block, a coding block, a prediction block, a transform block, a current block, a coding unit, a prediction unit, a transform unit, a unit, and a current unit. Here, the size may be defined as a minimum size and/or a maximum size so that the embodiments are applied, and may be defined as a fixed size at which the embodiments are applied. Further, in the embodiments, a first embodiment may be applied to a first size, and a second embodiment may be applied to a second size. That is, the embodiments may be compositely applied according to the size. Further, the embodiments of the present disclosure may be applied only to the case where the size is equal to or greater than the minimum size and is less than or equal to the maximum size. That is, the embodiments may be applied only to the case where a block size falls within a certain range.

Further, the embodiments of the present disclosure may be applied only when a condition that the size of a target block is greater than or equal to a minimum size and a condition that the size of the target block is less than or equal to a maximum size are satisfied, wherein each of the minimum size and the maximum size may be a size of one of blocks and units described above in the embodiment. That is, a block to be a target of the minimum size and a block to be a target of the maximum size may be different from each other. For example, the embodiments of the present disclosure may be applied only when a size of a target block is greater than a minimum size of a block and less than a maximum size of a block.

For example, the embodiments may be applied only to the case where the size of a target block is equal to or greater than 8×8. For example, the embodiments may be applied only to the case where the size of a target block is equal to or greater than 16×16. For example, the embodiments may be applied only to the case where the size of a target block is equal to or greater than 32×32. For example, the embodiments may be applied only to the case where the size of a target block is equal to or greater than 64×64. For example, the embodiments may be applied only to the case where the size of a target block is equal to or greater than 128×128. For example, the embodiments may be applied only to the case where the size of a target block is 4×4. For example, the embodiments may be applied only to the case where the size of a target block is less than or equal to 8×8. For example,

the embodiments may be applied only to the case where the size of a target block is less than or equal to 16×16 . For example, the embodiments may be applied only to the case where the size of a target block is equal to or greater 8×8 and less than or equal to 16×16 . For example, the embodiments

may be applied only to the case where the size of a target block is equal to or greater 16×16 and less than or equal to 64×64 . The embodiments of the present disclosure may be applied depending on a temporal layer. In order to identify a temporal layer to which the embodiments are applicable, a separate identifier may be signaled, and the embodiments may be applied to the temporal layer specified by the corresponding identifier. Here, the identifier may be defined as the lowest (bottom) layer and/or the highest (top) layer to which the embodiments are applicable, and may be defined as being indicating a specific layer to which the embodiments are applied. Further, a fixed temporal layer to which the embodiments are applied may also be defined.

For example, the embodiments may be applied only to the case where the temporal layer of a target image is the lowermost layer. For example, the embodiments may be applied only to the case where the temporal layer identifier of a target image is equal to or greater than 1. For example, the embodiments may be applied only to the case where the temporal layer of a target image is the highest layer.

A slice type or a tile group type to which the embodiments of the present invention to which the embodiments are applied may be defined, and the embodiments of the present invention may be applied depending on the corresponding slice type or tile group type.

In the above-described embodiments, it may be construed that, during the application of specific processing to a specific target, assuming that specified conditions may be required and the specific processing is performed under a specific determination, a specific coding parameter may be replaced with an additional coding parameter when a description has been made such that whether the specified conditions are satisfied is determined based on the specific coding parameter, or such that the specific determination is made based on the specific coding parameter. In other words, it may be considered that a coding parameter that influences the specific condition or the specific determination is merely exemplary, and it may be understood that, in addition to the specific coding parameter, a combination of one or more additional coding parameters functions as the specific coding parameter.

In the above-described embodiments, although the methods have been described based on flowcharts as a series of steps or units, the present disclosure is not limited to the sequence of the steps and some steps may be performed in a sequence different from that of the described steps or simultaneously with other steps. Further, those skilled in the art will understand that the steps shown in the flowchart are not exclusive and may further include other steps, or that one or more steps in the flowchart may be deleted without departing from the scope of the disclosure.

The above-described embodiments include examples in various aspects. Although all possible combinations for indicating various aspects cannot be described, those skilled in the art will appreciate that other combinations are possible in addition to explicitly described combinations. Therefore, it should be understood that the present disclosure includes other replacements, changes, and modifications belonging to the scope of the accompanying claims.

The above-described embodiments according to the present disclosure may be implemented as a program that can be

executed by various computer means and may be recorded on a computer-readable storage medium. The computer-readable storage medium may include program instructions, data files, and data structures, either solely or in combination. Program instructions recorded on the storage medium may have been specially designed and configured for the present disclosure, or may be known to or available to those who have ordinary knowledge in the field of computer software.

A computer-readable storage medium may include information used in the embodiments of the present disclosure. For example, the computer-readable storage medium may include a bitstream, and the bitstream may contain the information described above in the embodiments of the present disclosure.

The computer-readable storage medium may include a non-transitory computer-readable medium.

Examples of the computer-readable storage medium include all types of hardware devices specially configured to record and execute program instructions, such as magnetic media, such as a hard disk, a floppy disk, and magnetic tape, optical media, such as compact disk (CD)-ROM and a digital versatile disk (DVD), magneto-optical media, such as a floptical disk, ROM, RAM, and flash memory. Examples of the program instructions include machine code, such as code created by a compiler, and high-level language code executable by a computer using an interpreter. The hardware devices may be configured to operate as one or more software modules in order to perform the operation of the present disclosure, and vice versa.

As described above, although the present disclosure has been described based on specific details such as detailed components and a limited number of embodiments and drawings, those are merely provided for easy understanding of the entire disclosure, the present disclosure is not limited to those embodiments, and those skilled in the art will practice various changes and modifications from the above description.

Accordingly, it should be noted that the spirit of the present embodiments is not limited to the above-described embodiments, and the accompanying claims and equivalents and modifications thereof fall within the scope of the present disclosure.

What is claimed is:

1. A decoding method, comprising:

acquiring information about a target picture from a bitstream; and

performing decoding using the information about the target picture,

wherein the decoding comprises a determination of a reference picture list for the target picture and prediction for a target block in the target picture based on the reference picture list,

wherein the information about the target picture includes a slice header for a slice of the target picture,

wherein the bitstream includes first information, wherein the first information indicates whether reference

picture list syntax elements are capable of being present in slice headers of slices with a Network Abstraction Layer (NAL) unit type equal to a specific type,

wherein the reference picture list is configured for the target picture in a case that the first information indicates that the reference picture list syntax elements are capable of being present in the slice headers of the slices with the NAL unit type equal to the specific type,

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wherein the decoding comprises a determination of whether an override on a number of activated reference indexes in the slice header is to be performed or not, and

wherein the determination of whether the override is to be performed is performed regardless of a value of the first information.

2. The decoding method of claim 1, wherein the determination of the reference picture list in the slice header is performed when the NAL unit type of the slice is not the specific type.

3. The decoding method of claim 1, wherein: the bitstream includes a picture header for the target picture, and

the determination of the reference picture list in the slice header is performed when information related to the reference picture list is not present in the picture header for the target picture.

4. The decoding method of claim 1, wherein: the bitstream further includes second information, the override includes decoding on the second information, and

the second information is information indicating whether the override is to be performed on the number of activated reference indexes for the reference picture list.

5. The decoding method of claim 4, wherein, when the second information indicates that the override is to be performed, the information indicating the number of activated reference indexes for the reference picture list is decoded.

6. The decoding method of claim 5, wherein the specific type is an Instantaneous Decoding Refresh (IDR)_N_LP or IDR_W_Random Access Decodable Leading (RADL) type.

7. An encoding method, comprising: generating encoding information by performing encoding on a target picture; and

generating a bitstream including the encoding information,

wherein the encoding information includes information for a determination of a reference picture list for the target picture and information for prediction for a target block in the target picture based on the reference picture list,

wherein the encoding information includes a slice header for a slice of the target picture,

wherein the bitstream includes first information,

wherein the first information indicates whether reference picture list syntax elements are capable of being present in slice headers of slices with a Network Abstraction Layer (NAL) unit type equal to a specific type,

wherein the reference picture list is configured for the target picture in a case that the first information indicates that the reference picture list syntax elements are capable of being present in the slice headers of the slices with the NAL unit type equal to the specific type in decoding for the target picture,

wherein the encoding information further includes information required to determine whether an override is to be performed on a number of activated reference indexes in the slice header in the decoding for the target picture, and

wherein the determination of whether the override is to be performed is performed regardless of a value of the first information.

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8. The encoding method of claim 7, wherein the determination of the reference picture list in the slice header is performed when the NAL unit type of the slice is not the specific type.

9. The encoding method of claim 7, wherein:

the bitstream includes a picture header for the target picture, and

the determination of the reference picture list in the slice header is performed when information related to the reference picture list is not present in the picture header.

10. The encoding method of claim 7, wherein:

the bitstream further includes second information, and the second information is information indicating whether the override is to be performed on the number of activated reference indexes for the reference picture list.

11. The encoding method of claim 10, wherein, when the second information indicates that the override is to be performed, the information indicating the number of activated reference indexes for the reference picture list is encoded.

12. The encoding method of claim 11, wherein the specific type is an Instantaneous Decoding Refresh (IDR)_N_LP or IDR_W_Random Access Decodable Leading (RADL) type.

13. A non-transitory computer-readable storage medium storing a bitstream, generated by a processor performing an encoding method, wherein:

the bitstream includes information about a target picture and first information,

the information about the target picture is acquired from the bitstream,

decoding is performed using the information about the target picture,

the decoding includes a determination of a reference picture list for the target picture and prediction for a target block in the target picture based on the reference picture list,

the information about the target picture includes a slice header for a slice of the target picture,

the first information indicates whether reference picture list syntax elements are capable of being present in slice headers of slices with a Network Abstraction Layer (NAL) unit type equal to a specific type,

the reference picture list is configured for the target picture in a case that the first information indicates that the reference picture list syntax elements are capable of being present in the slice headers of the slices with the NAL unit type equal to the specific type,

the decoding comprises a determination of whether an override on a number of activated reference indexes in the slice header is to be performed or not, and

the determination of whether the override is to be performed is performed regardless of a value of the first information.

14. The non-transitory computer-readable storage medium of claim 13, wherein the determination of the reference picture list in the slice header is performed when the NAL unit type of the slice is not the specific type.

15. The non-transitory computer-readable storage medium of claim 13, wherein:

the bitstream includes a picture header for the target picture, and

the determination of the reference picture list in the slice header is performed when information related to the reference picture list is not present in the picture header of the target picture.

16. The non-transitory computer-readable storage medium of claim 13, wherein:

the bitstream further includes second information,

the override includes decoding on the second information,
and

the second information is information indicating whether
the override is to be performed on the number of
activated reference indexes for the reference picture
list.

17. The non-transitory computer-readable storage medium of claim 16, wherein, when the second information indicates that the override is to be performed, the information indicating the number of activated reference indexes for the reference picture list is decoded.

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